

Scalable Bayesian Estimation for Non-Gaussian Structural VARs*

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Abstract

Non-Gaussian features of macroeconomic data identify structural VARs without theory-based restrictions, but two computational problems arise: the conditional posterior of the impact matrix has no standard form, and, with k variables, draws must be relabeled among $k!2^k$ sign and permutation configurations. We propose a solution for both. We show that writing the inverse impact matrix as a diagonal matrix times two unit-triangular matrices yields standard conditional posteriors for the impact matrix in any model whose shocks are Gaussian conditional on a diagonal matrix of scales, including Student- t shocks, stochastic volatility, and Markov switching. We then show that the relabeling step is a linear assignment problem that the Hungarian algorithm solves exactly in under a millisecond for dozens of variables. In a ten-variable Student- t SVAR for the U.S., no shock's posterior median share of GDP fluctuations exceeds 30 percent at horizons up to 24 quarters, whereas max-share identification attributes nearly 80 percent to one shock.

Keywords: Non-Gaussianity, Shocks, identification, Bayesian inference, Gibbs sampler.

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1 Introduction

Structural Gaussian Vector Autoregressive (SVAR) models are widely used in the time series macroeconomic literature. Part of their popularity draws from the so-called orthogonal-reduced-form parametrization, which facilitates identification and Bayesian inference, and allows for prior beliefs that are flat on the dimension in which the data are uninformative (Uhlig 2005, Arias et al. 2018, 2025). However, the reduced-form-orthogonal parametrization raises challenges when conditional Gaussianity is combined with a stochastic covariance structure of the shocks. As advocated by Lewis (2025), this is an important class of models, because the implied non-Gaussian features, such as heavy tails and time-varying volatility, identify the shocks of a SVAR without requiring restrictions from economic theory. Unlike such restrictions, the properties that deliver identification are properties of the data: they can be tested, and models with and without them can be compared through their marginal data densities.

Two well-known obstacles are responsible for the existing computational challenges associated with non-Gaussian SVAR models. First, as in any SVAR model, the conditional posterior of the matrix that maps shocks into variables has no standard form. Second, non-Gaussianity identifies the shocks only up to their sign and ordering. Every posterior draw must therefore be relabeled before impulse responses or variance decompositions can be computed. Since the number of candidate labelings equals $k! 2^k$ for a model with k variables, this set becomes computationally untractable to explore even for moderate k .

This paper develops a new approach to tackle the above obstacles. First, we propose a novel parametrization that leads to a Gibbs sampler for the impact matrix. The sampler applies to any SVAR whose shocks are Gaussian conditional on a diagonal matrix of scales. Second, we show that the relabeling step can be rewritten as a linear assignment problem that can be solved exactly, and in a fraction of a millisecond even for models with dozens of variables.

Finally, we use the method to estimate a ten-variable SVAR with Student- t shocks on the data of [Angeletos et al. \(2020\)](#), who attribute close to 80 percent of the volatility of U.S. real GDP to a single shock. Once identification comes from the heavy tails of the data rather than from a max-share criterion, no shock has a posterior median contribution to the forecast error variance of GDP above 30 percent at any horizon up to 24 quarters.

We study SVARs whose structural shocks are Gaussian once one conditions on a diagonal matrix of time-varying scales. Different laws of motion for these scales deliver Student- t shocks, stochastic volatility, Markov-switching or regime-based heteroskedasticity, and combinations of these. The difficulty is the determinant of B , the matrix of impact effects of the shocks. It enters the likelihood raised to the power of the sample size and ties all entries of the matrix together. Our first contribution is a reparametrization that unties them. We write the inverse of B as the product of a diagonal matrix and two triangular matrices, one lower and one upper, each with ones on its diagonal. The determinant then depends only on the diagonal matrix, whose squared entries have Gamma conditional posteriors, while the two triangular matrices enter the likelihood only through a quadratic form, hence have Normal conditional posteriors. For a class of priors that includes a flat prior on B , every block of the sampler therefore has a conditional posterior of standard form, and no Metropolis-Hastings step is required. The sampler is invariant to the ordering of the variables. It needs no identifying restrictions but accommodates them: a recursive model is the special case in which the upper triangular matrix is the identity, and sign restrictions truncate the conditional posteriors without changing their form. Under Gaussian shocks, a flat prior on B implies a uniform prior on the orthogonal matrix of [Arias et al. \(2018\)](#), so the decomposition extends their orthogonal reduced form to models with time-varying scales.

Our second contribution concerns what to do with the draws. When the shocks are identified

from the data alone, the likelihood has $k!2^k$ modes that differ only by the sign and ordering of the shocks. A sampler that visits several of them produces marginal posteriors for the entries of B that are symmetric around zero and economically meaningless, even when the shocks are well identified (Hamilton et al. 2007). We relabel each draw by reordering and flipping the signs of its columns so that it comes as close as possible to a target matrix. This criterion generalizes the likelihood-preserving normalization of Waggoner & Zha (2003b), which handles signs, to signs and permutations. Checking every candidate relabeling is infeasible beyond a handful of variables. We show that the closest relabeling can be found without enumeration. The ordering that minimizes the distance solves a linear assignment problem, which can be solved using the Hungarian algorithm, which takes about a ten-thousandth of a second for twenty variables, where exhaustive search would require 2.5×10^{24} evaluations. Because the transformation preserves the likelihood and the prior, it does not change the posterior over sets of observationally equivalent parameter values, nor the posterior of any quantity that does not depend on the labeling of the shocks; it selects one representative from each such set. The same rule can be applied to bootstrap draws of GMM or maximum likelihood estimates.

Simulations show that both pieces matter. With 800 observations and $k = 10$ (20), the sampler, including the estimation of the target matrix, runs in about 3 minutes (13 minutes), to generate 10,000 draws. Regressing posterior means of impact effects and impulse responses on their true values gives slopes close to one. Other proposed normalizations in the literature, applied to the same draws, leave many entries of B centered at zero because they do not target a single mode.

We then return to the question of Angeletos et al. (2020): is there one shock behind U.S. business cycles? We estimate their ten-variable model on their quarterly data from 1955 to 2017 with mutually independent Student- t shocks and a flat prior on B , so that identification

comes only from the heavy tails. The posterior degrees of freedom are low for nine of the ten shocks, tests on the estimated residuals reject Gaussianity for seven of the ten equations, and no pair of estimated structural shocks rejects independence. No shock has a median contribution to the forecast error variance of GDP above 30 percent at any horizon up to 24 quarters. Two shocks stand out. One raises real activity, lowers unemployment, leaves TFP unchanged, and moves inflation little; it resembles the main business cycle shock of [Angeletos et al. \(2020\)](#) but explains a smaller share of GDP. The other matters at longer horizons, moves TFP, as a technology shock does ([Kydland & Prescott 1982](#)), and moves output and inflation in opposite directions, as a technology shock does in New Keynesian models ([Smets & Wouters 2007](#)). The two sets of results differ because the Student- t likelihood uses distributional information beyond second moments, whereas a max-share shock can be a combination of several structural shocks ([Wolf 2020](#), [Francis & Kindberg-Hanlon 2022](#), [Dou et al. 2025](#)).

[Waggoner & Zha \(2003a\)](#) developed a Gibbs sampler for homoskedastic, Gaussian SVARs with over-identifying restrictions. Papers that estimate SVARs with Student- t shocks either impose a triangular impact matrix ([Clark & Ravazzolo 2015](#), [Chiu et al. 2017](#), [Karlsson & Mazur 2020](#)), which restricts the timing of the shocks and is over-identifying once the data identify the model, or use Metropolis-Hastings steps ([Lanne & Luoto 2020](#), [Brunnermeier et al. 2021](#), [Anttonen et al. 2024](#)). [Jarociński \(2024\)](#) normalizes draws with a criterion similar to ours but enumerates all permutations. Our reparameterization is similar to other changes of coordinates that have made SVAR inference tractable, such as the orthogonal reduced form of [Rubio-Ramirez et al. \(2010\)](#) and [Arias et al. \(2018\)](#) and the correlation-matrix parametrization of [Archakov & Hansen \(2021\)](#). The assignment formulation has precedents outside macroeconomics: [Ilmonen et al. \(2010\)](#) use the Hungarian algorithm to compute a distance between mixing matrices in independent component analysis, and [Cron & West](#)

(2011) use it to resolve label switching in finite mixture models. If some shocks are Gaussian, the non-Gaussian block remains identified and our procedure recovers it, while the Gaussian block is identified only up to rotation (Guay 2021, Maxand 2020, Anttonen et al. 2026).

2 Our Approach

In this section, we describe a conditionally Gaussian class of models, propose our reparametrization, explain how it yields a Gibbs sampler, and detail our normalization of sign and permutation equivalence. Sections A-C of the Online Appendix provide the technical derivations.

2.1 The Model

We study models of the form

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^p \Pi_l \mathbf{y}_{t-l} + B \boldsymbol{\epsilon}_t, \quad (1)$$

$$\boldsymbol{\epsilon}_t = D_t^{\frac{1}{2}} \mathbf{e}_t, \quad (2)$$

$$\mathbf{e}_t \stackrel{\text{i.i.d.}}{\sim} N(\mathbf{0}, I). \quad (3)$$

The vector $\mathbf{y}_t$ collects the k endogenous variables of the model, whose evolution depends on p lags of the endogenous variables and on the vector of constants $\mathbf{c}$. The $k \times m$ matrix $\Pi = [\Pi_1, \dots, \Pi_p, \mathbf{c}]$ contains the autoregressive parameters and the constant terms, with $m = kp + 1$. $\boldsymbol{\epsilon}_t$ is of dimension $k \times 1$ and captures structural shocks, whose conditional variances at time t are stored in the positive diagonal matrix $D_t = \text{diag}(d_{1t}, \dots, d_{kt})$.¹ $\boldsymbol{\epsilon}_t$ are mutually independent conditioning on D_t , although independence may not need to hold unconditionally. We assume that $\mathbf{e}_t$ is independent of $\{D_s\}_{s=1}^T$, so that conditional on D the innovations remain i.i.d. Gaussian; this rules out, for example, leverage-type dependence

¹Positivity of the diagonal elements makes the matrix square root unique.

between the scales and the Gaussian innovations. We combine $\{D_t\}_{t=1}^T$ in the $kT \times kT$ block diagonal matrix $D = \text{diag}(D_1, \dots, D_T)$. When $D = I_{kT}$ the model simplifies to the standard SVAR model with Gaussian shocks.

Model (1)-(3) summarizes a very wide class of models, in that alternative specifications of D_t can conveniently model popular departures from the homoskedastic Gaussian model (see [Lewis 2025](#) for a survey). Some choices include:

1. **Symmetric Student- t shocks.** When

$$d_{it} \stackrel{\text{i.i.d.}}{\sim} \text{i}\Gamma\left(\frac{v_i}{2}, \frac{v_i - 2}{2}\right), \quad v_i > 2, \quad (4)$$

$E(d_{it}) = 1$ and $\epsilon_{it} = d_{it}^{1/2} e_{it}$ has a unit-variance Student- t distribution with v_i degrees of freedom ([Geweke 1993](#)). Independence of d_{it} across i and t produces structural shocks that are mutually and serially independent after D_t is integrated out. Models along this line have been extensively used to identify structural shocks via non-Gaussianity, including from [Chiu et al. \(2017\)](#), [Karlsson & Mazur \(2020\)](#), [Lanne & Luoto \(2020\)](#), [Brunnermeier et al. \(2021\)](#), [Karlsson et al. \(2023\)](#), [Anttonen et al. \(2024, 2026\)](#).

2. **Stochastic Volatility of autoregressive form.** Consider

$$d_{it} = \exp(h_{it}), \quad h_{it} = \mu_i + \rho_i(h_{i,t-1} - \mu_i) + \sigma_i u_{it}^h, \quad u_{it}^h \stackrel{\text{i.i.d.}}{\sim} N(0, 1), \quad (5)$$

with $|\rho_i| < 1$. If the volatility innovations u_{it}^h are statistically dependent across shocks, the resulting structural shocks ϵ_t are mutually uncorrelated, although not independent. Nonproportional volatility dynamics supply identifying information through the autocovariances of squared innovations ([Kim et al. 1998](#), [Stock & Watson 2007](#), [Lewis 2021](#), [Bertsche & Braun 2022](#), [Chan & Eisenstat 2018](#), [Chan 2023](#)).

Several alternative or complementary specifications are possible, as well as extensions.²

²Replacing (4) with

$$d_{it} \stackrel{\text{i.i.d.}}{\sim} \Gamma(1, 1), \quad (6)$$

One key challenge addressed in our paper is how to draw from the conditional posterior distribution of B in a potentially medium to large specification of the model ($k \geq 10$). We refer to this object of interest as $p(B|Y, \Theta_{-B})$, where Θ captures all parameters of the model (including D) and Θ_{-B} represents all parameters of the model except B . Inference on B is needed not only for structural analysis for the causal effect of the structural shocks. It is also relevant for forecasting whenever the shocks have shock-specific distributions or scale dynamics, that is, whenever $D \neq I_{kT}$ in (1)-(3), since the predictive distribution then depends on B beyond the reduced-form covariance BB' ; this is frequently the case (for example in [Clark & Ravazzolo 2015](#) and [Chan 2023](#)).

We focus our attention on inference on B , starting from the known fact that $p(B|Y, \Theta_{-B})$ is not a common form. Inference on the remaining parameters is already feasible using existing methods even when k is large. We also work with specifications of D which imply that $p(D|Y, \Theta_{-D})$ is computationally unchallenging to draw from. For instance, when ϵ_t are independent t -distributed shocks, $p(D|Y, \Theta_{-D})$ is inverse Gamma ([Geweke 1993](#)).

makes ϵ_t Laplace rather than t -distributed. Known volatility regimes can be modeled using

$$D_t = D^{(r(t))} = \text{diag}(d_{1,r(t)}, \dots, d_{k,r(t)}), \quad r(t) \in \{1, \dots, R\}, \quad (7)$$

see [Rigobon \(2003\)](#) and [Lewis \(2022\)](#). Markov-switching volatility can be modeled using the state process

$$D_t = D^{(s_t)}, \quad \Pr(s_t = j \mid s_{t-1} = \ell) = p_{\ell j}, \quad s_t \in \{1, \dots, R\}, \quad (8)$$

see [Lanne et al. \(2010\)](#), [Lütkepohl & Woźniak \(2020\)](#) and [Herwartz & Lütkepohl \(2014\)](#). A smooth transition in the evolution of the reduced form covariance matrix is also possible ([Lütkepohl & Netsunajev 2017](#) and [Bruns & Piffer 2024](#)). Some of the above modeling features have been used in the literature jointly, for instance [Clark & Ravazzolo \(2015\)](#) and [Chiu et al. \(2017\)](#), who combine t -distributed shocks and stochastic volatility. Extensions of the model include allowing for asymmetries in t -distributed shocks ([Karlsson & Mazur 2020](#), [Karlsson et al. 2023](#), [Anttonen et al. 2026](#)), modeling the distribution of shocks nonparametrically via Dirichlet processes ([Braun 2023](#)), and more.

2.2 Sampling from $p(B|Y, \Theta_{-B})$

Deriving a convenient sampler from $p(B|Y, \Theta_{-B})$ raises challenges. Following [Cogley & Sargent \(2005\)](#), many papers in the literature parametrize the model using

$$A = B^{-1}, \quad (9)$$

and draw from $p(A|Y, \Theta_{-A})$ under the assumption that A (and hence B) is triangular ([Clark & Ravazzolo 2015](#), [Chiu et al. 2017](#), [Karlsson & Mazur 2020](#), [Karlsson et al. 2023](#)). Depending on the specification of D , the triangular structure on A can introduce over-identifying restrictions and will bias both the forecasting and the structural properties of the model. Papers that avoid zero restrictions on (A, B) have so far followed two alternative strategies. In the literature on stochastic volatility, some use the sampler by [Waggoner & Zha \(2003a\)](#), which was developed for homoskedastic models with overidentifying restrictions and applied more broadly ([Villani 2009](#), [Chan et al. 2024](#)). Instead, many contributions that do not employ stochastic volatility resort to MCMC procedures (as in [Lanne & Luoto 2020](#), [Brunnermeier et al. 2021](#), [Anttonen et al. 2024, 2026](#)).

One of our contributions is to develop a new way of drawing from $p(B|Y, \Theta_{-B})$ using a Gibbs sampler, hence using only distributions of common form. This means that no MCMC step is required. Our core insight hinges on a new reparametrization, which, to our knowledge, has not been used in Bayesian econometrics. Reparametrize the model as

$$A = \Lambda L U, \quad (10)$$

or equivalently as $B = U^{-1}L^{-1}\Lambda^{-1}$, with Λ a $k \times k$ diagonal matrix with entries λ_i , and (L, U) lower- and upper-triangular $k \times k$ matrices, respectively, both restricted to have ones on the main diagonals. We work under the maintained assumption that all leading principal minors of A are non zero, as with the LDU decomposition.³ The determinant

³To see the relation between decomposition (10) and the LDU decomposition, apply the latter to get

of A (equivalently, the inverse of the determinant of B) enters the likelihood as $|\det A|^T$, which complicates posterior derivations unless additional assumptions are imposed (Sims & Zha 1998, Waggoner & Zha 2003a, Kocięcki et al. 2025). However, our reparametrization conveniently yields

$$|\det(A)| = \prod_{i=1}^k |\lambda_i|. \quad (11)$$

This is important for two reasons. First, the determinant of A only depends on the diagonal entries of Λ , because (L, U) drop out due to the unitary entries on the main diagonal. Second, $\prod_{i=1}^k |\lambda_i|$ resembles the kernel (i.e., the non-normalized density) of a Gamma distribution. It is these two properties that make a Gibbs sampler feasible for $p(B|Y, \Theta_{-B})$, because they lead to a Gamma conditional posterior for Λ and a Normal conditional posterior for L and U . This is not feasible under a standard LDU decomposition of A . What is required is a prior that does not conflict with this feature of the likelihood function.⁴

We work with the class of priors

$$p(A) \propto |\det(A)|^\alpha. \quad (12)$$

 $A = \tilde{L}\tilde{D}\tilde{U}$. Then set $\Lambda = \tilde{D}$, $U = \tilde{U}$ and $L = \tilde{D}^{-1}\tilde{L}\tilde{D}$. Like the LDU decomposition, our parametrization requires that all leading principal minors of A are nonzero. This condition is necessary and sufficient for a unique factorization of the form (10) and implies $\lambda_i \neq 0$ for every i . Non-singularity of A is not enough for decomposition (10) to exist and be unique. Matrices outside this chart form a measure-zero set under a continuous prior. If A lies outside this chart, some permutation of its rows lies inside it. Since a row permutation of A relabels the shocks, this is an observationally equivalent representation of the same model, so no explicit pivoting is required.

⁴Other reparametrizations have proved helpful in the literature. Rubio-Ramirez et al. (2010) apply sign restrictions by setting $A = Qh(\Sigma)$, where Σ is the reduced-form covariance matrix of a Gaussian VAR, $h(\cdot)$ is a unique decomposition, and Q is an orthogonal matrix. The list of contributions using the same parametrization is large. Archakov & Hansen (2021) propose a reparametrization of the correlation matrix which preserves its positive definiteness (see, for instance, its application in time-varying SVAR models, Arias et al. 2026). Wu & Koop (2026) use an eigenvalue–eigenvector decomposition of Σ to simplify sampling in large-scale VARs, while preserving order invariance.

This class was popular in the Bayesian simultaneous-equations literature and nests the Dreze and Jeffreys priors. For $\alpha = 0$, the prior density is flat with respect to dA . Our baseline sets $\alpha = -2k$, which is the density in A -coordinates induced by a prior that is flat with respect to dB , $p(B) \propto 1$. Matrix inversion contributes the volume element $|\det(A)|^{-2k}$ when the entries of B are functionally unconstrained (Kocięcki 2010). The distinction between these two cases is merely a prior choice.

Prior (12) implies the following priors on the diagonal of Λ (the $k \times 1$ vector $\boldsymbol{\lambda}$) and the vectorized free entries of L and U (the $k(k-1)/2$ vectors $\boldsymbol{l}$ and $\boldsymbol{u}$):

$$p(\boldsymbol{\lambda}, \boldsymbol{l}, \boldsymbol{u}) = \underbrace{p(\boldsymbol{\lambda})}_{\prod_{i=1}^k |\lambda_i|^{k-1+\alpha}} \cdot \underbrace{p(\boldsymbol{l})}_1 \cdot \underbrace{p(\boldsymbol{u})}_1. \quad (13)$$

One key result of the paper is to show that these priors imply conditional posterior distributions for (L, U, Λ) of common form, namely Gamma for the squared entries $x_i = \lambda_i^2$, with the sign of each λ_i drawn separately, and Normal for $(\boldsymbol{l}, \boldsymbol{u})$:

$$\begin{aligned} p(\boldsymbol{l}|Y, \Theta_{-L}) &\sim N(\boldsymbol{\mu}_L^*, V_L^*), & p(\boldsymbol{u}|Y, \Theta_{-U}) &\sim N(\boldsymbol{\mu}_U^*, V_U^*), \\ p(x_i|Y, \Theta_{-x_i}) &\sim \Gamma(h_{\lambda,i}^*, r_{\lambda,i}^*), & \lambda_i = \varepsilon_i \sqrt{x_i}, & \varepsilon_i = \begin{cases} +1, & \text{with probability } \frac{1}{2} \\ -1, & \text{with probability } \frac{1}{2} \end{cases}. \end{aligned} \quad (14)$$

Details of the posterior distributions can be found in Section A of the Online Appendix, including proof that the method is invariant to the ordering of the variables. To the best of our knowledge, this insight has never been exploited in the literature.⁵

Our sampler works also under Gaussianity. Define $\boldsymbol{u}_t = B\boldsymbol{\epsilon}_t$ as the VAR innovations, and $\Sigma_t = BD_tB'$ its conditional covariance matrix. For $D_t = I_k$, $\forall t$, this reduces to

⁵Another prior that preserves the convenient form of the likelihood function is a Gamma (or flat) prior on the squared entries of Λ , a joint Normal (or flat) prior on the free entries of L , and a joint Normal (or flat) prior on the free entries of U . Some priors (eg. Baumeister & Hamilton 2015, Bruns & Piffer 2023) are not compatible with Gibbs sampler, unless a reweighting importance step is introduced.

$\Sigma = BB'$. The orthogonal-reduced form parametrization by [Arias et al. \(2018\)](#) can be used to write $B = h(\Sigma)Q$, with $h(\Sigma)$ the Cholesky factorization of Σ and $Q = (h(BB'))^{-1}B$ the orthogonal matrix consistent with (Σ, B) . Our prior (12) implies

$$p(\Sigma, Q) = p(\Sigma) \cdot p(Q|\Sigma), \quad p(\Sigma) \propto |\det(\Sigma)|^{-\left(k + \frac{1}{2} + \frac{\alpha}{2}\right)}, \quad p(Q|\Sigma) \propto 1. \quad (15)$$

Accordingly, our prior features a uniform prior on orthogonal matrices, as advocated by [Arias et al. \(2018\)](#) and [Arias et al. \(2025\)](#).⁶ However, our sampler never requires drawing from the conditional posterior of Q . Rather, it indirectly explores the conditional and marginal distribution of Q by working on an alternative parametrization. Whether the likelihood function is potentially flat along certain directions is irrelevant for the validity of the sampler. Compared to the orthogonal-reduced-form parametrization, our approach generalizes to $D_t \neq I_k$.

2.3 Identification and normalization of sign/permutation

The Gibbs sampler proposed in the previous section is valid irrespective of whether the model specification for D_t makes the model identified or not. However, identification plays a major role in the interpretation of the parameters.

Our sampler is compatible with identifying restrictions. For example, considering the Gaussian case of $D_t = I_k$, identification can be achieved via a Cholesky ordering. This requires no more than setting $U = I$. External instruments can be handled by ordering the instrument first in a recursive system, as in [Plagborg-Møller & Wolf \(2021\)](#). As an alternative, one can combine the sampler with sign restrictions on linear or nonlinear functions of the parameters of the model. Since sign restrictions truncate the joint posterior

⁶Note that absent identifying restrictions, there is no difference between what [Arias et al. \(2018\)](#) refer to as conditionally versus unconditionally uniform priors on Q . See also [Uhlig \(2005\)](#) and [Read & Zhu \(2026\)](#) for a discussion on this point.

distribution, the conditional posterior distributions for (Λ, L, U) remain valid, up to a truncation (Section A of the Online Appendix). Methods like the Elliptical Slice Sampler or others can be used to draw from the truncated distributions, as in [Arias et al. \(2026\)](#) and [Read & Zhu \(2026\)](#). However, the general model specification from (1)-(3) lends itself most naturally to statistical identification, in which it is the properties of the data that identify the model statistically. There, the above restrictions become overidentifying. We now turn to discuss this class of models, where we make the second methodological contribution of the paper. We work under the assumption that no additional identifying restrictions are introduced.

There exist several specifications under which model (1)-(3) is statistically identified. One is the case in which the structural shocks are statistically independent across shocks and uncorrelated across time, with at most one structural shock being Gaussian ([Comon 1994](#), [Lanne et al. 2017](#)). Another is the case of structural shocks being mutually uncorrelated, but statistically dependent, as for instance in some stochastic volatility models ([Bertsche & Braun 2022](#), [Lewis 2025](#)) or under diagonality conditions on higher-order cross-cumulants of the shocks ([Mesters & Zwiernik 2024](#)). Under these and other specifications, the model is identified up to sign and permutations of the columns of B (or equivalently, the rows of A). For any $k \times k$ matrix B , there are $k! 2^k$ observationally equivalent signed permutations. The literature on this topic is vast, see for instance [Herwartz & Lütkepohl \(2014\)](#), [Brunnermeier et al. \(2021\)](#) and [Lewis \(2022\)](#).⁷

Suppose $\alpha = -2k$, which coincides with the prior $p(B) \propto 1$. The posterior on B inherits the same multimodality as the likelihood function in B . Because B is identified up to sign and permutation, the posterior on B inevitably mixes up shocks. Even if this posterior has

⁷A shock-specific scale normalization, such as $E(d_{it}) = 1$, is needed to identify B up to sign and permutation.

a unique componentwise median, that median can lie in a region with negligible posterior mass. The implication is that the marginal posterior of each entry of B could be very wide and symmetric around 0, when it is effectively mixing up shocks that only differ up to the sign and the ordering of the shocks, hence in ways that are economically irrelevant.

[Appendix D](#) of the Online Appendix provides an illustration in a bivariate model.

We refer to the above challenge as the ‘mode-switching’ problem, on which we focus our attention. This problem is not to be confused with the ‘label-switching’ problem. The latter comes from the fact that since the shocks are identified purely from the data, their interpretation as economic shocks requires further investigation (for instance by inspecting the impulse responses to the shock). This is in contrast to shocks identified via economically motivated restrictions, where the labeling of the shocks naturally comes from the restrictions. The labeling problem *per se* does not hinge on a multimodality of the likelihood function, but on the interpretability of the shocks. If the statistical properties of the data delivered a unimodal likelihood, the shocks would still require labeling ([Lewis 2025](#)).

Our second contribution is to propose an efficient way of handling the mode-switching problem stemming from the statistical identification of shocks up to sign and permutation. The key insight is to leverage a target matrix $\hat{B}$ that approximates one (potentially arbitrary) mode of the posterior distribution. When such a matrix is available, it can be used as a target matrix to flip the sign and/or ordering of each posterior draw of $p(B|Y, \Theta_{-B})$ to minimize the distance from the target. We first formalize this approach as a minimum distance criterion. We then show that existing results from the combinatorial calculus literature can be applied to speed up computations remarkably. To the best of our knowledge, this point has not been acknowledged for models with statistical identification. Again, we stress that the procedure aims at fixing the mode-switching problem, and is silent about the labeling problem. Our normalization can also be useful in a frequentist rather than a Bayesian

setting.

Let P_s be a signed permutation matrix of dimension $k \times k$. A signed permutation matrix is a square matrix $P_s \in \mathbb{R}^{k \times k}$ that has exactly one nonzero element in each row and each column, where each nonzero element equals either $+1$ or -1 . Equivalently, it can be written as $P_s = \Delta P$, where P is a (standard) permutation matrix and Δ is a diagonal matrix with diagonal entries equal to ± 1 . In our context, postmultiplying a matrix B by P_s changes the order and the signs of the columns of B . Each sampler iteration draws (Λ, L, U) , hence B . The Gibbs sampler proceeds for B . However, rather than store B , we propose to store BP_s , with P_s the signed permutation matrix that solves

$$\min_{P_s} \text{trace} \left[(BP_s - \hat{B})' (\hat{B} \hat{B}')^{-1} (BP_s - \hat{B}) \right]. \quad (16)$$

In other words, for each iteration we choose the signed permutation matrix P_s that minimizes the weighted distance between BP_s and the target matrix $\hat{B}$, and apply the transformation

$$(B, \{D_t\}_{t=1}^T) \mapsto (BP_s, \{P' D_t P\}_{t=1}^T). \quad (17)$$

This transformation is likelihood-preserving. Since our prior (12) satisfies $p(P'_s A) = p(A)$, it is also posterior-preserving. Any fixed signed permutation leaves the posterior unchanged; the draw-dependent choice of P_s then folds each equivalence class onto one representative. The transformation changes only the labels of the shocks, not the posterior mass attached to a draw, and we apply it at every iteration of the sampler after burn-in (Algorithm 2 in Section G of the Online Appendix; see Kaufmann & Schumacher 2019 for a similar procedure in factor models).

The general minimum distance principle from equation (16) has already been explored in the literature. As shown in Section E of the Online Appendix, (16) generalizes the Likelihood Preserving (LP) normalization of Waggoner & Zha (2003b) — originally designed for Gaussian SVARs. We therefore refer to this approach as *generalized LP normalization*.

Jarociński (2024) proposes a similar approach that evaluates an approximate likelihood function to select P_s . What is new in our approach is how we make (16) operational.

One might compute the distance (16) for all $k! 2^k$ possible signed permutation matrices in each Gibbs iteration. This option is not scalable, as it is barely feasible for even moderate k . As an example, when $k = 5$ it requires evaluating 3,840 possible permutations in each iteration. Instead, as we discuss in Section E of the Online Appendix, we can show that the minimization problem from (16) amounts to solving

$$\min_P \text{tr}\{P \cdot -|\hat{B}^{-1}B|\}, \quad (18)$$

where tr signifies ‘trace’, and $|\hat{B}^{-1}B|$ is the absolute values for all entries in $\hat{B}^{-1}B$ matrix (taken element-wise). Being able to rewrite (16) as (18) is a crucial advantage because it rewrites the key minimization problem as a linear assignment problem. Crucially, the literature on combinatorial optimization has already derived solutions for this type of assignment problem. The Online Appendix shows how to apply the Hungarian algorithm from combinatorial optimization to solve (18) and transform the optimal P into the optimal P_s . Importantly, all these operations can be executed at negligible computational cost – even for large models. As an example, for a 20-variable VAR, the assignment takes approximately 0.0001 seconds on a standard laptop, whereas exhaustive evaluation involves $20! 2^{20} \approx 2.5 \times 10^{24}$ signed permutations.

Our normalization procedure requires a target matrix $\hat{B}$. The multimodality that follows from statistical identification is a property of the likelihood. Let $\{\hat{A}_{ML}^{(j)}\}_{j=1}^{k! 2^k}$ denote its modes when written as a function of A , and let $\{\hat{B}_{ML}^{(j)}\}_{j=1}^{k! 2^k}$ denote the corresponding modes when written as a function of B . Matrix inversion maps these sets exactly: each $\hat{A}_{ML}^{(j)}$ is the inverse of one $\hat{B}_{ML}^{(j)}$. No Jacobian enters this statement.

For statistically identified models, we set $\alpha = -2k$ in (12), which is equivalent to $p(B) \propto 1$. Conditional on the remaining parameters, the posterior density for B then has the same

B -dependent kernel as the conditional likelihood, while the joint posterior preserves the likelihood’s signed-permutation orbit, provided the prior on the shock-specific parameters (here $\mathbf{v}$ and D) is exchangeable across shocks. Section B of the Online Appendix states this condition formally. We therefore set $\hat{B}$ equal to one element of $\{\hat{B}_{ML}^{(j)}\}_{j=1}^{k!2^k}$.⁸ The density of the same posterior measure with respect to dA contains the factor $|\det(A)|^{-2k}$, and its modes therefore need not equal the inverses of the B -coordinate posterior-density modes. This coordinate dependence does not alter the normalization problem: the determinant factor is identical across all signed permutations. Section B establishes both results formally. For comparability across specifications, we suggest adjusting the estimated modal point by postmultiplying it by the sign-permutation matrix that makes it closest to the identity matrix. This fixes a common convention without affecting the assignment algorithm nor the results.

Our method differs from other normalizations used in non-Gaussian SVARs to handle the mode-switching problem. Compared with Lanne et al. (2017) and Gouriéroux et al. (2020), our method explicitly targets a single representative $\hat{B}$. Relative to Brunnermeier et al. (2021), the baseline specification uses a signed-permutation-invariant flat prior rather than an informative prior. Unlike Jarociński (2024), our method does not evaluate a target function for every permutation. Keweloh & Wang (2026) use a refinement of the Lanne et al. (2017) normalization scheme that exploits a preliminary estimate of B , just as our approach. This normalization will likely often work well in practice, but there is no guarantee that it will always normalize the parameter draw to be the closest to the preliminary estimate, which our normalization guarantees by construction. In Appendix F, we show this by constructing an example. Computationally, there is no gain in speed by using the Keweloh

⁸We found it helpful to estimate this element using the three-step maximum-likelihood procedure of Lanne et al. (2017). With analytical derivatives, this procedure remains fast even for large k .

& Wang (2026) approach instead of ours.

Although our approach is new to the VAR literature, it has some precedents. When structural shocks have normalized variances, the distance measure of our generalized LP normalization is equivalent to the minimum distance index (MDI) proposed by Ilmonen et al. (2010) and used for comparing ICA algorithms (Matteson & Tsay 2017, Moneta & Pallante 2022). Interestingly, Ilmonen et al. (2010) recognized that minimizing the MDI leads to a linear assignment problem, and proposed the Hungarian algorithm to solve it, as we do. Contrary to this literature, we use the distance to implement a normalization procedure, not to evaluate estimator quality.

We conclude this section by acknowledging two related strands of the literature. The first one shows that if a subset of the structural shocks is Gaussian, then statistical identification still achieves identification of the remaining shocks, but not the shocks within the Gaussian block (Guay 2021, Maxand 2020, Anttonen et al. 2026).⁹ subsection H.2 of the Online Appendix shows a simulation within the context of our methodology. Our sampler and normalization work successfully. In general, we caution against using the sampler without inspection of the statistical support for non-Gaussianity provided by the data. Last, part of the literature combines statistical identification with identifying restrictions (Keweloh & Wang 2026). While our sampler can potentially be combined with identifying restrictions, we leave this application for future research.

⁹Lanne et al. (2026) extend this partial-identification logic beyond the independence assumption, showing that shocks satisfying weaker moment-diagonality conditions (absence of co-skewness or excess co-kurtosis) remain identified even when dependent on the rest.

3 Illustration using simulated data

In this section we use a set of simulation exercises to illustrate our methodology. While the method from [section 2](#) can accommodate several specifications of D_t , for simplicity we use a single specification for all simulations and applications. We work with a model that features independent t distributed structural shocks with shock-specific degrees of freedom v_i .

We refer to [Section C](#) in the Online Appendix for a full discussion. In short, following [Geweke \(1993\)](#), we model independent t distributed structural shocks using an inverse Gamma prior on d_{it} ,

$$p(d_{it}|h_i, r_i) = \frac{r_i^{h_i}}{\Gamma(h_i)} \cdot d_{it}^{-h_i-1} e^{-r_i \cdot \frac{1}{d_{it}}}, \quad h_i = \frac{v_i}{2}, \quad r_i = \frac{v_i - 2}{2}, \quad (19)$$

with $v_i > 2, \forall i$. d_{it} are a-priori independent across (i, t) , and independent of (ϕ, B) . [\(19\)](#) ensures that the variance of ϵ_{it} equals unity. For the autoregressive parameters ϕ we use a Normal prior centered at zero, calibrated according to [Kadiyala & Karlsson \(1997\)](#) and [Canova \(2007\)](#), using a training sample to set the hyperparameters, in the spirit of the Minnesota prior. For the impact matrix B , we specify a flat prior.

Sampling from the conditional posterior of the degrees of freedom is done via a Griddy Gibbs sampler, which requires specifying a finite support for v_i (other methods are possible). We set the lower bound at 3. For the upper bound, we take into account the fact that the sample size constrains the ability of the model to detect relatively high values of v_i , as the likelihood function becomes progressively flat as v_i increases. Depending on the application, we set the upper bound to 38.5 (bivariate simulation and application) or 65 (simulation on larger models), see [Section C.3](#) of the Online Appendix for a detailed discussion. Given the support of v_i we either work with a flat prior in this support, or with a truncated Normal prior. When applying our normalization, we apply the transformation

$$(B, \{D_t\}_{t=1}^T, \mathbf{v}) \mapsto (BP_s, \{P'D_tP\}_{t=1}^T, P'\mathbf{v}). \quad (20)$$

when saving draws, while leaving the progression of the Gibbs sampler unaffected.

3.1 Bivariate model

In the bivariate illustration we generate the data from model

$$\mathbf{y}_t = \sum_{l=1}^6 \tilde{\Pi}_l \mathbf{y}_{t-l} + \tilde{B} \boldsymbol{\epsilon}_t, \quad p(\boldsymbol{\epsilon}_t) = \prod_{i=1}^2 p(\epsilon_{it}), \quad \epsilon_{it} \sim t(\tilde{v}_i). \quad (21)$$

The model simulates the dynamics of two variables, which we label as “output” (ordered first) and the “price level”. The driving shocks are a demand shock and a supply shock.

Following [Canova et al. \(2024\)](#), we set the parameters of the data-generating process by first setting the true impulse responses, and then choosing $(\tilde{\Pi}, \tilde{B})$ to match those responses. Let $\tilde{\psi}_{ij,h}$ be the true impulse response of variable i to shock j at horizon h . We specify

$$\tilde{\psi}_{ij,h} = a_{ij} \cdot \exp\left(-\frac{h}{c_{ij}^2}\right), \quad (22)$$

where a_{ij} governs the impact effect of shock j on variable i and c_{ij} captures the persistence of the response. Each response over $H + 1$ horizons is fully determined by two parameters, (a_{ij}, c_{ij}) , which we set to generate the following patterns. The first shock is a demand shock. A positive one-standard deviation demand shock raises output and the price level on impact by 0.6 and 0.7, respectively. Both responses then decline to zero, reaching half their impact effect 5 and 3 horizons later, respectively. The second shock is a supply shock. A positive one-standard deviation supply shock raises output by 0.4 on impact and lowers the price level by 0.7. Both responses then decline to zero, reaching half their impact effect 8 horizons later. Note that the data generating process features the demand shock as the first shock, and the true signs are such that both positive shocks *increase* output.

We specify impulse responses up to horizon $H = 6$, matching the number of lags in the true model. As discussed by [Canova et al. \(2024\)](#), there is a unique solution in $(\tilde{\Pi}, \tilde{B})$ such that the VAR-implied responses up to horizon H coincide with the responses from equation (22).

We set $\tilde{v}_i = 8$ for $i = 1, 2$. See Section H of the Online Appendix for additional details on the data-generating process. The prior on the degrees of freedom is truncated, with mean equal to the midpoint of the support, and standard deviation equal to 10.

We generate a single dataset from the data-generating process. We initialize the simulation at the unconditional mean (zero) and create 350 observations. We discard the first 100, retain the next 50 as a training sample, and use the remaining 200 for estimation. We then estimate a VAR with 6 lags (no constant) on the simulated data. We compute the target matrix $\hat{B}$ required for the normalization by maximizing the likelihood function. We initialize the sampler as described in Section G of the Online Appendix. The sampler runs for 65,000 draws, of which 15,000 as burn-in draws. On a standard computer with an Intel i7-7700K 4.2 GHz Quad Core processor and 64 GB RAM, this takes about two minutes.

Figure 1: Impulse response functions

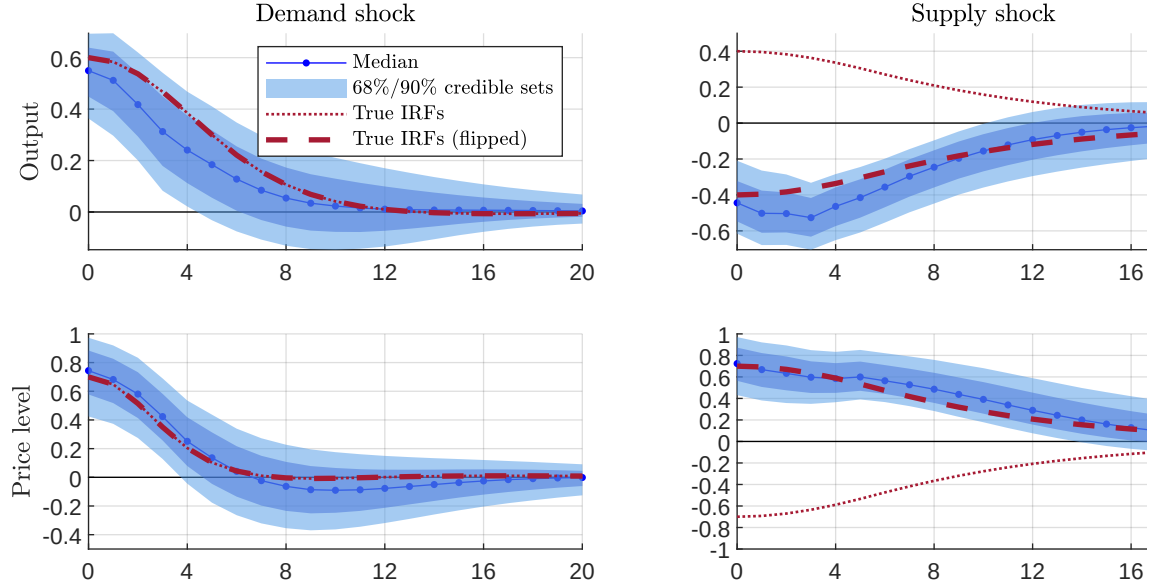


Figure 1 presents our methodology's performance in estimating impulse responses. The dotted blue line and shaded regions represent the pointwise median and 68/90% credible sets, respectively. The red dotted line shows the true impulse response from the data-generating

process. The red dashed line shows an alternative set of responses identical to the true ones, except that the supply shock's response is flipped in sign. The figure shows that the model correctly recovers a sign-permutation of the true impulse responses, namely the response associated with permutation $P_s = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Figure 2: Posterior distributions on (Λ, L, U, B)

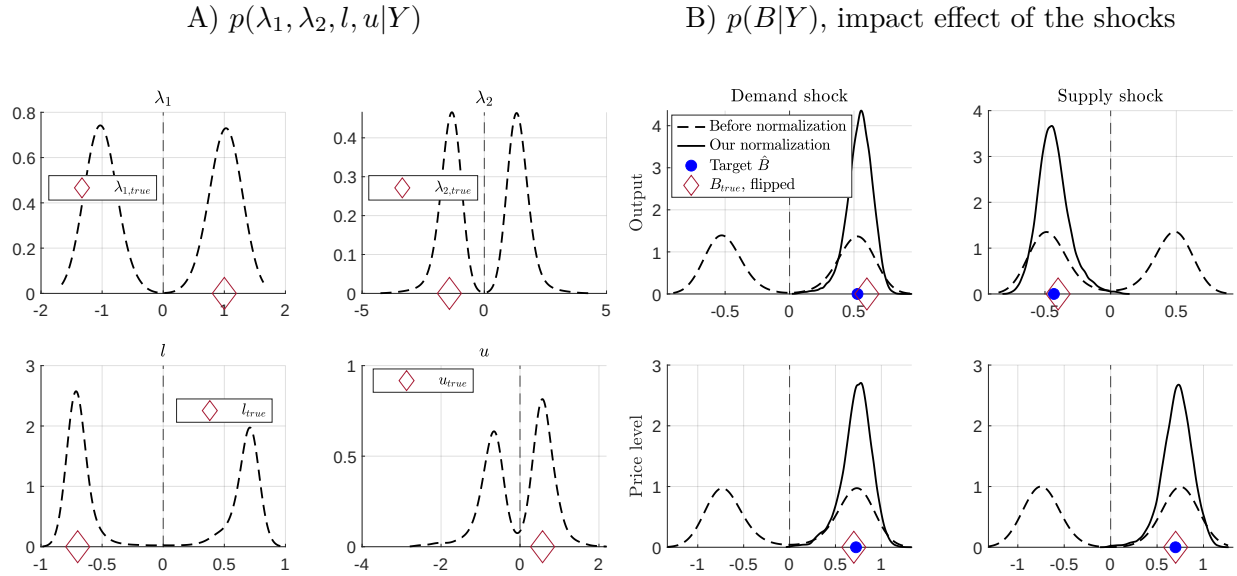


Figure 2 illustrates how our sampler succeeds in recovering the true impact responses. The dashed lines in the left panel show the marginal posterior distribution of the free entries of (Λ, L, U) , while the dashed lines in the right panel show the marginal posterior distribution of each entry of B before the normalization is applied. In exploring the conditional posteriors on (Λ, L, U) the sampler is effectively uncovering the posterior on B . Note that the marginal posteriors on Λ and on the non-normalized draws of B are symmetric around zero. The red diamonds report the true values of (Λ, L, U, B) , while the blue dot in the lower panel shows the entries of the target matrix $\hat{B}$ used for our normalization. $\hat{B}$ features the demand shock as the first shock, and a positive supply shock will *decrease* output. The solid black line shows the marginal posterior of B after applying the normalization. The normalized posterior on B successfully shows no multimodality, and correctly recovers an (arbitrary)

sign-permutation of the true impact effect of the shocks.¹⁰

Figure H-3 in Section H of the Online Appendix shows that our relatively uninformative prior is updated towards the truth, detecting heavy tails, although degrees of freedom remain challenging to estimate, as previously discussed in the literature (Fonseca et al. 2008, Hosszejni 2021).

3.2 Larger model

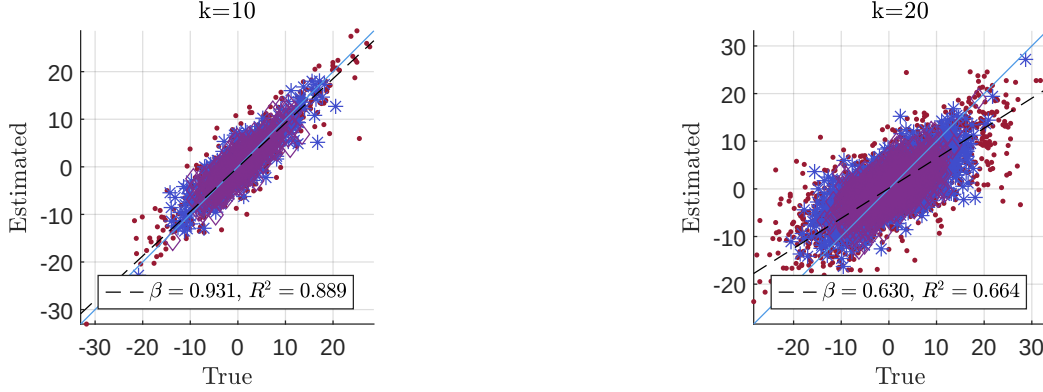
The previous section illustrated our methodology in a bivariate setting using a single dataset. In this section, we generate data for 100 periods as training sample and then 800 periods for the estimation sample. We then consider models of $k = 10, 20, 30$ variables. Both the sample size and the number of variables imply a medium to large scale application.

We proceed as follows, starting from the same model as in the bivariate simulation. We keep the true degrees of freedom at 8 for all shocks. We randomly draw the true parameters for B from independent Normal distributions with mean 0 and standard deviation 10. We set the number of lags of both the generating process and the estimated model to 1 and set the true autoregressive parameters to $\Pi = 0.90 \cdot I_k$. We simulate the data and estimate the model using, in turn, the normalizations from subsection 2.3, Lanne et al. (2017) and Gouriéroux et al. (2020). We repeat the exercise over 5 pseudo datasets, holding k constant. We replicate the analysis for $k = 10, 20, 30$. For the degrees of freedom we use a flat prior in the support from 3 to 65, as suggested by the sample size.

The dimensionality of the model requires estimating between $10^2 = 100$ and $30^2 = 900$ free

¹⁰The target matrix plays an important role in the successful performance of the normalization. Figure H-2 in the Online Appendix shows this by reporting the results when the target matrix is set to the identity matrix. As should be expected, an arbitrary target matrix will not tackle the mode-switching problem successfully.

Figure 3: Posterior mean estimates of impulse responses: estimated versus true



entries B , depending on k , in addition to the elements in $(\Pi, \mathbf{v})$. Yet, the computational time of the combined sampler and normalization approach is competitive: across pseudo datasets, the algorithm takes on average 3 minutes for $k = 10$, 13 minutes for $k = 20$ and 1 hour for $k = 30$. Most of the computational advantage comes from the sampler rather than the normalization: replacing our normalization with the normalizations by [Lanne et al. \(2017\)](#) and [Gouriéroux et al. \(2020\)](#) only mildly makes inference slower.

We inspect the accuracy of the estimates as done in [Chan et al. \(2024\)](#). For each estimated model, we compute the posterior pointwise mean of the estimated B , and use our minimum distance criterion from (16) to compute the sign-permutation of B_{true} that sits closest to the estimated pointwise mean of B . This ensures a meaningful comparison despite the sampler being potentially successful at estimating any of the $k!2^k$ possible sign-permutations of B_{true} , in a way that can change across replications. We then compare the estimated response to the truth at horizon 0 (impact effect), 4 and 8 after the shocks. For a given dataset, this procedure delivers $3k^2$ points. Last, we pool the points across the 5 pseudo datasets.

[Figure 3](#) reports the scatter plots for $k = 10, 20$, when applying our normalization. The figure reports the mean estimates on the y axis and the true values on the x axis. The impact effects are reported in dots, the responses at horizon 4 and 8 in stars and diamonds,

respectively. The blue line shows the diagonal, while the black dashed line shows the least-squares fit of the estimated values on the true values. The figure reports the estimated slope and R^2 . Ideally, all points lie on the diagonal, with estimated slope and R^2 equal to 1. The results from our normalization are encouraging, with the slope estimated at 0.931 and $R^2 = 0.889$ for $k = 10$. Both statistics decrease for $k = 20$, but remain competitive. Figure J-16 in the Online Appendix shows that our normalizations performs better than Lanne et al. (2017) and Gouriéroux et al. (2020) for all $k = 10, 20, 30$, although performance inevitably gets worse for $k = 30$.

Section H.2 in the Online Appendix reports additional simulation results for the case in which a subset of the shocks is close to Gaussian. The non-Gaussian block is estimated accurately, as should be expected.

4 What Shocks Drive Real GDP? An application to a medium scale model

We now apply our framework to assess which shocks matter most for U.S. real GDP. Angeletos et al. (2020) use a structural VAR with Gaussian shocks to identify a “Main Business Cycle” shock that explains much of the volatility in real activity at business-cycle frequencies. They find that (i) a single shock can account for nearly 80 percent of GDP volatility when GDP is used to measure real activity,¹¹ (ii) the shock resembles a weakly inflationary demand shock based on its impulse responses, (iii) it is largely uncorrelated with TFP, and (iv) it does not drive long-run fluctuations in real activity.

We revisit and extend these findings under non-Gaussian shocks. Specifically, we apply a

¹¹When targeting unemployment, instead, their identified shock still explains more than half of the business-cycle variation in GDP.

Student- t SVAR and ask whether the business cycle can be attributed to a single shock or if multiple shocks share responsibility for GDP volatility. Our specification follows the baseline [Angeletos et al. \(2020\)](#) model, using their exact dataset, which includes 10 variables (GDP, investment, consumption, hours worked, unemployment rate, labor share, Federal funds rate, inflation, labor productivity, and TFP), all in log levels except for the Federal funds rate. All variables are real and in per capita terms where applicable. The sample is quarterly from 1955Q1 to 2017Q4, and, as in the original study, we include two lags.

We use a Minnesota prior for the autoregressive coefficients and a flat prior on B . We use a flat prior for the degrees of freedom with positive support between 3 and 38.5. The upper bound support was selected taking into account the sample size, to avoid entering the parameter space where the likelihood is effectively flat (see Section C.3 in the Online Appendix). We generate 65,000 posterior draws and discard the first 15,000. We estimate the target matrix $\hat{B}$ for our normalization numerically via maximum likelihood maximization. The full sampler (including the computation of the target matrix) runs in 11 minutes on an Intel i7-7700K 4.2 GHz Quad Core processor and 64 GB RAM.

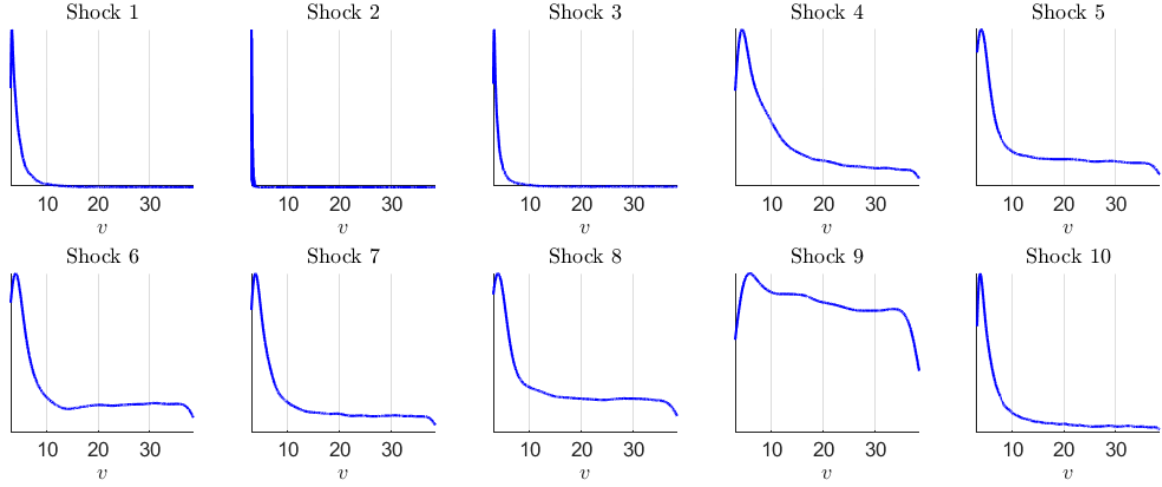
The structural interpretation is conditional on a linear, fundamental VAR with mutually independent, symmetric Student- t shocks and independent latent scales. Persistent or common volatility, dependence among structural shocks, multiple Gaussian shocks, or nonlinear dynamics can generate non-Gaussian reduced-form innovations without satisfying these identifying assumptions. Unlike restrictions from economic theory these assumptions concern the distribution of the data and can be checked: independence of the shocks and departures from Gaussianity can be tested directly, as we do below, and specifications that add or remove a property, such as stochastic volatility or heavy tails, can be compared through their marginal data densities. Relatedly, [Kolesár & Plagborg-Møller \(2025\)](#) show that, unlike VAR- or proxy-based identification, identification strategies that exploit non-

Gaussianity or heteroskedasticity of latent shocks are highly sensitive to departures from linearity in the true data-generating process; our forecast-error variance decomposition results should therefore be interpreted conditional on the maintained linear specification.

Figure 4 reports the posterior distributions of the (normalized) degrees of freedom under the maintained Student- t specification. Relative to the flat prior, the posteriors move sharply toward the lower boundary for nine shocks out of ten, indicating very heavy tails. This is particularly the case for shocks 2 and 3. We stress that the figure shows a within-model diagnostic of tail thickness. It does not test shock independence, determine the number of non-Gaussian shocks under a more general specification, or assign posterior probability to exact Gaussianity, which the support $[3, 38.5]$ excludes. The estimated degrees of freedom provide a general measure of identification strength. However, identification of the exact degrees of freedom is weak when non-Gaussianity is only mild (Hoesch et al. 2024). We run two diagnostics to further inspect the results from the figure. First, we run the test by Kilian & Demiroglu (2000) to assess non-Gaussianity in each reduced form innovation of the model. Second, we run the test by Amengual et al. (2022) to assess the statistical independence across our estimated structural shocks. Both tests use the estimated reduced form and structural shocks associated with the pointwise mean values of Π and A . The results show that 7 reduced form innovations out of 10 display non-Gaussianity, and all 10 structural shocks feature statistical independence (Table I-2-Table I-3 of the Online Appendix). Figure I-7 in the Appendix reports the marginal posteriors of the normalized entries of B . None of the entries features multimodality, suggesting that the normalization procedure is working well. Figure I-8-I-9 show that some multimodality is still present when applying alternative normalizations.

Next, we turn to our main question: Is there one key shock driving GDP? To answer, we compute forecast error variance decompositions (see Lütkepohl 2005). We compute the

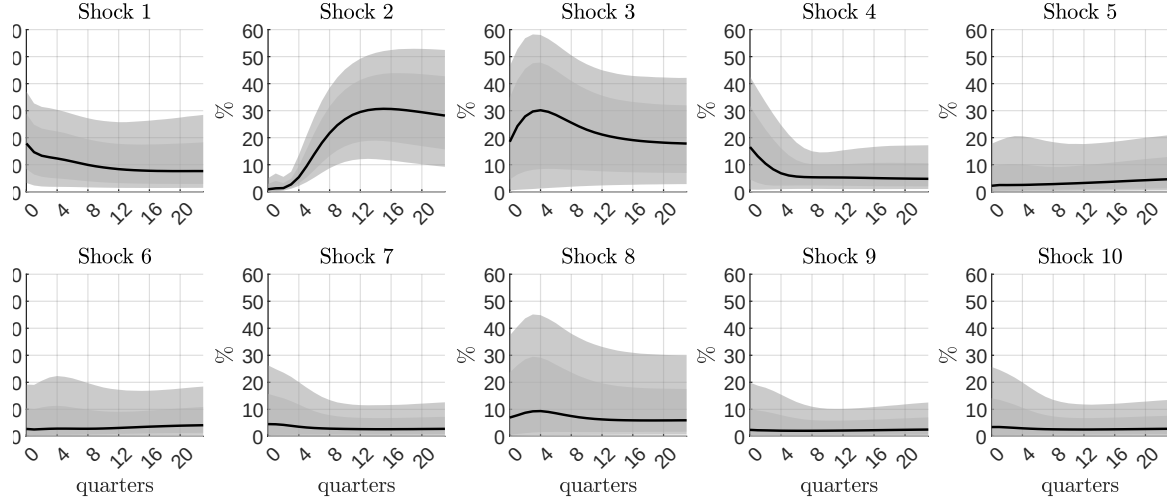
Figure 4: Posterior degrees of freedom under the maintained Student- t specification



time-domain variance decomposition because it is more common in the literature than the frequency-domain version used for identification in [Angeletos et al. \(2020\)](#). [Angeletos et al. \(2020\)](#) also report time-domain decompositions, making the results directly comparable. We consider horizons up to 24 quarters in [Figure 5](#). For every shock and horizon, the pointwise posterior median FEVD is below 30 percent. This is a statement about each shock's marginal posterior. The two largest median contributions come from shock 2 at longer horizons and shock 3 at shorter horizons. What do these shocks represent?

[Figure 6](#) shows the impulse responses to these shocks, normalized to a unit standard deviation impulse. The top two rows show the responses to shock 3, which increases real activity, lowers unemployment, but has no effect on TFP and only a small (if any) effect on inflation. Thus, this shock closely resembles the main business cycle shock of [Angeletos et al. \(2020\)](#), with the caveat that we find it is less important than suggested in that work (albeit still one of the two key drivers of GDP in the US). The second shock we identify, which is important at longer horizons, moves real activity and inflation in opposite directions and moves TFP. With the sign convention of the figure, it lowers GDP, investment, consumption and hours, raises unemployment and inflation, and lowers TFP; its mirror image is a positive

Figure 5: Forecast error variance decomposition of real GDP



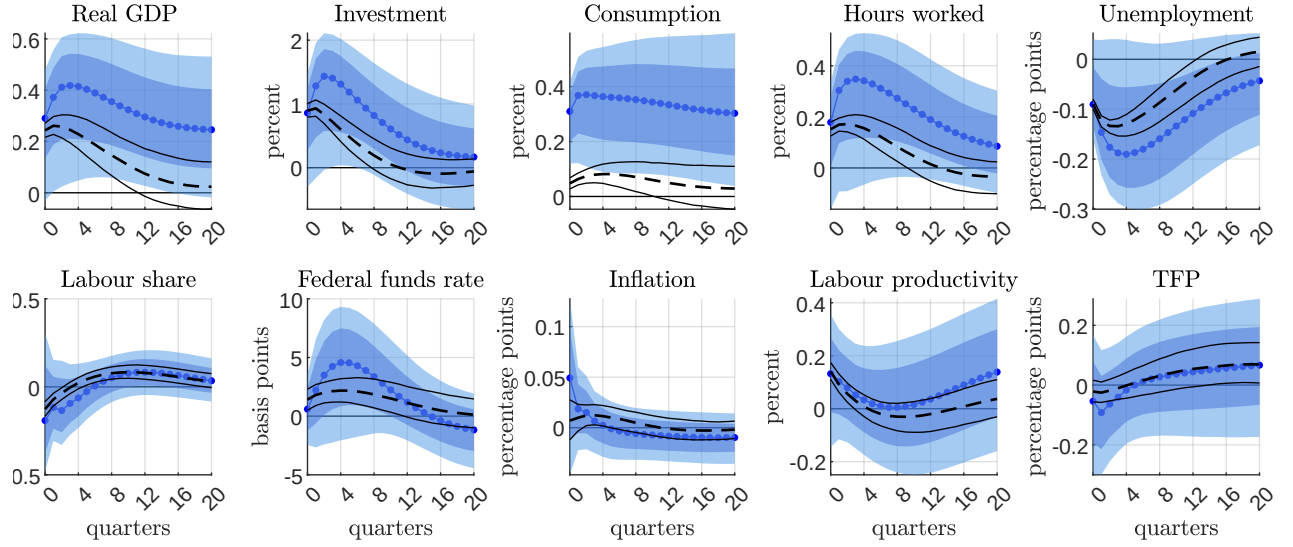
technology shock (Kydland & Prescott 1982). In fact, in the canonical Smets & Wouters (2007) model, a positive TFP shock raises output and lowers inflation, just as we find. The full forecast error variance decompositions associated with the shocks from Figure 6 are reported in Figure I-10 of the Online Appendix. In the Appendix, we show that the results do not change when using a looser prior on the autoregressive parameters (I-11–I-12), and a model with four rather than two lags (I-13–I-14).

5 Conclusions

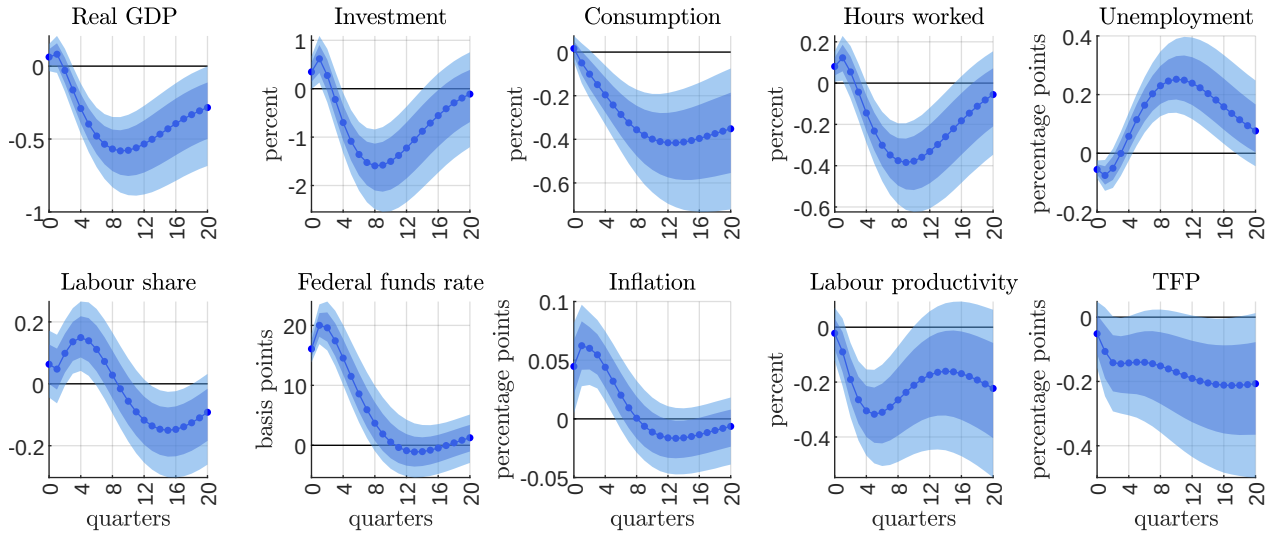
Non-Gaussian features of the data identify the shocks of an SVAR without restrictions from economic theory, but two computational problems have kept this approach confined to small systems: the conditional posterior of the impact matrix B has no standard form, and every posterior draw must be relabeled among $k! 2^k$ observationally equivalent sign and permutation configurations. This paper removes both. Writing $B^{-1} = \Lambda LU$ separates the determinant of the likelihood from its quadratic form, so that under priors of the form $p(A) \propto |\det A|^\alpha$ a Gibbs sampler becomes feasible. Casting the relabeling step as a linear

Figure 6: Impulse responses

Shock 3: demand shock



Shock 2: supply shock



assignment problem lets the Hungarian algorithm find the closest signed permutation of each draw exactly, in a fraction of a millisecond, for models with dozens of variables.

6 Disclosure statement

The authors report there are no competing interests to declare.

7 Data Availability Statement

Deidentified data have been made available at the following URL: XX.

References

- Amengual, D., Fiorentini, G. & Sentana, E. (2022), ‘Moment tests of independent components’, *SERIEs: Journal of the Spanish Economic Association* **13**, 429–474.
- Angeletos, G.-M., Collard, F. & Dellas, H. (2020), ‘Business-cycle anatomy’, *American Economic Review* **110**(10), 3030–3070.
- Anttonen, J. J., Lanne, M. & Luoto, J. (2024), ‘Statistically identified structural VAR model with potentially skewed and fat-tailed errors’, *Journal of Applied Econometrics* **39**(3), 422–437.
- Anttonen, J., Lanne, M. & Luoto, J. (2026), ‘Bayesian inference on fully and partially identified potentially non-Gaussian structural vector autoregressions’, *Empirical Economics* **71**(2), 24.
- Archakov, I. & Hansen, P. R. (2021), ‘A new parametrization of correlation matrices’, *Econometrica* **89**(4), 1699–1715.
- Arias, J. E., Rubio-Ramírez, J. F. & Shin, M. (2026), Large SVARs, Working Paper 26-04, Federal Reserve Bank of Philadelphia.
- Arias, J. E., Rubio-Ramírez, J. F. & Waggoner, D. F. (2018), ‘Inference based on Structural

- Vector Autoregressions identified with sign and zero restrictions: Theory and applications', *Econometrica* **86**(2), 685–720.
- Arias, J. E., Rubio-Ramírez, J. F. & Waggoner, D. F. (2025), 'Uniform priors for impulse responses', *Econometrica* **93**(2), 695–718.
- Baumeister, C. & Hamilton, J. D. (2015), 'Sign restrictions, structural vector autoregressions, and useful prior information', *Econometrica* **83**(5), 1963–1999.
- Bertsche, D. & Braun, R. (2022), 'Identification of structural vector autoregressions by stochastic volatility', *Journal of Business & Economic Statistics* **40**(1), 328–341.
- Braun, R. (2023), 'The importance of supply and demand for oil prices: Evidence from non-Gaussianity', *Quantitative Economics* **14**(4), 1163–1198.
- Brunnermeier, M., Palia, D., Sastry, K. A. & Sims, C. A. (2021), 'Feedbacks: financial markets and economic activity', *American Economic Review* **111**(6), 1845–1879.
- Bruns, M. & Piffer, M. (2023), 'A new posterior sampler for bayesian structural vector autoregressive models', *Quantitative Economics* **14**(4), 1221–1250.
- Bruns, M. & Piffer, M. (2024), 'Tractable Bayesian estimation of smooth transition vector autoregressive models', *The Econometrics Journal* **27**(3), 343–361.
- Canova, F. (2007), *Methods for applied macroeconomic research*, Vol. 13, Princeton University Press.
- Canova, F., Kocięcki, A. & Piffer, M. (2024), 'Flexible prior beliefs on impulse responses in Bayesian vector autoregressive models', *CEPR Discussion Paper No. 18992*.
- Chan, J. C. (2023), 'Comparing stochastic volatility specifications for large Bayesian vars', *Journal of Econometrics* **235**(2), 1419–1446.

- Chan, J. C. & Eisenstat, E. (2018), ‘Bayesian model comparison for time-varying parameter VARs with stochastic volatility’, *Journal of Applied Econometrics* **33**(4), 509–532.
- Chan, J. C., Koop, G. & Yu, X. (2024), ‘Large order-invariant Bayesian VARs with stochastic volatility’, *Journal of Business & Economic Statistics* **42**(2), 825–837.
- Chiu, C.-W. J., Mumtaz, H. & Pinter, G. (2017), ‘Forecasting with VAR models: Fat tails and stochastic volatility’, *International Journal of Forecasting* **33**(4), 1124–1143.
- Clark, T. E. & Ravazzolo, F. (2015), ‘Macroeconomic forecasting performance under alternative specifications of time-varying volatility’, *Journal of Applied Econometrics* **30**(4), 551–575.
- Cogley, T. & Sargent, T. J. (2005), ‘Drifts and volatilities: monetary policies and outcomes in the post WWII US’, *Review of Economic Dynamics* **8**(2), 262–302.
- Comon, P. (1994), ‘Independent component analysis, a new concept?’, *Signal Processing* **36**(3), 287–314.
- Cron, A. J. & West, M. (2011), ‘Efficient Classification-Based Relabeling in Mixture Models’, *The American Statistician* **65**(1), 16–20.
- Dou, L., Ho, P. & Lubik, T. A. (2025), Max-share misidentification. Working paper.
- Fonseca, T. C. O., Ferreira, M. A. R. & Migon, H. S. (2008), ‘Objective Bayesian analysis for the Student- t regression model’, *Biometrika* **95**(2), 325–333.
- Francis, N. & Kindberg-Hanlon, G. (2022), ‘Signing out confounding shocks in variance-maximizing identification methods’, *AEA Papers and Proceedings* **112**, 476–80.
- Geweke, J. (1993), ‘Bayesian treatment of the independent Student- t linear model’, *Journal of Applied Econometrics* **8**(S1), S19–S40.
- Gouriéroux, C., Monfort, A. & Renne, J.-P. (2020), ‘Identification and estimation in non-

- fundamental structural VARMA models’, *The Review of Economic Studies* **87**(4), 1915–1953.
- Guay, A. (2021), ‘Identification of structural vector autoregressions through higher unconditional moments’, *Journal of Econometrics* **225**, 27–46.
- Hamilton, J. D., Waggoner, D. F. & Zha, T. (2007), ‘Normalization in Econometrics’, *Econometric Reviews* **26**(2-4), 221–252.
- Herwartz, H. & Lütkepohl, H. (2014), ‘Structural vector autoregressions with Markov switching: Combining conventional with statistical identification of shocks’, *Journal of Econometrics* **183**(1), 104–116.
- Hoesch, L., Lee, A. & Mesters, G. (2024), ‘Locally robust inference for non-Gaussian SVAR models’, *Quantitative Economics* **15**(2), 523–570.
- Hosszejni, D. (2021), ‘Bayesian estimation of the degrees of freedom parameter of the Student- t distribution—A beneficial re-parameterization’, *arXiv preprint arXiv:2109.01726* .
- Ilmonen, P., Nordhausen, K., Oja, H. & Ollila, E. (2010), A new performance index for ICA: properties, computation and asymptotic analysis, in V. Vigneron, V. Zarzoso, E. Moreau, R. Gribonval & E. Vincent, eds, ‘Latent Variable Analysis and Signal Separation’, Springer, pp. 229–236.
- Jarociński, M. (2024), ‘Estimating the FED’s unconventional policy shocks’, *Journal of Monetary Economics* **144**, 103548.
- Kadiyala, K. R. & Karlsson, S. (1997), ‘Numerical methods for estimation and inference in Bayesian VAR-models’, *Journal of Applied Econometrics* **12**(2), 99–132.
- Karlsson, S. & Mazur, S. (2020), Flexible fat-tailed vector autoregression, Working Paper 2020:5, Örebro University School of Business.

- Karlsson, S., Mazur, S. & Nguyen, H. (2023), ‘Vector autoregression models with skewness and heavy tails’, *Journal of Economic Dynamics and Control* **146**, 104580.
- Kaufmann, S. & Schumacher, C. (2019), ‘Bayesian estimation of sparse dynamic factor models with order-independent and ex-post mode identification’, *Journal of Econometrics* **210**(1), 116–134.
- Keweloh, S. A. & Wang, S. (2026), ‘Uncertain Short-Run Restrictions and Statistically Identified Structural Vector Autoregressions’, *Journal of Applied Econometrics* **41**(1), 12–25.
- Kilian, L. & Demiroglu, U. (2000), ‘Residual-based tests for normality in autoregressions: Asymptotic theory and simulation evidence’, *Journal of Business & Economic Statistics* **18**(1), 40–50.
- Kim, S., Shephard, N. & Chib, S. (1998), ‘Stochastic volatility: Likelihood inference and comparison with arch models’, *The Review of Economic Studies* **65**(3), 361–393.
- Kocięcki, A. (2010), ‘A prior for impulse responses in Bayesian structural VAR models’, *Journal of Business & Economic Statistics* **28**(1), 115–127.
- Kocięcki, A., Ca’ Zorzi, M. & Rubaszek, M. (2025), ‘Bayesian analysis of recursive SVAR models with overidentifying restrictions’, *Przegląd Statystyczny. Statistical Review* **72**(1), 1–17.
- Kolesár, M. & Plagborg-Møller, M. (2025), ‘Dynamic causal effects in a nonlinear world: the good, the bad, and the ugly’, *Journal of Business & Economic Statistics* **43**(4), 737–754.
- Kydland, F. E. & Prescott, E. C. (1982), ‘Time to build and aggregate fluctuations’, *Econometrica: Journal of the Econometric Society* pp. 1345–1370.
- Lanne, M., Liu, K. & Luoto, J. (2026), ‘Identifying structural vector autoregressions via

- Non-Gaussianity of potentially dependent structural shocks', *The Econometrics Journal* .
forthcoming.
- Lanne, M. & Luoto, J. (2020), 'Identification of economic shocks by inequality constraints in Bayesian structural vector autoregression', *Oxford Bulletin of Economics and Statistics* **82**(2), 425–452.
- Lanne, M., Lütkepohl, H. & Maciejowska, K. (2010), 'Structural vector autoregressions with Markov switching', *Journal of Economic Dynamics and Control* **34**(2), 121–131.
- Lanne, M., Meitz, M. & Saikkonen, P. (2017), 'Identification and estimation of non-Gaussian structural vector autoregressions', *Journal of Econometrics* **196**(2), 288–304.
- Lewis, D. J. (2021), 'Identifying shocks via time-varying volatility', *Review of Economic Studies* **88**, 3086–3124.
- Lewis, D. J. (2022), 'Robust inference in models identified via heteroskedasticity', *Review of Economics and Statistics* **104**, 510–524.
- Lewis, D. J. (2025), 'Identification Based on Higher Moments in Macroeconometrics', *Annual Review of Economics* **17**, 665–693.
- Lütkepohl, H. (2005), *New introduction to multiple time series analysis*, Springer Science & Business Media.
- Lütkepohl, H. & Netsunajev, A. (2017), 'Structural vector autoregressions with smooth transition in variances', *Journal of Economic Dynamics and Control* **84**, 43–57.
- Lütkepohl, H. & Woźniak, T. (2020), 'Bayesian inference for structural vector autoregressions identified by markov-switching heteroskedasticity', *Journal of Economic Dynamics and Control* **113**, 103862.

- Matteson, D. S. & Tsay, R. S. (2017), ‘Independent Component Analysis via Distance Covariance’, *Journal of the American Statistical Association* **112**(518), 623–637.
- Maxand, S. (2020), ‘Identification of independent structural shocks in the presence of multiple Gaussian components’, *Econometrics and Statistics* **16**, 55–68.
- Mesters, G. & Zwiernik, P. (2024), ‘Non-independent component analysis’, *The Annals of Statistics* **52**(6), 2506–2528.
- Moneta, A. & Pallante, G. (2022), ‘Identification of Structural VAR Models via Independent Component Analysis: A Performance Evaluation Study’, *Journal of Economic Dynamics and Control* **144**, 104530.
- Plagborg-Møller, M. & Wolf, C. K. (2021), ‘Local Projections and VARs Estimate the Same Impulse Responses’, *Econometrica* **89**(2), 955–980.
- Read, M. & Zhu, D. (2026), ‘Fast posterior sampling in tightly identified SVARs using soft sign restrictions’, *arXiv preprint arXiv:2603.27088*.
- Rigobon, R. (2003), ‘Identification through heteroskedasticity’, *The Review of Economics and Statistics* **85**(4), 777–792.
- Rubio-Ramirez, J. F., Waggoner, D. F. & Zha, T. (2010), ‘Structural vector autoregressions: Theory of identification and algorithms for inference’, *The Review of Economic Studies* **77**(2), 665–696.
- Sims, C. A. & Zha, T. (1998), ‘Bayesian methods for dynamic multivariate models’, *International Economic Review* **39**(4), 949–968.
- Smets, F. & Wouters, R. (2007), ‘Shocks and frictions in US business cycles: A Bayesian DSGE approach’, *American Economic Review* **97**(3), 586–606.

- Stock, J. H. & Watson, M. W. (2007), ‘Why has U.S. inflation become harder to forecast?’, *Journal of Money, Credit and Banking* **39**(s1), 3–33.
- Uhlig, H. (2005), ‘What are the effects of monetary policy on output? Results from an agnostic identification procedure’, *Journal of Monetary Economics* **52**(2), 381–419.
- Villani, M. (2009), ‘Steady-state priors for vector autoregressions’, *Journal of Applied Econometrics* **24**(4), 630–650.
- Waggoner, D. F. & Zha, T. (2003a), ‘A Gibbs sampler for structural vector autoregressions’, *Journal of Economic Dynamics and Control* **28**(2), 349–366.
- Waggoner, D. F. & Zha, T. (2003b), ‘Likelihood preserving normalization in multiple equation models’, *Journal of Econometrics* **114**(2), 329–347.
- Wolf, C. K. (2020), ‘SVAR (mis)identification and the real effects of monetary policy shocks’, *American Economic Journal: Macroeconomics* **12**(4), 1–32.
- Wu, P. & Koop, G. (2026), ‘Fast, order-invariant Bayesian inference in VARs using the eigendecomposition of the error covariance matrix’, *Journal of Business & Economic Statistics* . forthcoming.

SUPPLEMENTARY MATERIAL

A Derivations of the conditional posterior $p(B|Y, \Theta_{-B})$ via our parametrization

A.1 Derivations

We rewrite here model (1)-(3) from the paper:

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^p \Pi_l \mathbf{y}_{t-l} + B \boldsymbol{\epsilon}_t, \quad (\text{A-1})$$

$$\boldsymbol{\epsilon}_t = D_t^{\frac{1}{2}} \mathbf{e}_t, \quad (\text{A-2})$$

$$\mathbf{e}_t \stackrel{\text{i.i.d.}}{\sim} N(\mathbf{0}, I). \quad (\text{A-3})$$

We leave D_t unspecified at this stage, as this section aims to derive $p(B|Y, \Theta_{-B})$, with Θ_{-B} gathering all parameters of the model except B , including D .

Posterior sampling on B is achieved indirectly via sampling on the parametrization $A = B^{-1}$.

Define $\mathbf{x}_t = (\mathbf{y}'_{t-1}, \dots, \mathbf{y}'_{t-p}, 1)'$ an $m \times 1$ vector containing the lagged variables and the constant term, and define

$$\mathbf{z}_t = \mathbf{y}_t - \Pi \mathbf{x}_t. \quad (\text{A-4})$$

Then model (A-1)-(A-3) coincides with $\mathbf{z}_t \sim N(\mathbf{0}, A^{-1} D_t A^{-1})$, hence the likelihood function is

$$p(Z|\Pi, A, D) = (2\pi)^{-\frac{Tk}{2}} \cdot |\det(A)|^T \cdot \left(\prod_{t=1}^T |\det(D_t)|^{-\frac{1}{2}} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' A' D_t^{-1} A \mathbf{z}_t}, \quad (\text{A-5})$$

with $Z = [\mathbf{z}_1, \dots, \mathbf{z}_t, \dots, \mathbf{z}_T]$.

Define $p_A(A)$ as the prior distribution on A . $p_A(A)$ is of interest to us irrespective of whether prior beliefs are expressed directly on A or on another parametrization, with $p_A(A)$ the

distribution then implied on A . Combining (A-5) with $p_A(A)$ gives the conditional posterior for A :

$$p(A|Y, \phi, D, \mathbf{v}) \propto p_A(A) \cdot |\det(A)|^T \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' A' D_t^{-1} A \mathbf{z}_t}. \quad (\text{A-6})$$

We work with

$$p_A(A) \propto |\det(A)|^\alpha, \quad (\text{A-7})$$

which nests two interesting special cases:

$$p_A(A) \propto |\det(A)|^{-2k}, \quad (\text{A-8})$$

$$p_A(A) \propto 1. \quad (\text{A-9})$$

(A-8), or $\alpha = -2k$, is the density in A -coordinates implied by a prior that is flat with respect to dB , with $|\det(A)|^{-2k}$ the volume element in the transformation from B to A when the entries of B are functionally unconstrained (Kocięcki 2010). (A-9), or $\alpha = 0$, is instead flat with respect to dA .

As explained in the paper, we introduce a new parametrization for A :

$$A = \Lambda LU. \quad (\text{A-10})$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_i, \dots, \lambda_k)$ is a $k \times k$ diagonal matrix, and L and U are lower and upper triangular $k \times k$ matrices with unit diagonal entries. The free entries of (L, U) are $(\mathbf{l}, \mathbf{u})$, respectively. We use the notation

$$\text{vec}(L) = \mathbf{s} + S_L \mathbf{l}, \quad (\text{A-11})$$

$$\text{vec}(U) = \mathbf{s} + S_U \mathbf{u}, \quad (\text{A-12})$$

with $(\mathbf{l}, \mathbf{u})$ of dimension $k(k-1)/2 \times 1$, (S_L, S_U) of dimension $k^2 \times k(k-1)/2$, $\mathbf{s}$ of dimension $k^2 \times 1$, and $(S_L, S_U, \mathbf{s})$ having zero or one entries as appropriate. Note that because (L, U) are triangular matrices with unit diagonal entries, the determinant of A evaluated in the

parametrization (Λ, L, U) is only a function of Λ , namely

$$|\det(A)| = \prod_{i=1}^k |\lambda_i|. \quad (\text{A-13})$$

Prior $p_A(A)$ implies a prior on (Λ, L, U) . To compute it we need the volume element for the transformation from A to (Λ, L, U) , which equals¹

$$v_{A \rightarrow \Lambda, L, U} = \prod_{i=1}^k |\lambda_i|^{k-1}. \quad (\text{A-19})$$

¹The differential of (A-10) is

$$dA = d\Lambda \cdot LU + \Lambda dL U + \Lambda L dU. \quad (\text{A-14})$$

Premultiplying by $L^{-1}\Lambda^{-1}$ and postmultiplying by U^{-1} gives

$$C \equiv L^{-1}\Lambda^{-1} dA U^{-1} = L^{-1}(\Lambda^{-1} d\Lambda) L + L^{-1} dL + dU U^{-1}. \quad (\text{A-15})$$

Since L, U have unit diagonal, $L^{-1}dL$ and $dU U^{-1}$ are strictly triangular and unit-triangular in $d\mathbf{l}$ and $d\mathbf{u}$ respectively, contributing unit Jacobians. Conjugation of the diagonal matrix $\Lambda^{-1}d\Lambda = \text{diag}(d\lambda_1/\lambda_1, \dots, d\lambda_k/\lambda_k)$ by L preserves its diagonal, so $L^{-1}(\Lambda^{-1}d\Lambda)L$ contributes $d\lambda_i/\lambda_i$ to the i -th diagonal entry of C . The map $(d\boldsymbol{\lambda}, d\mathbf{l}, d\mathbf{u}) \mapsto C$ is thus triangular, giving

$$dC = \left(\prod_{i=1}^k \lambda_i^{-1} \right) d\boldsymbol{\lambda} d\mathbf{l} d\mathbf{u}. \quad (\text{A-16})$$

Separately, $C = P dA Q$ with $P = L^{-1}\Lambda^{-1}$, $Q = U^{-1}$ fixed given (Λ, L, U) , so $\text{vec}(C) = (Q' \otimes P) \text{vec}(dA)$ and $\det(Q' \otimes P) = (\det Q)^k (\det P)^k$. With $\det(L) = \det(U) = 1$,

$$dC = \left(\prod_{i=1}^k \lambda_i^{-1} \right)^k dA. \quad (\text{A-17})$$

Equating (A-16) and (A-17) gives the volume element for the transformation from A to (Λ, L, U) :

$$dA = |\det(\Lambda)|^{k-1} d\boldsymbol{\lambda} d\mathbf{l} d\mathbf{u}. \quad (\text{A-18})$$

Hence $p_A(A)$ implies

$$p(\Lambda, L, U) \propto v_{A \rightarrow \Lambda, L, U} \cdot p_A(\Lambda LU) \propto \prod_{i=1}^k |\lambda_i|^{k-1+\alpha}, \quad (\text{A-20})$$

$$p(\Lambda) \propto \prod_{i=1}^k |\lambda_i|^{k-1+\alpha}, \quad (\text{A-21})$$

$$p(L) \propto 1, \quad (\text{A-22})$$

$$p(U) \propto 1, \quad (\text{A-23})$$

$$p(\Lambda, L, U) = p(\Lambda) p(L) p(U). \quad (\text{A-24})$$

As for the posterior, (A-6) implies the following conditional posterior jointly for (Λ, L, U) :

$$\begin{aligned} p(\Lambda, L, U | Y, \phi, D) &\propto \underbrace{\left(\prod_{i=1}^k |\lambda_i|^{k-1} \right) \cdot p_A(\Lambda LU)}_{\propto p(\Lambda, L, U) \propto \prod_{i=1}^k |\lambda_i|^{k-1+\alpha}} \cdot \left(\prod_{i=1}^k |\lambda_i|^T \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' L' \Lambda D_t^{-1} \Lambda L U \mathbf{z}_t}, \\ & \quad (\text{A-25}) \end{aligned}$$

$$= \left(\prod_{i=1}^k |\lambda_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' L' \Lambda D_t^{-1} \Lambda L U \mathbf{z}_t}. \quad (\text{A-26})$$

Our approach to sample from $p(\Lambda, L, U | Y, \phi, D)$ consists of showing that the conditional posteriors of Λ , L and U all have a common form. Starting with L , under either (A-8) or (A-9) we can derive

$$p(L | Y, \phi, D, \mathbf{v}, \Lambda, U) \propto e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' L' \Lambda D_t^{-1} \Lambda L U \mathbf{z}_t}, \quad (\text{A-27})$$

$$= e^{-\frac{1}{2} \text{vec}(L)' \sum_{t=1}^T \left(U \mathbf{z}_t \mathbf{z}_t' U' \otimes \Lambda D_t^{-1} \Lambda \right) \text{vec}(L)}, \quad (\text{A-28})$$

$$= e^{-\frac{1}{2} \text{vec}(L)' \sum_{t=1}^T \left(U \mathbf{z}_t \mathbf{z}_t' U' \otimes \Lambda^2 D_t^{-1} \right) \text{vec}(L)}, \quad (\text{A-29})$$

$$p(\mathbf{l} | Y, \phi, D, \mathbf{v}, \Lambda, U) \propto e^{-\frac{1}{2} (\mathbf{s} + S_L \mathbf{l})' W_L (\mathbf{s} + S_L \mathbf{l})}, \quad (\text{A-30})$$

$$\propto e^{-\frac{1}{2} (\mathbf{l}' S_L' W_L S_L \mathbf{l} + 2 \mathbf{l}' S_L' W_L \mathbf{s})}, \quad (\text{A-31})$$

hence,

$$\mathbf{l}|Y, \boldsymbol{\phi}, D, \mathbf{v}, \Lambda, U \sim N(\boldsymbol{\mu}_L^*, V_L^*), \quad (\text{A-32})$$

$$V_L^* = (S_L' W_L S_L)^{-1} \quad (\text{A-33})$$

$$\boldsymbol{\mu}_L^* = -V_L^* S_L' W_L \mathbf{s}, \quad (\text{A-34})$$

$$W_L = \sum_{t=1}^T \left(U \mathbf{z}_t \mathbf{z}_t' U' \otimes \Lambda^2 D_t^{-1} \right), \quad (\text{A-35})$$

The derivations for U are similar: under either (A-8) or (A-9) we can derive

$$p(U|Y, \boldsymbol{\phi}, D, \mathbf{v}, \Lambda, L) \propto e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' L' \Lambda D_t^{-1} \Lambda L U \mathbf{z}_t}, \quad (\text{A-36})$$

$$= e^{-\frac{1}{2} \text{vec}(U)' \sum_{t=1}^T \left(\mathbf{z}_t \mathbf{z}_t' \otimes L' \Lambda D_t^{-1} \Lambda L \right) \text{vec}(U)}, \quad (\text{A-37})$$

$$= e^{-\frac{1}{2} \text{vec}(U)' \sum_{t=1}^T \left(\mathbf{z}_t \mathbf{z}_t' \otimes L' \Lambda^2 D_t^{-1} L \right) \text{vec}(U)}, \quad (\text{A-38})$$

$$p(\mathbf{u}|Y, \boldsymbol{\phi}, D, \mathbf{v}, \Lambda, L) \propto e^{-\frac{1}{2} (\mathbf{s} + S_U \mathbf{u})' W_U (\mathbf{s} + S_U \mathbf{u})}, \quad (\text{A-39})$$

$$\propto e^{-\frac{1}{2} (\mathbf{u}' S_U' W_U S_U \mathbf{u} + 2 \mathbf{u}' S_U' W_U \mathbf{s})}, \quad (\text{A-40})$$

hence

$$\mathbf{u}|Y, \boldsymbol{\phi}, D, \mathbf{v}, \Lambda, L \sim N(\boldsymbol{\mu}_U^*, V_U^*), \quad (\text{A-41})$$

$$V_U^* = (S_U' W_U S_U)^{-1}, \quad (\text{A-42})$$

$$\boldsymbol{\mu}_U^* = -V_U^* S_U' W_U \mathbf{s}, \quad (\text{A-43})$$

$$W_U = \sum_{t=1}^T \left(\mathbf{z}_t \mathbf{z}_t' \otimes L' \Lambda^2 D_t^{-1} L \right). \quad (\text{A-44})$$

It remains to derive the conditional posterior for Λ , which is

$$p(\Lambda|Y, \phi, D, \mathbf{v}, L, U) \propto \left(\prod_{i=1}^k |\lambda_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' L' \Lambda D_t^{-1} \Lambda L U \mathbf{z}_t}, \quad (\text{A-45})$$

$$= \left(\prod_{i=1}^k |\lambda_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' L' D_t^{-0.5} \Lambda^2 D_t^{-0.5} L U \mathbf{z}_t}, \quad (\text{A-46})$$

$$= \left(\prod_{i=1}^k |\lambda_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{c}_t' \Lambda^2 \mathbf{c}_t}, \quad (\text{A-47})$$

$$= \left(\prod_{i=1}^k |\lambda_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{i=1}^k \sum_{t=1}^T c_{it}^2 \lambda_i^2}, \quad (\text{A-48})$$

$$= \left(\prod_{i=1}^k |\lambda_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \lambda_i^2 \sum_{t=1}^T c_{it}^2}, \quad (\text{A-49})$$

where $\mathbf{c}_t = D_t^{-0.5} L U \mathbf{z}_t$.

$p(\Lambda|Y, \phi, D, L, U)$ is not yet of common form. However, let us define $x_i = \lambda_i^2$. Although this transformation is not 1-1 since λ_i may be both positive and negative, from standard probability we know that if λ_i has pdf $p(\lambda_i)$ then the pdf of x_i is $g(x_i) = \frac{1}{2} x_i^{-\frac{1}{2}} p(\sqrt{x_i}) + \frac{1}{2} x_i^{-\frac{1}{2}} p(-\sqrt{x_i})$, for $x_i > 0$. Since in our case $p(\sqrt{x_i}) = p(-\sqrt{x_i})$, hence $g(x_i) = x_i^{-\frac{1}{2}} p(\sqrt{x_i})$.

It follows that

$$p(x_1, x_2, \dots, x_k | Y, \phi, D, \mathbf{v}, L, U) \propto \prod_{i=1}^k x_i^{-\frac{1}{2}} |x_i^{\frac{1}{2}}|^{T+k-1+\alpha} \cdot e^{-\frac{1}{2} x_i \sum_{t=1}^T c_{it}^2}, \quad (\text{A-50})$$

$$\propto \prod_{i=1}^k x_i^{\frac{T+k+\alpha}{2}-1} \cdot e^{-\frac{1}{2} x_i \sum_{t=1}^T c_{it}^2}, \quad (\text{A-51})$$

hence

$$x_i | Y, \phi, D, \mathbf{v}, L, U \sim \Gamma(h_{\lambda,i}^*, r_{\lambda,i}^*), \quad (\text{A-52})$$

$$r_{\lambda,i}^* = \frac{\sum_{t=1}^T c_{it}^2}{2}, \quad (\text{A-53})$$

$$\mathbf{c}_t = D_t^{-0.5} L U \mathbf{z}_t, \quad (\text{A-54})$$

$$h_{\lambda,i}^* = \frac{T+k+\alpha}{2}, \quad (\text{A-55})$$

Note that we are using the following shape-rate parametrization of the Gamma distribution

$$p(x|h, r) = \frac{r^h}{\Gamma(h)} \cdot x^{h-1} e^{-r \cdot x}. \quad (\text{A-56})$$

Note also that, whenever $2h_{\lambda,i}^* = T + k + \alpha$ is a positive integer, as it is for the flat priors on A and on B used in the paper (the latter requiring $T > k$), drawing from (A-52) is equivalent to drawing from standard Normal distributions as follows:

$$\mathbf{z}_i = (z_{i,1}, \dots, z_{i,2h_{\lambda,i}^*})' \sim \mathcal{N}(\mathbf{0}, I_{2h_{\lambda,i}^*}), \quad (\text{A-57})$$

$$x_i \mid Y, \phi, D, \mathbf{v}, L, U = \frac{1}{2r_{\lambda,i}^*} \mathbf{z}_i' \mathbf{z}_i. \quad (\text{A-58})$$

Having drawn x_i we set $\lambda_i = \sqrt{x_i}$ with probability $\frac{1}{2}$, or $\lambda_i = -\sqrt{x_i}$ with probability $\frac{1}{2}$, see Waggoner & Zha (2003a), p. 357, for an analogous treatment.

A.2 Discussion

Part of the literature finds it more natural to write model (1) - (3) as

$$A\mathbf{y}_t = \sum_{l=1}^p G_l \mathbf{y}_{t-l} + \mathbf{g} + \sqrt{D_t} \mathbf{e}_t, \quad \mathbf{e}_t \sim N(\mathbf{0}, I_k), \quad (\text{A-59})$$

or, compactly, $A\mathbf{y}_t = C\mathbf{x}_t + \sqrt{D_t} \mathbf{e}_t$, with $C = [G_1, \dots, G_p, \mathbf{g}]$ and $\mathbf{x}_t$ as defined above. $[G_1, \dots, G_p]$ captures the structural (rather than the reduced form) parameters of the model, with $G_l = A \Pi_l$. We stress that our sampler works only when conditioning on Π rather than $C = A \Pi$. In fact, the likelihood function under the alternative parametrization is

$$p(Y|C, A, D) = (2\pi)^{-\frac{Tk}{2}} \cdot |\det(A)|^T \cdot \left(\prod_{t=1}^T |\det(D_t)|^{-\frac{1}{2}} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{w}_t' D_t^{-1} \mathbf{w}_t}, \quad (\text{A-60})$$

with $\mathbf{w}_t \equiv A\mathbf{y}_t - C\mathbf{x}_t$. Unlike $\mathbf{z}_t = \mathbf{y}_t - \Pi\mathbf{x}_t$ in (A-5), $\mathbf{w}_t$ depends on A , so the quadratic form is no longer a pure sandwich in A . Expanding and stacking over t gives

$$\sum_{t=1}^T \mathbf{w}_t' D_t^{-1} \mathbf{w}_t = \text{vec}(A)' M \text{vec}(A) - 2 \text{vec}(A)' \mathbf{m} + \text{const}, \quad (\text{A-61})$$

$$M = \sum_{t=1}^T \mathbf{y}_t \mathbf{y}_t' \otimes D_t^{-1}, \quad \mathbf{m} = \sum_{t=1}^T \mathbf{y}_t \otimes D_t^{-1} C \mathbf{x}_t. \quad (\text{A-62})$$

Combining (A-60)-(A-62) with the prior $p_A(A) \propto |\det(A)|^\alpha$ gives the conditional posterior of A :

$$p(A|Y, C, D) \propto |\det(A)|^{T+\alpha} \cdot \exp \left\{ -\frac{1}{2} \left[\text{vec}(A)' M \text{vec}(A) - 2 \text{vec}(A)' \mathbf{m} \right] \right\}. \quad (\text{A-63})$$

Relative to (A-6), the quadratic form in $\text{vec}(A)$ in (A-63) now carries an additional linear term $-2 \text{vec}(A)' \mathbf{m}$ induced by C . With our prior, the reparametrization into Λ, L, U still delivers Normal conditional posterior on L, U . However, our key result for the conditional posterior on Λ does not carry through because

$$p(\lambda_i|Y, C, D, L, U) \propto |\lambda_i|^{T+k-1+\alpha} \cdot \exp \left\{ -\frac{1}{2} a_i \lambda_i^2 + \beta_i \lambda_i \right\}, \quad (\text{A-64})$$

with

$$a_i = \sum_{t=1}^T (\boldsymbol{\iota}_i' D_t^{-1} \boldsymbol{\iota}_i) (\boldsymbol{\iota}_i' L U \mathbf{y}_t)^2, \quad (\text{A-65})$$

$$\beta_i = \sum_{t=1}^T (\boldsymbol{\iota}_i' D_t^{-1} \boldsymbol{\iota}_i) (\boldsymbol{\iota}_i' L U \mathbf{y}_t) (\boldsymbol{\iota}_i' C \mathbf{x}_t), \quad (\text{A-66})$$

where $\boldsymbol{\iota}_i$ is a $k \times 1$ vector of zeros, except for entry i , which is set to 1. Hence, as long as $\beta_i \neq 0$, which is the generic case when $C \neq 0$, (A-64) is not symmetric in $\lambda_i \rightarrow -\lambda_i$. This breaks the change of variables $x_i = \lambda_i^2$ used to obtain a Gamma conditional posterior with an independent sign draw.

A similar reasoning explains why the Gibbs sampler can be derived with our parametrization, but not with a standard LDU parameterization. Consider the alternative decomposition $A = L \tilde{D} U$, with L, U unit triangular as in $\Lambda L U$ and $\tilde{D} = \text{diag}(\tilde{d}_1, \dots, \tilde{d}_k)$ diagonal. Since L, U have unit diagonal entries, $|\det(A)| = \prod_{i=1}^k |\tilde{d}_i|$, exactly as in (A-13), so an analogous prior on A implies, as in (A-25),

$$p(L, \tilde{D}, U|Y, \phi, D) \propto \left(\prod_{i=1}^k |\tilde{d}_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{z}_t' U' \tilde{D} L' D_t^{-1} L \tilde{D} U \mathbf{z}_t}. \quad (\text{A-67})$$

Define $\mathbf{c}_t = U\mathbf{z}_t$ and $\Omega_t = L'D_t^{-1}L$. Unlike the ΛLU case, in which the matrix between the two copies of Λ is D_t^{-1} and hence diagonal, here the intervening matrix $\Omega_t = L'D_t^{-1}L$ is not diagonal because L is unit lower triangular. The conditional posterior for $\tilde{D}$ is then

$$p(\tilde{D}|Y, \phi, D, L, U) \propto \left(\prod_{i=1}^k |\tilde{d}_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \mathbf{c}_t' \tilde{D} \Omega_t \tilde{D} \mathbf{c}_t}, \quad (\text{A-68})$$

$$= \left(\prod_{i=1}^k |\tilde{d}_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{t=1}^T \sum_{i=1}^k \sum_{j=1}^k \tilde{d}_i \tilde{d}_j \Omega_{t,ij} c_{it} c_{jt}}, \quad (\text{A-69})$$

$$= \left(\prod_{i=1}^k |\tilde{d}_i|^{T+k-1+\alpha} \right) \cdot e^{-\frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \tilde{d}_i \tilde{d}_j Q_{ij}}, \quad (\text{A-70})$$

with $Q_{ij} \equiv \sum_{t=1}^T \Omega_{t,ij} c_{it} c_{jt}$. Because Ω_t is not diagonal, $Q_{ij} \neq 0$ for $i \neq j$ in general, so the exponent in (A-70) is a genuine quadratic form in $(\tilde{d}_1, \dots, \tilde{d}_k)$ rather than the separable sum $\sum_i \tilde{d}_i^2 \sum_t c_{it}^2$ obtained in the ΛLU case at the corresponding step. Consequently (A-70) cannot be written as a product of k terms each depending on a single $\tilde{d}_i$, and the change of variables $x_i = \tilde{d}_i^2$ used to obtain (A-52) does not apply: the cross terms $\tilde{d}_i \tilde{d}_j Q_{ij}$ mix the signs of $\tilde{d}_i$ and $\tilde{d}_j$, so they are not expressible as functions of (x_i, x_j) alone. The conditional posterior for $\tilde{D}$ therefore has no Gamma form, and no analogous independent sign draw is available.

Sign restrictions are compatible with our sampler. Define $f(\Theta)$ a function that returns 1 if candidate sign restrictions are satisfied and zero otherwise, where the restrictions can be introduced on nonlinear functions of the model parameters (including impulse responses). Since $f(\Theta)$ effectively truncates the joint posterior distributions, the conditional posterior distributions for (Λ, L, U) remain valid, up to an appropriate truncation term. It remains to draw from these distributions. One option is to follow [Arias et al. \(2026\)](#) and use an Elliptical Slice Sampler. An alternative approach is to follow [Read & Zhu \(2026\)](#). When doing so, one should assess how sign restrictions interact with the multimodality of the likelihood and the mode-switching problem.

We conclude by showing that the method is order invariant, as long as the joint prior on (A, Π) is equivariant to the reordering, which is the case for the prior used in the paper (Chan et al. 2024, Arias et al. 2023). Let P be a $k \times k$ permutation matrix, and define the reordered variables $\tilde{\mathbf{y}}_t = P\mathbf{y}_t$. Premultiplying the model by P and inserting $P'P = I_k$ before each $\mathbf{y}_{t-l}$ gives

$$\tilde{\mathbf{y}}_t = \sum_{l=1}^p \left(P\Pi_l P' \right) \tilde{\mathbf{y}}_{t-l} + P\mathbf{c} + (PB)\boldsymbol{\epsilon}_t, \quad (\text{A-71})$$

$$= \sum_{l=1}^p \tilde{\Pi}_l \tilde{\mathbf{y}}_{t-l} + \tilde{\mathbf{c}} + \tilde{B}\boldsymbol{\epsilon}_t, \quad (\text{A-72})$$

with $\tilde{\Pi}_l = P\Pi_l P'$, $\tilde{\mathbf{c}} = P\mathbf{c}$, $\tilde{B} = PB$, for every lag $l = 1, \dots, p$. Since $|PBD_t B' P'| = |BD_t B'|$ and $(PBD_t B' P')^{-1} = P(BD_t B')^{-1} P'$, the quadratic form transforms as

$$\left(\tilde{\mathbf{y}}_t - \sum_{l=1}^p \tilde{\Pi}_l \tilde{\mathbf{y}}_{t-l} - \tilde{\mathbf{c}} \right)' (\tilde{B}D_t \tilde{B}')^{-1} \left(\tilde{\mathbf{y}}_t - \sum_{l=1}^p \tilde{\Pi}_l \tilde{\mathbf{y}}_{t-l} - \tilde{\mathbf{c}} \right) \quad (\text{A-73})$$

$$= \left(\mathbf{y}_t - \sum_{l=1}^p \Pi_l \mathbf{y}_{t-l} - \mathbf{c} \right)' (BD_t B')^{-1} \left(\mathbf{y}_t - \sum_{l=1}^p \Pi_l \mathbf{y}_{t-l} - \mathbf{c} \right), \quad (\text{A-74})$$

using $P'P = I_k$ term by term. Hence

$$\mathcal{L}(\{\tilde{\Pi}_l\}_{l=1}^p, \tilde{\mathbf{c}}, \tilde{B}, \{D_t\}) = \mathcal{L}(\{\Pi_l\}_{l=1}^p, \mathbf{c}, B, \{D_t\}), \quad (\text{A-75})$$

i.e. the likelihood is invariant to reordering the variables, provided one simultaneously permutes both rows *and* columns of every Π_l ($l = 1, \dots, p$), the entries of $\mathbf{c}$, and the rows of B , exactly as in the one-lag case. For this order invariance to carry over from the likelihood to the posterior, the prior must satisfy $p(\{\Pi_l\}, \mathbf{c}, B) = p(\{P\Pi_l P'\}, P\mathbf{c}, PB)$. This is satisfied for prior (12) in the paper, since $|\det(PA)| = |\det(A)|$. The Normal prior on $\boldsymbol{\phi}$ also satisfies it, because the Minnesota-type calibration treats equations symmetrically: reordering the variables reorders the equation-specific training-sample scales and the own-lag versus cross-lag shrinkage in the same way, so the prior of the reordered system is the reordered prior.

B Flat priors under $A = B^{-1}$

Writing the sampler in terms of $A = B^{-1}$ does not by itself change posterior inference: transforming the prior density with the appropriate Jacobian leaves the posterior measure unchanged. Specifying a flat density does change the prior measure. A density that is flat with respect to dA is not flat with respect to dB , and posterior-density modes computed in the two coordinate systems need not map by inversion. These differences do not affect sign and permutation equivalence or the normalization used to address it.

B.1 Matrix inversion and reference measures

Let A and B be unrestricted nonsingular $k \times k$ matrices, with $B = A^{-1}$. Differentiating the inverse gives

$$dB = -A^{-1}(dA)A^{-1}, \quad (\text{B-76})$$

and vectorization yields

$$\text{vec}(dB) = -\left(A^{-T} \otimes A^{-1}\right) \text{vec}(dA). \quad (\text{B-77})$$

Since $\det(C \otimes D) = \det(C)^k \det(D)^k$ for $k \times k$ matrices C and D , the absolute Jacobian of matrix inversion is

$$\left| \frac{\partial \text{vec}(B)}{\partial \text{vec}(A)} \right| = |\det(A)|^{-2k}. \quad (\text{B-78})$$

A density $p_B(B)$ with respect to dB therefore induces the density

$$p_A(A) = p_B(A^{-1})|\det(A)|^{-2k} \quad (\text{B-79})$$

with respect to dA . In particular, $p_B(B) \propto 1$ implies $p_A(A) \propto |\det(A)|^{-2k}$.

Equation (B-79) changes density per unit coordinate volume, not posterior probabilities.

Let $\mathcal{C}_A$ be a measurable set of nonsingular matrices and let $\mathcal{C}_B = \{A^{-1} : A \in \mathcal{C}_A\}$. The

corresponding posterior probabilities satisfy

$$\int_{\mathcal{C}_A} \pi_A(A | Y) dA = \int_{\mathcal{C}_B} \pi_B(B | Y) dB, \quad (\text{B-80})$$

where π_A and π_B are densities of the same posterior measure. Thus, when the posterior is proper, transforming a prior from B to A leaves inference unchanged, while imposing a flat density separately in the two coordinate systems changes the prior measure. Equation (B-78) applies to Lebesgue measure on all k^2 entries. Functional restrictions on A or B require the Jacobian for the corresponding free parameters.

B.2 Likelihood and posterior-density modes

Let $L_B(B)$ denote the likelihood in B -coordinates. In A -coordinates,

$$L_A(A) = L_B(A^{-1}).$$

Because matrix inversion is one-to-one and continuous in both directions, it maps likelihood modes exactly:

$$\hat{A}_L = \hat{B}_L^{-1}. \quad (\text{B-81})$$

No Jacobian enters (B-81) because the equation rewrites the same function using a different argument.

Posterior-density modes need not obey the same relation. Under a prior that is flat with respect to dB , the posterior densities of the same measure are

$$\pi_B(B | Y) \propto L_B(B), \quad (\text{B-82})$$

$$\pi_A(A | Y) \propto L_A(A) |\det(A)|^{-2k}. \quad (\text{B-83})$$

At the inverse likelihood mode $\hat{A}_L$, the score of the density with respect to dA is

$$\nabla_A \log \pi_A(\hat{A}_L | Y) = -2k \hat{A}_L^{-T} \neq 0. \quad (\text{B-84})$$

The inverse of a B -coordinate posterior-density mode is therefore generally not a mode of the posterior density measured with respect to dA . This difference reflects the coordinate volume used to define a density; it does not represent a change in the posterior measure.

The conditional posterior used in our sampler makes the displacement in (B-84) explicit. Conditional on D and the remaining parameters, define

$$Q(A) = \sum_{t=1}^T \mathbf{z}'_t A' D_t^{-1} A \mathbf{z}_t, \quad (\text{B-85})$$

so that $Q(cA) = c^2 Q(A)$. Under $p_A(A) \propto |\det(A)|^\alpha$, the conditional log density from (A-6) is

$$\ell_\alpha(A) = (T + \alpha) \log |\det(A)| - \frac{1}{2} Q(A) + \text{constant}. \quad (\text{B-86})$$

Suppose $\hat{A}_{0,D}$ is a stationary point of the conditional likelihood and $T + \alpha > 0$. Its first-order condition is $\nabla Q(\hat{A}_{0,D})/2 = T \hat{A}_{0,D}^{-T}$. Evaluating the score of (B-86) at $c\hat{A}_{0,D}$ gives

$$\nabla_A \ell_\alpha(c\hat{A}_{0,D}) = \left(\frac{T + \alpha}{c} - cT \right) \hat{A}_{0,D}^{-T}. \quad (\text{B-87})$$

It follows that

$$\hat{A}_{\alpha,D} = \sqrt{\frac{T + \alpha}{T}} \hat{A}_{0,D} \quad (\text{B-88})$$

is a stationary point of the conditional posterior density. The Hessian at $\hat{A}_{\alpha,D}$ has the same local classification as the Hessian at $\hat{A}_{0,D}$. For the prior that is flat with respect to dB , $\alpha = -2k$, and hence

$$\hat{A}_{\text{flat } B,D} = \sqrt{\frac{T - 2k}{T}} \hat{A}_{0,D}, \quad T > 2k. \quad (\text{B-89})$$

Equation (B-89) is an exact conditional result with D and the other parameter blocks held fixed. The restriction $T > 2k$ permits an interior stationary point along the scale direction; it does not establish posterior propriety. Both flat priors are improper, so propriety must be assessed separately.

B.3 Signed permutations and the sampler

Let P_s be a signed permutation matrix and $P = |P_s|$ its unsigned permutation. The observationally equivalent representation BP_s corresponds to $P'_s A$ in A -coordinates. Shock-specific quantities must be relabeled at the same time: D_t becomes $P'D_tP$, and $\mathbf{v}$ becomes $P'\mathbf{v}$. Since D_t is diagonal, this joint transformation satisfies

$$(P'_s A)'(P'D_tP)^{-1}(P'_s A) = A'D_t^{-1}A. \quad (\text{B-90})$$

With exchangeable priors on the shock-specific quantities, the joint relabeling therefore leaves the likelihood and posterior measure unchanged. Moreover,

$$|\det(P'_s A)|^\alpha = |\det(A)|^\alpha \quad (\text{B-91})$$

for every real α , because $|\det(P_s)| = 1$. A determinant-power prior therefore assigns the same density to all $k!2^k$ observationally equivalent representations. It can move the modes within each representation, as in (B-88), but it cannot select a sign or ordering of the shocks.

The same Jacobians determine the conditional distribution used by the sampler. Combining (B-79) with the volume element for $A = \Lambda LU$ in (A-18) gives the Gamma shape

$$h_{\lambda,i}^* = \frac{T + k + \alpha}{2}. \quad (\text{B-92})$$

The shape is $(T - k)/2$ under a prior that is flat with respect to dB and $(T + k)/2$ under a prior that is flat with respect to dA , as reported in the paper. The flat- B shape is positive when $T > k$ and the rate is nonzero. These terms ensure that the sampler targets the intended posterior measure. The generalized LP normalization serves a different purpose: it selects one consistently signed and ordered representative from each observationally equivalent set. It neither changes the posterior measure nor selects a coordinate-density mode.

C Model used in Sections 3–4 of the paper

Model (1)-(3) in the paper is very general and can accommodate different forms of departures from a standard Gaussian multivariate SVAR model. Section 3-4 of the paper illustrate our methodology using the following specification of model (1)-(3):

$$\mathbf{y}_t = \sum_{l=1}^p \Pi_l \mathbf{y}_{t-l} + \mathbf{c} + u_t, \quad (\text{C-93})$$

$$= \Pi \mathbf{x}_t + u_t, \quad (\text{C-94})$$

$$u_t = B \boldsymbol{\epsilon}_t, \quad (\text{C-95})$$

$$p(\boldsymbol{\epsilon}_t | \boldsymbol{\sigma}, \mathbf{v}) = \prod_{i=1}^k p(\epsilon_{it} | \sigma_i, v_i), \quad (\text{C-96})$$

$$\epsilon_{it} \sim t(\sigma_i, v_i), \quad (\text{C-97})$$

$$p(\epsilon_{it} | \sigma_i, v_i) = \sigma_i^{-1} \cdot v_i^{-\frac{1}{2}} \cdot \frac{\Gamma\left(\frac{v_i+1}{2}\right)}{\pi^{\frac{1}{2}} \Gamma\left(\frac{v_i}{2}\right)} \cdot \left(1 + \frac{\epsilon_{it}^2}{v_i \cdot \sigma_i^2}\right)^{-\frac{v_i+1}{2}}. \quad (\text{C-98})$$

Specifically, the structural shocks $\boldsymbol{\epsilon}_t$ are *i.i.d.* with zero mean. Individual components of $\boldsymbol{\epsilon}_t$, i.e. ϵ_{it} , are mutually independent and possess a univariate t -distribution, possibly with different degrees of freedom. Following, for example, Geweke (1993), the probability density function of each shock is parametrized according to equation (C-98), with (σ_i, v_i) the shock-specific scale and degrees of freedom, and $\mathbf{v} = (v_1, \dots, v_k)'$, $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_k)'$. As in Brunnermeier et al. (2021) (see their footnote 11), we set the scale parameter to:

$$\sigma_i = \sqrt{\frac{v_i - 2}{v_i}}, \quad (\text{C-99})$$

which implies that the variance of each structural t -distributed shock is normalized to unity,

$$V(\epsilon_{it}) = 1. \quad (\text{C-100})$$

Under this normalization, the $k \times k$ matrix B captures the impact effect of a one-standard-deviation shock.²

²This normalization implies that the set of observationally equivalent models differ only up to sign and permutation of the shocks, but not up to the scale of the shocks (Lanne et al. 2017). We exploit this feature

As in [Geweke \(1993\)](#), we use the following, alternative specification of the model:

$$\mathbf{y}_t = \sum_{l=1}^p \Pi_l \mathbf{y}_{t-l} + \mathbf{c} + U_t, \quad (\text{C-101})$$

$$= \Pi \mathbf{x}_t + U_t, \quad (\text{C-102})$$

$$U_t = B \sqrt{D_t} \mathbf{e}_t, \quad (\text{C-103})$$

$$\mathbf{e}_t \sim N(\mathbf{0}, I), \quad (\text{C-104})$$

$$D_t = \text{diag}(d_{1t}, \dots, d_{it}, \dots, d_{kt}), \quad (\text{C-105})$$

$$p(d_{it}|h_{d,i}, r_{d,i}) = \frac{r_{d,i}^{h_{d,i}}}{\Gamma(h_{d,i})} \cdot d_{it}^{-h_{d,i}-1} e^{-r_{d,i} \cdot \frac{1}{d_{it}}}, \quad (\text{C-106})$$

$$h_{d,i} = \frac{v_i}{2}, \quad r_{d,i} = \frac{v_i - 2}{2}. \quad (\text{C-107})$$

The k stochastic terms in the new specification, $\mathbf{e}_t$, are Gaussian with the identity covariance matrix. The latent variables d_{it} are treated as unknown parameters with an inverse Gamma prior that is independent across i and t , parametrized according to [\(C-106\)](#), and $\sqrt{D_t} = \text{diag}(d_{1t}^{\frac{1}{2}}, \dots, d_{it}^{\frac{1}{2}}, \dots, d_{kt}^{\frac{1}{2}})$. It is assumed that D_t is independent of $\mathbf{e}_s$ for every t and s . The shape and rate parameters $(h_{d,i}, r_{d,i})$ of $p(d_{it}|h_{d,i}, r_{d,i})$ are set as in [\(C-107\)](#), which comply with the normalization $V(\epsilon_{it}) = 1$ made in the original model specification.

Rewrite the parametrization of the model as

$$\boldsymbol{\phi} = \text{vec}(\Pi), \quad (\text{C-108})$$

$$D = \text{diag}(D_1, \dots, D_t, \dots, D_T). \quad (\text{C-109})$$

The vector $\boldsymbol{\phi}$ is of dimensions $km \times 1$. The array D is sparse and of dimensions $kT \times kT$.

The joint prior distribution for the alternative specification of the model is

$$p(\boldsymbol{\phi}, B, D, \mathbf{v}) = p(\boldsymbol{\phi}) \cdot p(B) \cdot p(D|\mathbf{v}) \cdot p(\mathbf{v}). \quad (\text{C-110})$$

Our approach requires the prior on $\boldsymbol{\phi}$ to be Normal or flat. While we set $\boldsymbol{\phi}$ as a-priori independent of $(B, D, \mathbf{v})$, this modelling assumption can be removed. As discussed above,

in order to build the generalized LP normalization, see [subsection 2.3](#) of the paper and [Appendix E](#).

the prior $p(D|\mathbf{v})$ is given by (C-106), with hyperparameters calibrated according to (C-107). The prior on $\mathbf{v}$ can be chosen freely, since it enters only through the grid. The prior on B must belong to the determinant-power class (12) or to the compatible alternatives discussed in Section A.2; an arbitrary prior on B would not preserve the standard-form conditionals. Posterior sampling is achieved by means of a Gibbs sampler. The conditional posterior of B was discussed in Section A. The conditional posteriors for the parameters $(\phi, D, \mathbf{v})$ are standard in the literature and are reported below for completeness.

C.1 Conditional posterior $p(\phi|Y, B, D, \mathbf{v})$

The likelihood function of model (C-101)-(C-104) can be written as

$$p(Y|\phi, B, D) = (2\pi)^{-\frac{Tk}{2}} \cdot |\det(B)|^{-T} \cdot |\det(D)|^{-\frac{1}{2}} \cdot e^{-\frac{1}{2}(\tilde{\mathbf{y}}-W\phi)' \left((I_T \otimes B) D (I_T \otimes B') \right)^{-1} (\tilde{\mathbf{y}}-W\phi)}, \quad (\text{C-111})$$

with

$$Y = [\mathbf{y}_1, \dots, \mathbf{y}_t, \dots, \mathbf{y}_T], \quad (\text{C-112})$$

$$\tilde{\mathbf{y}} = \text{vec}(Y), \quad (\text{C-113})$$

$$X = [\mathbf{x}_1, \dots, \mathbf{x}_t, \dots, \mathbf{x}_T], \quad (\text{C-114})$$

$$W = (X' \otimes I_k). \quad (\text{C-115})$$

Using (C-111), we can derive

$$p(\phi|Y, B, D, \mathbf{v}) \propto p(\phi) \cdot p(Y|\phi, B, D), \quad (\text{C-116})$$

$$\propto e^{-\frac{1}{2}(\phi-\mu)' V^{-1}(\phi-\mu)} \cdot e^{-\frac{1}{2}(\tilde{\mathbf{y}}-W\phi)' \left((I_T \otimes B) D (I_T \otimes B') \right)^{-1} (\tilde{\mathbf{y}}-W\phi)}, \quad (\text{C-117})$$

$$= e^{-\frac{1}{2}(\phi-\mu)' V^{-1}(\phi-\mu)} \cdot e^{-\frac{1}{2}(\tilde{\mathbf{y}}-W\phi)' \Omega^{-1}(\tilde{\mathbf{y}}-W\phi)}, \quad (\text{C-118})$$

$$\propto e^{-\frac{1}{2}(\phi' V^{-1} \phi - 2\phi' V^{-1} \mu + \phi' W' \Omega^{-1} W \phi - 2\phi' W' \Omega^{-1} \tilde{\mathbf{y}})}, \quad (\text{C-119})$$

$$\phi|Y, B, D, \mathbf{v} \sim N(\boldsymbol{\mu}^*, V^*), \quad (\text{C-120})$$

$$V^* = (V^{-1} + W'\Omega^{-1}W)^{-1}, \quad (\text{C-121})$$

$$\boldsymbol{\mu}^* = V^*[V^{-1}\boldsymbol{\mu} + W'\Omega^{-1}\tilde{\mathbf{y}}], \quad (\text{C-122})$$

$$\Omega = (I_T \otimes B)D(I_T \otimes B'). \quad (\text{C-123})$$

C.2 Conditional posterior $p(D|Y, \phi, B, \mathbf{v})$

The conditional posterior for D is also standard. After defining

$$\mathbf{g}_t = B^{-1}[\mathbf{y}_t - \Pi\mathbf{x}_t], \quad (\text{C-124})$$

model (C-101)-(C-104) coincides with $\mathbf{g}_t \sim N(\mathbf{0}, D_t)$, hence

$$p(D|Y, \phi, B, \mathbf{v}) = \prod_{t=1}^T \prod_{i=1}^k p(d_{it}|\mathbf{y}_t, \mathbf{x}_t, \Pi, B, v_i), \quad (\text{C-125})$$

$$p(d_{it}|\mathbf{y}_t, \mathbf{x}_t, \Pi, B, v_i) \propto p(d_{it}|h_{d,i}, r_{d,i}) \cdot p(\mathbf{g}_t|\Pi, B, \mathbf{v}, D_t), \quad (\text{C-126})$$

$$\propto \left[\frac{r_{d,i}^{h_{d,i}}}{\Gamma(h_{d,i})} \cdot d_{it}^{-h_{d,i}-1} e^{-r_{d,i} \cdot \frac{1}{d_{it}}} \right] \cdot d_{it}^{-\frac{1}{2}} \cdot e^{-\frac{1}{2} \frac{g_{it}^2}{d_{it}}}, \quad (\text{C-127})$$

$$\propto d_{it}^{-(h_{d,i}+\frac{1}{2})-1} \cdot e^{-(r_{d,i}+\frac{g_{it}^2}{2})\frac{1}{d_{it}}}, \quad (\text{C-128})$$

$$= d_{it}^{-h_{d,i}^*-1} \cdot e^{-r_{d,it}^* \frac{1}{d_{it}}}, \quad (\text{C-129})$$

$$d_{it}|\mathbf{y}_t, \mathbf{x}_t, \Pi, B, \mathbf{v} \sim i\Gamma(h_{d,i}^*, r_{d,it}^*), \quad (\text{C-130})$$

$$h_{d,i}^* = h_{d,i} + \frac{1}{2}, \quad (\text{C-131})$$

$$r_{d,it}^* = r_{d,i} + \frac{g_{it}^2}{2}. \quad (\text{C-132})$$

C.3 Conditional posterior $p(\mathbf{v}|Y, \phi, B, D)$

The conditional posterior for $\mathbf{v}$ is

$$p(\mathbf{v}|Y, \Pi, B, D) \propto p(D|\mathbf{v}) \cdot p(\mathbf{v}), \quad (\text{C-133})$$

$$\propto \left[\prod_{t=1}^T \prod_{i=1}^k \frac{r_{d,i}^{h_{d,i}}}{\Gamma(h_{d,i})} \cdot d_{it}^{-h_{d,i}-1} e^{-r_{d,i} \cdot \frac{1}{d_{it}}} \right] \cdot p(\mathbf{v}), \quad (\text{C-134})$$

$$= \left[\prod_{i=1}^k \left(\frac{r_{d,i}^{h_{d,i}}}{\Gamma(h_{d,i})} \right)^T \cdot \prod_{t=1}^T \left(d_{it}^{-h_{d,i}-1} e^{-r_{d,i} \cdot \frac{1}{d_{it}}} \right) \right] \cdot p(\mathbf{v}), \quad (\text{C-135})$$

$$= \left[\prod_{i=1}^k \left(\frac{r_{d,i}^{h_{d,i}}}{\Gamma(h_{d,i})} \right)^T \cdot \left(\prod_{t=1}^T d_{it} \right)^{-h_{d,i}-1} e^{-r_{d,i} \cdot \left(\sum_{t=1}^T \frac{1}{d_{it}} \right)} \right] \cdot p(\mathbf{v}). \quad (\text{C-136})$$

We assume prior independence across $\mathbf{v}$, i.e. $p(\mathbf{v}) = \prod_{i=1}^k p(v_i)$. Since $(h_{d,i}, r_{d,i})$ only depend on entry i of $\mathbf{v}$, prior independence implies independence in conditional posterior:

$$p(\mathbf{v}|Y, \Pi, B, D) = \prod_{i=1}^k p(v_i|Y, \Pi, B, D), \quad (\text{C-137})$$

$$p(v_i|Y, \Pi, B, D) \propto \left(\frac{r_{d,i}^{h_{d,i}}}{\Gamma(h_{d,i})} \right)^T \cdot \left(\prod_{t=1}^T d_{it} \right)^{-h_{d,i}-1} e^{-r_{d,i} \cdot \left(\sum_{t=1}^T \frac{1}{d_{it}} \right)} \cdot p(v_i). \quad (\text{C-138})$$

Last, we use a Griddy-Gibbs sampler and discretize the support for v_i .

We set the upper bound of the grid support for each v_i using a data-driven criterion based on the statistical detectability of excess kurtosis. For an i.i.d. sample of size T , the sampling variance of the sample excess kurtosis under the Gaussian null is asymptotically $24/T$. Since a Student- $t(v)$ distribution has excess kurtosis $6/(v-4)$ for $v > 4$, we define v^* as the value of v at which the implied excess kurtosis equals the 5 percent critical value $z_{0.975} \sqrt{24/T}$ of a test of Gaussianity based on this statistic, i.e.

$$v^* = 4 + \frac{6}{z_{0.975} \sqrt{24/T}}, \quad (\text{C-139})$$

where $z_{0.975} = 1.96$. At $v = v^*$ such a test rejects with probability of roughly one half, and with lower probability for larger v ; v^* is therefore a signal-to-noise benchmark rather than a sharp detectability limit. In this region the likelihood becomes increasingly flat in v , which

is the source of the weak identification and slow mixing of the degrees-of-freedom parameter in Student- t models (e.g. [Fonseca et al. 2008](#) and [Hosszejni 2021](#)). Rather than using v^* itself, we set the upper bound of the support to

$$\bar{v} = 3 v^*, \tag{C-140}$$

which leaves enough room above v^* for the posterior to represent genuine uncertainty about near-Gaussian shocks, while avoiding the far larger, effectively uninformative range required to approximate the Gaussian limit (e.g. $v = 100$). The rule gives $\bar{v} = 38.5$ for $T = 200$, as in the bivariate simulation, and $\bar{v} = 65$ for $T = 800$, as in the larger simulations. In the application we use the same bound of 38.5 as in the bivariate simulation. The support $[3, \bar{v}]$ is discretized on an evenly spaced grid, and the conditional posterior (C-138) is evaluated at each grid point, weighted by the corresponding trapezoidal quadrature weight, normalized, and sampled from.

D Example of statistical identification and mode-switching problem

This section offers an intuitive illustration of how departures from Gaussianity provide higher moments, which support identification. Many possible specifications could be used, and we work with independent t distributed shocks. The results in this section are already known in the literature, see for instance [Hoesch et al. \(2024\)](#) and [Lewis \(2025\)](#).

Consider the following bivariate data generating process:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}, \tag{D-141}$$

where $\boldsymbol{\epsilon}_t = (\epsilon_{1t}, \epsilon_{2t})$ are independently t -distributed structural shocks with degrees of freedom $\boldsymbol{v} = (v_1, v_2)'$. Normalize the variance of these shocks to 1 (which implies that we assume

the variance to be finite and thus $v_1, v_2 > 2$). Model (D-141) can be rewritten as

$$\mathbf{y}_t = B_c Q \boldsymbol{\epsilon}_t, \quad (\text{D-142})$$

where $\Sigma = BB'$ is the covariance matrix of $\mathbf{y}_t$, B_c the Cholesky decomposition of Σ , and $Q = B_c^{-1}B$ an orthogonal matrix (so that $QQ' = I$). Finally, define the individual elements of B as $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$. Our goal is to identify $(B, \mathbf{v})$, or equivalently, $(B_c, Q, \mathbf{v})$.

Equation $\Sigma = BB'$ implies the following second moment restrictions on the elements of B :

$$E(y_{1t}^2) = b_{11}^2 + b_{12}^2, \quad (\text{D-143a})$$

$$E(y_{1t}y_{2t}) = b_{11}b_{21} + b_{12}b_{22}, \quad (\text{D-143b})$$

$$E(y_{2t}^2) = b_{21}^2 + b_{22}^2. \quad (\text{D-143c})$$

Because these three equations involve four unknowns, B_c is identified, but B is not (Kilian & Lütkepohl 2017). While these restrictions hold under both Gaussianity and more generally with a t -distribution, these are the *only* model restrictions that can be used for the identification of B under Gaussianity. In fact, the Gaussian special case of the model (which holds for $v_i = \infty$, $i = 1, 2$) implies that higher moments are constant in B , given that the full distribution of a Gaussian process is pinned down by the first two moments. Thus, in the Gaussian setting, only second moment restrictions are available from the model, and we would need to impose further restrictions to identify B (e.g., $b_{12} = 0$ to obtain $B = B_c$).

Non-Gaussianity helps with the identification of B . To do so, in this example we assume that higher moments exist, and show that information in these moments can be exploited. It is important to note, however, that identification via non-Gaussianity does *not* rely on the existence of higher moments, see e.g. Kagan et al. (1973), ch. 10, Eriksson & Koivunen (2004). We assume their existence only to streamline the exposition in this example. Since we assume symmetric distributions (Gaussian or t), third moments vanish, so any arguments

based on moments have to use at least fourth moments. Other specifications are also possible.

In this bivariate example, the relevant expressions for the fourth moments (derived at the end of this section) are:

$$E(y_{1t}^4) = b_{11}^4 \kappa_1 + 6b_{11}^2 b_{12}^2 + b_{12}^4 \kappa_2, \quad (\text{D-144a})$$

$$E(y_{1t}^3 y_{2t}) = b_{11}^3 b_{21} \kappa_1 + 3b_{11} b_{12}^2 b_{21} + 3b_{11}^2 b_{12} b_{22} + b_{12}^3 b_{22} \kappa_2, \quad (\text{D-144b})$$

$$E(y_{1t}^2 y_{2t}^2) = b_{11}^2 b_{21}^2 \kappa_1 + b_{12}^2 b_{21}^2 + 4b_{11} b_{12} b_{21} b_{22} + b_{11}^2 b_{22}^2 + b_{12}^2 b_{22}^2 \kappa_2, \quad (\text{D-144c})$$

$$E(y_{1t} y_{2t}^3) = b_{11} b_{21}^3 \kappa_1 + 3b_{11} b_{21} b_{22}^2 + 3b_{12} b_{21}^2 b_{22} + b_{12} b_{22}^3 \kappa_2, \quad (\text{D-144d})$$

$$E(y_{2t}^4) = b_{21}^4 \kappa_1 + 6b_{21}^2 b_{22}^2 + b_{22}^4 \kappa_2, \quad (\text{D-144e})$$

with $\kappa_i = \frac{3(v_i-2)}{v_i-4}$ (note that κ_i coincides with both the fourth moment and kurtosis of ϵ_{it} , namely $E(\epsilon_{it}^4)$, thanks to $E(\epsilon_{it}) = 0$ and $E(\epsilon_{it}^2) = 1$). When the data generating process is Gaussian, $v_1 = v_2 = \infty$ implies that $\kappa_1 = \kappa_2 = 3$, and all equations (D-144) become constant in the orthogonal matrix Q (i.e., in alternative choices of B such that $BB' = \Sigma$). Hence, the fourth moments provide no additional information to identify B . This is because under Gaussianity, the right-hand sides of (D-144) can be written exclusively as functions of the right-hand sides of the second-moment conditions in (D-143). Since those second moments are invariant to orthogonal rotations, so must be the fourth moments under Gaussianity. Instead, under a t distribution, where $\kappa_i \neq 3$, the right-hand sides of (D-144) vary with Q , given B_c . As a result, the fourth moments provide additional information beyond what is contained in the second moments.

To illustrate the above point, we provide a numerical example by setting

$$B_{\text{true}} = \begin{pmatrix} 1 & -1.25 \\ 2 & 0.5 \end{pmatrix}. \quad (\text{D-145})$$

Although these parameter values are arbitrary, one motivation might be the following: If $\mathbf{y}_t$

includes output and prices, and ϵ_t includes a demand shock and a supply shock (in that order), then a positive demand shock increases output and prices, while a positive supply shock decreases output and increases prices. To highlight identification issues that persist even as sample size grows, we work in population, assuming knowledge of the second and fourth moments of the data $\mathbf{y}_t$ (with $v_{\text{true}} = 6$, moments of order six and higher do not exist).

Equations (D-143) and (D-144) form a system of eight polynomial equations (up to fourth order in the entries of B) in six parameters, $(b_{11}, b_{12}, b_{21}, b_{22}, \kappa_1, \kappa_2)$. Reparametrize the model and set

$$Q = Q(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}, \quad (\text{D-146})$$

a Givens rotation that produces an orthogonal matrix with $\det(Q(\theta)) = 1$. This assumption is standard in the literature (Canova & De Nicoló 2002) and loses no generality aside from fixing the determinant (a restriction we come back to below). Next, we impose $\mathbf{v} = (v, v)'$, reducing the unknown parameter space from $(b_{11}, b_{12}, b_{21}, b_{22}, \kappa_1, \kappa_2)$ to (θ, v) , and we treat B_c as known (since B_c is identified by second moments alone).³ The value of θ consistent with B_{true} is $\theta_{\text{true}} = 0.29\pi$, and we set $v_{\text{true}} = 6$. We then use (D-143)-(D-144) to compute the fourth moments implied by $(\theta_{\text{true}}, v_{\text{true}})$.⁴ Checking whether these fourth moments identify the model parameters amounts to asking if the system (D-144) has a solution other than $(\theta, v) = (\theta_{\text{true}}, v_{\text{true}}) = (0.29\pi, 6)$. By construction, any θ is consistent with the second moments alone.

³We set $v_1 = v_2$ only to simplify this illustrative example. With $v_1 \neq v_2$ held fixed under their labels, the column-swap modes disappear and only the sign modes remain; if (v_1, v_2) are unknown and relabeled jointly with the columns of B , the swap modes reappear as joint relabeling modes. The identification argument itself does not rely on $v_1 = v_2$.

⁴These are $E(y_{1t}^4) = 30.01$, $E(y_{1t}^3 y_{2t}) = 13.64$, $E(y_{1t}^2 y_{2t}^2) = 27.84$, $E(y_{1t} y_{2t}^3) = 41.06$ and $E(y_{2t}^4) = 102.38$.

Each of the three panels in [Figure D-1](#) shows the fourth moments from equation (D-144) evaluated over a grid of θ , conditioning on a value of v shown in each panel. The moments are shown in percentage deviation from the true moments associated with $(\theta_{true}, v_{true})$. Panel A reports the case when evaluating moments (D-144) for $v = \infty$, hence $\kappa_i = 3$, $i = 1, 2$ (Gaussianity). Two results emerge from Panel A. First, that the fourth moments are constant in θ , showing that fourth moments do not provide any information when the estimated model is restricted to be Gaussian. Second, that a Gaussian model fails to match the fourth moments in the data. As an example, the true moments $E(y_{1t}^4) = 30.01$ and $E(y_{1t}^2) = 2.56$ imply the population kurtosis of y_{1t} equal to $\frac{E(y_{1t}^4)}{(E(y_{1t}^2))^2} = 4.57$. Evaluating (D-144) using $\kappa_i = 3$, $i = 1, 2$ gives $E(y_{1t}^4) = 19.70$ and kurtosis $\frac{E(y_{1t}^4)}{(E(y_{1t}^2))^2} = 3$ for *any* value of θ , with $E(y_{1t}^4)$ being 34% below the true value (red thick line), showing that the Gaussian model is misspecified. Since the likelihood function of model under Gaussianity is uniquely pinned down by the first and second moments, higher moments provide no additional information, except for uncovering that the true fourth moments cannot be reconciled with a Gaussian model.

Things change dramatically when evaluating (D-144) using $\kappa_i = 6$ for $i = 1, 2$, which coincides with the true value. The middle panel of [Figure D-1](#) shows that the fourth moments in (D-144) are no longer constant in θ , hence fourth moments provide additional information for identification. In addition, it shows that there are now exactly four values of θ that match the five moments from (D-144): $(0.29\pi, 0.78\pi, 1.26\pi, 1.78\pi)'$. We return to this residual identification issue below when we discuss normalization. In the middle panel, we specifically evaluate the moments at the true degrees of freedom. This step matters only if the data's fourth moments let us infer the correct degrees of freedom in the first place. Panel C of [Figure D-1](#) demonstrates that using $v \neq v_{true}$ (specifically $v = 9$ here) prevents an exact match of all moments. Hence, the model's fourth moments — and thus the likelihood

function – are informative about both the degrees of freedom in the t -distribution and the elements of B .

Table D-1 clarifies how the four values of θ identified in Panel B of Figure D-1 relate to matrix B . Following Lanne et al. (2017), the likelihood function of the t -distributed model has $2! \cdot 2^2 = 8$ modes, which differ from B_{true} by the sign or the ordering of the shocks. Matrix a in Table D-1 shows B_{true} . The remaining matrices on the left differ from B_{true} by flipping the sign of one or both columns, while those on the right swap the columns. From Lanne et al. (2017), we know that the model’s likelihood peaks at these eight solutions. Economically, all solutions produce the same impulse responses, once we account for shock ordering and sign. The four θ -values from Panel B of Figure D-1 correspond to matrices a, f, g , and d in Table D-1. To generate all solutions listed in the table, we can generalize the example to

$$\mathbf{y}_t = B_c Q(\theta) P \boldsymbol{\epsilon}_t,$$

where P is a permutation matrix. Under this specification, $\theta = 0.29\pi$ yields matrix a (no column permutation) or matrix b (with column permutation), but the implied Q for matrix b has determinant -1 and thus cannot be generated by a Givens rotation. The same reasoning applies to the remaining θ -solutions.

Table D-1 also shows why it is important to address the inference challenges posed by identification up to the sign and ordering of shocks. Although the eight modes in Table D-1 are economically identical, if the posterior sampler jumps among them, the resulting posterior approximation can be highly misleading about the genuine uncertainty (Hamilton et al. 2007). To avoid this problem, prior work has proposed normalization schemes (Waggoner & Zha 2003b, Jarociński 2024). In the paper, we adopt their insight of selecting parameter draws closest to a particular mode during sampling, and we demonstrate how to implement this strategy efficiently, even in large-scale systems.

Table D-1: Modes of the likelihood of the model in Panel B from [Figure D-1](#)

	B	θ	permute		B	θ	permute
a)	$\begin{pmatrix} 1 & -1.25 \\ 2 & 0.5 \end{pmatrix}$	0.29π	no	b)	$\begin{pmatrix} -1.25 & 1 \\ 0.5 & 2 \end{pmatrix}$	0.29π	yes
c)	$\begin{pmatrix} -1 & -1.25 \\ -2 & 0.5 \end{pmatrix}$	0.78π	yes	d)	$\begin{pmatrix} 1.25 & 1 \\ -0.5 & 2 \end{pmatrix}$	1.78π	no
e)	$\begin{pmatrix} 1 & 1.25 \\ 2 & -0.5 \end{pmatrix}$	1.78π	yes	f)	$\begin{pmatrix} -1.25 & -1 \\ 0.5 & -2 \end{pmatrix}$	0.78π	no
g)	$\begin{pmatrix} -1 & 1.25 \\ -2 & -0.5 \end{pmatrix}$	1.26π	no	h)	$\begin{pmatrix} 1.25 & -1 \\ -0.5 & -2 \end{pmatrix}$	1.26π	yes

Note: $B_{true} = \begin{pmatrix} 1 & -1.25 \\ 2 & 0.5 \end{pmatrix}$ corresponds to matrix *a*). The remaining matrices *c*, *e*, *g* on the left differ from B_{true} up to the sign of the first or second column, or both. Matrices *b*, *d*, *f*, *h* differ from B_{true} regarding the ordering of the columns, and sign of the columns.

We conclude this section by providing the derivations for the population moments ([D-144](#)) used above. The model is given by

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}, \quad (\text{D-147})$$

where

$$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, \quad (\text{D-148})$$

and $\epsilon_t = (\epsilon_{1t}, \epsilon_{2t})$ are independently t -distributed structural shocks with degrees of freedom $\mathbf{v} = (v_1, v_2)'$ and variances normalized to 1. $E(\epsilon_{it}) = 0$ implies $E(y_{it}) = 0$, $i = 1, 2$. Then

$$E(y_{1t}^4) = E(b_{11}\epsilon_{1t} + b_{12}\epsilon_{2t})^4, \quad (\text{D-149})$$

$$= E(b_{11}^4\epsilon_{1t}^4 + 4b_{11}^3\epsilon_{1t}^3b_{12}\epsilon_{2t} + 6b_{11}^2\epsilon_{1t}^2b_{12}^2\epsilon_{2t}^2 + 4b_{11}\epsilon_{1t}b_{12}^3\epsilon_{2t}^3 + b_{12}^4\epsilon_{2t}^4), \quad (\text{D-150})$$

$$= b_{11}^4E(\epsilon_{1t}^4) + 6b_{11}^2b_{12}^2 + b_{12}^4E(\epsilon_{2t}^4), \quad (\text{D-151})$$

with the last step deriving from the independence assumption between ϵ_{1t} and ϵ_{2t} , and $E(\epsilon_{it}) = 0$, $E(\epsilon_{it}^2) = 1$. We define κ_i as the kurtosis of ϵ_{it} . Since ϵ_{it} has mean 0 and variance normalized to 1, it holds that

$$E(\epsilon_{it}^4) = \kappa_i = \frac{3(\nu_i - 2)}{\nu_i - 4}. \quad (\text{D-152})$$

Hence

$$E(y_{1t}^4) = b_{11}^4 \kappa_1 + 6b_{11}^2 b_{12}^2 + b_{12}^4 \kappa_2, \quad (\text{D-153})$$

while similar derivations give

$$E(y_{2t}^4) = b_{21}^4 \kappa_1 + 6b_{21}^2 b_{22}^2 + b_{22}^4 \kappa_2. \quad (\text{D-154})$$

Next:

$$E(y_{1t}^3 y_{2t}) = E(b_{11}\epsilon_{1t} + b_{12}\epsilon_{2t})^3 (b_{21}\epsilon_{1t} + b_{22}\epsilon_{2t}), \quad (\text{D-155})$$

$$= E(b_{11}^3 \epsilon_{1t}^3 + 3b_{11}^2 \epsilon_{1t}^2 b_{12} \epsilon_{2t} + 3b_{11} \epsilon_{1t} b_{12}^2 \epsilon_{2t}^2 + b_{12}^3 \epsilon_{2t}^3) (b_{21}\epsilon_{1t} + b_{22}\epsilon_{2t}), \quad (\text{D-156})$$

$$= E(b_{11}^3 \epsilon_{1t}^3 b_{21}\epsilon_{1t} + 3b_{11}^2 \epsilon_{1t}^2 b_{12}\epsilon_{2t} b_{21}\epsilon_{1t} + 3b_{11} \epsilon_{1t} b_{12}^2 \epsilon_{2t}^2 b_{21}\epsilon_{1t} + b_{12}^3 \epsilon_{2t}^3 b_{21}\epsilon_{1t} + b_{11}^3 \epsilon_{1t}^3 b_{22}\epsilon_{2t} + 3b_{11}^2 \epsilon_{1t}^2 b_{12}\epsilon_{2t} b_{22}\epsilon_{2t} + 3b_{11} \epsilon_{1t} b_{12}^2 \epsilon_{2t}^2 b_{22}\epsilon_{2t} + b_{12}^3 \epsilon_{2t}^3 b_{22}\epsilon_{2t}), \quad (\text{D-157})$$

$$= E(b_{11}^3 \epsilon_{1t}^4 b_{21} + 3b_{11}^2 \epsilon_{1t}^2 b_{12}^2 \epsilon_{2t}^2 b_{21} + 3b_{11}^2 \epsilon_{1t}^2 b_{12} \epsilon_{2t}^2 b_{22} + b_{12}^3 \epsilon_{2t}^4 b_{22}), \quad (\text{D-158})$$

$$= b_{11}^3 b_{21} \kappa_1 + 3b_{11} b_{12}^2 b_{21} + 3b_{11}^2 b_{12} b_{22} + b_{12}^3 b_{22} \kappa_2. \quad (\text{D-159})$$

$$E(y_{1t} y_{2t}^3) = E(b_{11}\epsilon_{1t} + b_{12}\epsilon_{2t}) (b_{21}\epsilon_{1t} + b_{22}\epsilon_{2t})^3, \quad (\text{D-160})$$

$$= E(b_{11}\epsilon_{1t} + b_{12}\epsilon_{2t}) (b_{21}^3 \epsilon_{1t}^3 + 3b_{21}^2 \epsilon_{1t}^2 b_{22} \epsilon_{2t} + 3b_{21} \epsilon_{1t} b_{22}^2 \epsilon_{2t}^2 + b_{22}^3 \epsilon_{2t}^3), \quad (\text{D-161})$$

$$= E(b_{11}\epsilon_{1t} b_{21}^3 \epsilon_{1t}^3 + 3b_{11}\epsilon_{1t} b_{21}^2 \epsilon_{1t}^2 b_{22} \epsilon_{2t} + 3b_{11}\epsilon_{1t} b_{21} \epsilon_{1t} b_{22}^2 \epsilon_{2t}^2 + b_{11}\epsilon_{1t} b_{22}^3 \epsilon_{2t}^3 + b_{12}\epsilon_{2t} b_{21}^3 \epsilon_{1t}^3 + 3b_{12}\epsilon_{2t} b_{21}^2 \epsilon_{1t}^2 b_{22} \epsilon_{2t} + 3b_{12}\epsilon_{2t} b_{21} \epsilon_{1t} b_{22}^2 \epsilon_{2t}^2 + b_{12}\epsilon_{2t} b_{22}^3 \epsilon_{2t}^3), \quad (\text{D-162})$$

$$= E(b_{11} b_{21}^3 \epsilon_{1t}^4 + 3b_{11} \epsilon_{1t}^2 b_{21}^2 b_{22}^2 \epsilon_{2t}^2 + 3b_{12} \epsilon_{2t}^2 b_{21}^2 \epsilon_{1t}^2 b_{22} + b_{12} b_{22}^3 \epsilon_{2t}^4), \quad (\text{D-163})$$

$$= b_{11} b_{21}^3 \kappa_1 + 3b_{11} b_{21} b_{22}^2 + 3b_{12} b_{21}^2 b_{22} + b_{12} b_{22}^3 \kappa_2, \quad (\text{D-164})$$

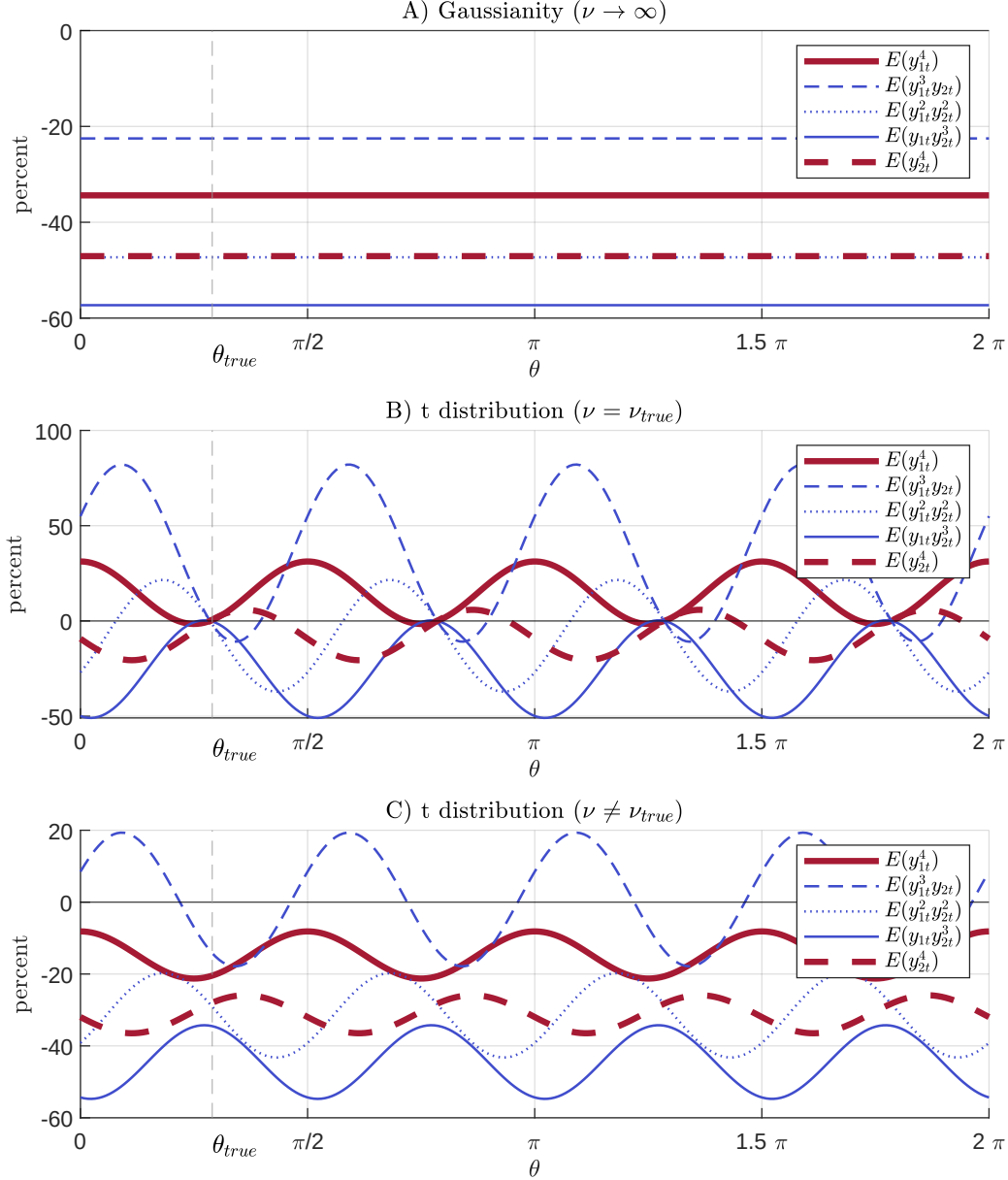
$$E(y_{1t}^2 y_{2t}^2) = E(b_{11}\epsilon_{1t} + b_{12}\epsilon_{2t})^2 (b_{21}\epsilon_{1t} + b_{22}\epsilon_{2t})^2, \quad (\text{D-165})$$

$$= E(b_{11}^2 \epsilon_{1t}^2 + 2b_{11}\epsilon_{1t}b_{12}\epsilon_{2t} + b_{12}^2 \epsilon_{2t}^2)(b_{21}^2 \epsilon_{1t}^2 + 2b_{21}\epsilon_{1t}b_{22}\epsilon_{2t} + b_{22}^2 \epsilon_{2t}^2), \quad (\text{D-166})$$

$$\begin{aligned} &= E(b_{11}^2 \epsilon_{1t}^2 b_{21}^2 \epsilon_{1t}^2 + b_{12}^2 \epsilon_{2t}^2 b_{21}^2 \epsilon_{1t}^2 + \\ &\quad + 4b_{11}\epsilon_{1t}b_{12}\epsilon_{2t}b_{21}\epsilon_{1t}b_{22}\epsilon_{2t} + \\ &\quad + b_{11}^2 \epsilon_{1t}^2 b_{22}^2 \epsilon_{2t}^2 + b_{12}^2 \epsilon_{2t}^2 b_{22}^2 \epsilon_{2t}^2), \end{aligned} \quad (\text{D-167})$$

$$= b_{11}^2 b_{21}^2 \kappa_1 + b_{12}^2 b_{21}^2 + 4b_{11}b_{12}b_{21}b_{22} + b_{11}^2 b_{22}^2 + b_{12}^2 b_{22}^2 \kappa_2. \quad (\text{D-168})$$

Figure D-1: Illustrative example



Note: Fourth moments are computed using equations (D-144) evaluated for $B = B_{chol} \cdot Q$, with B_{chol} set equal to the Cholesky decomposition of $B_{true} \cdot B'_{true}$ and Q set as in equation (D-146) over a grid on θ . The figure shows the moments in percentage deviation from the true values associated with (B_{true}, ν_{true}) . The top panel evaluates equations (D-144) for $\kappa_i = 3$, $i = 1, 2$, which is associated with a Gaussian model ($\nu_1 = \nu_2 = \infty$). The middle and lower panel set κ_i implied by $\nu = \nu_{true} = 6$ or $\nu = 9$.

E Identification up to sign and ordering

The non-Gaussian, statistically independent nature of the structural shocks combined with the normalization of the variance to unity implies that the model is identified up to sign and permutation of the shocks (Lanne et al. 2017). In order to avoid the associated multimodality of the posterior of A and B , we have to be sure that we uncover the posterior uncertainty surrounding only one (possibly arbitrarily chosen) mode. We achieve this via a normalization rule that builds on Waggoner & Zha (2003b) (WZ hereafter). WZ work with the Gaussian SVAR model with variance of the structural shocks normalized to 1, and develop the Likelihood Preserving (LP) normalization that addresses the indeterminacy of the model up to the sign of the shocks. We extend their method to address indeterminacy up to sign and ordering (or permutation) of the shocks in a SVAR with independent t -distributed shocks. We refer to this as the generalized LP normalization. In this section we first define the generalized LP normalization and then relate it to the original specification by WZ. Last, we show that existing combinatorial optimization techniques allow for a very fast computation of the matrix needed to operationalize the normalization rule. This makes the normalization rule feasible also for large models.

Let P denote the permutation matrix and P_s the signed permutation matrix. Our criterion to choose P_s is

$$\min_{P_s} \text{tr}\{(BP_s - \hat{B})' \hat{A} \hat{A} (BP_s - \hat{B})\}, \quad (\text{E-169})$$

where $\hat{\theta}$ denotes the Maximum Likelihood (ML) estimator of θ .⁵ By multiplying matrices in (E-169) and using the fact that $\hat{B} = \hat{A}^{-1}$, one can show that (E-169) is equivalent to any

⁵Note that using the notation in Proposition 4 in WZ, the minimizing function can be written as $\|BP_s - \hat{B}\|_{\hat{\Omega}^{-1}}$, where $\hat{\Omega} = \hat{A}^{-1} \hat{A}'^{-1} = \hat{B} \hat{B}'$ is the ML estimate of the covariance of the reduced form disturbances. The weighting function $\hat{\Omega}^{-1}$ is important for the distance to be invariant under changing the measurement units in the data. In particular, the solution to (E-169) remains the same if instead of the original data y_t we use $H y_t$, where H is any nonsingular matrix - see Proposition 5 in WZ.

of the following maximization problems:

$$\max_{P_s} \text{tr}\{P_s \hat{A} B\} = \max_{P_s} \text{tr}\{\hat{A} B P_s\} = \max_{P_s} \text{tr}\{\hat{A} A^{-1} P_s\} = \max_{P_s} \text{tr}\{P_s' A'^{-1} \hat{A}'\}. \quad (\text{E-170})$$

In order to appreciate the similarity of our criterion to the original LP normalization, we show that if P_s were the diagonal matrix with ± 1 on the diagonal, then the solution to (E-169) would be exactly the LP normalization. Since WZ use the SVAR with its transposed form, their $(A, \hat{A})$ are our $(A', \hat{A}')$. Provided that P_s is a diagonal matrix with ± 1 on the diagonal, the solution to the last formula in (E-170) amounts to multiplying each diagonal element of $A^{-1} \hat{A}$ by -1 if it is negative and by 1 , if it is positive. Following WZ's notation, let e_i denote the i -th column of I_k . Then the i -th diagonal element of $A^{-1} \hat{A}$ may be written as $e_i' A^{-1} \hat{A} e_i = e_i' A^{-1} \hat{a}_i$, where $\hat{a}_i$ denotes the i -th column of $\hat{A}$. An A draw is LP normalized if $e_i' A^{-1} \hat{a}_i > 0$ for each $i = 1, \dots, k$. Multiplication of the i -th diagonal element of $A^{-1} \hat{A}$ by -1 is equivalent to multiplication of the i -th column of A by -1 since $-1 \cdot e_i' A^{-1} \hat{a}_i = e_i' I_k^* A^{-1} \hat{a}_i = e_i' (A I_k^*)^{-1} \hat{a}_i$, where I_k^* is the $k \times k$ identity matrix except its i -th diagonal element is set to -1 . This is exactly the LP normalization for Gaussian SVARs, rewritten in WZ's notation.

In our general setup, we need to compute the signed permutation matrix P_s that solves one of the equivalent problems in (E-170). To this end, let us denote $G = \hat{A} A^{-1}$ and $g_{i,j}$ the (i, j) -th element of G . We focus on $\max_{P_s} \text{tr}\{P_s G\}$. The problem is to choose the permutation of rows of G (possibly multiplied by -1) such that $\text{tr}\{P_s G\}$ attains its maximum. At the maximum $\text{tr}\{P_s G\} = p_1 g_{\pi(1),1} + p_2 g_{\pi(2),2} + \dots + p_k g_{\pi(k),k}$, where $\pi(i)$ denotes permutation of the row index and each $p_i = \pm 1$. We first note that at the maximum each term $p_i g_{\pi(i),i}$ must be nonnegative. To see it, assume, by contradiction, that at least one $p_i g_{\pi(i),i} < 0$. Then by setting $p_i^* = -1 \cdot p_i$ we have $p_1 g_{\pi(1),1} + \dots + p_i^* g_{\pi(i),i} + \dots + p_k g_{\pi(k),k} <$

$p_1 g_{\pi(1),1} + \dots + p_i^* g_{\pi(i),i} + \dots + p_k g_{\pi(k),k}$, i.e. contradiction. Hence at the maximum:

$$\begin{aligned}
p_1 g_{\pi(1),1} + p_2 g_{\pi(2),2} + \dots + p_k g_{\pi(k),k} &= |p_1 g_{\pi(1),1}| + |p_2 g_{\pi(2),2}| + \dots + |p_k g_{\pi(k),k}|, \\
&= |p_1| |g_{\pi(1),1}| + |p_2| |g_{\pi(2),2}| + \dots + |p_k| |g_{\pi(k),k}|, \\
&= |g_{\pi(1),1}| + |g_{\pi(2),2}| + \dots + |g_{\pi(k),k}|. \tag{E-171}
\end{aligned}$$

This suggests the modified problem:

$$\max_P \text{tr}\{P \cdot |G|\} = \max_P \text{tr}\{P \cdot |\hat{A}A^{-1}|\}, \tag{E-172}$$

where P is the usual permutation matrix and $|G|$ means absolute values taken element-wise for all entries in G matrix. At the maximum $\text{tr}\{P \cdot |G|\} = |g_{\pi(1),1}| + |g_{\pi(2),2}| + \dots + |g_{\pi(k),k}|$, hence though the space of permutation matrices is a subset of the space of signed permutation matrices, the maximum of the original optimization problem (E-170) is attained by the modified (i.e. constrained) optimization problem (E-172). When working with (E-172), finding the signed permutation corresponding to this maximum only requires picking $p_i = 1$ or $p_i = -1$ such that each $p_i g_{\pi(i),i}$ is positive (we omit considering the measure zero event such that $g_{\pi(i),i} = 0$ for some i , which does not appear in practice).

All in all, instead of evaluating the objective function (E-169) (or E-170) for every P_s , which is computationally infeasible even for medium-scale models, we only need to solve

$$\max_P \text{tr}\{P \cdot |G|\} = \min_P \text{tr}\{P \cdot -|G|\}. \tag{E-173}$$

However, the formulation in (E-173) shows that we encounter the linear assignment problem from combinatorial optimization. The classic method to solve it is the so-called Hungarian algorithm and its modern refinements, which are computationally really fast. Trying many versions of this algorithm, it turned out that the built-in MATLAB function ‘matchpairs’ is the fastest. To appreciate its computational efficiency, take a generic number of variables k . Set $\hat{A} = 10 \cdot I_k$, draw the entries of B from independent $N(0, 1)$, compute $G = \hat{A}B$, evaluate

the time it takes to solve (E-173), and repeat. The following summarizes, on average over 100,000 repetitions, how long it took to solve (E-173): 0.000056 seconds for $k = 5$; 0.00007 seconds for $k = 10$; 0.0001 seconds for $k = 20$ and 0.0013 seconds for $k = 100$ (computation done on an Intel Xeon E5-1603 v4 and 2.80 GHz). As the illustration shows, the execution time of this optimization technique does not increase substantially even for large k , hence making the normalization rule practical even for large models.

With our method, once we find the permutation matrix P that solves (E-173), we have to consider the diagonal elements in $PG = P\hat{A}A^{-1} = P\hat{A}B$. If the i -th diagonal element in $P\hat{A}B$ is negative we change the 1 in the i -th row of P to -1 , otherwise we do nothing. Doing so we accomplish the task of finding the *signed* permutation matrix that solves (E-169). This completes the computation of P_s , which is required at every iteration of the posterior sampler.

The following algorithm summarizes the steps required for the generalized LP normalization:

Algorithm 1: generalized LP normalization:

Given a target matrix $\hat{B}$, define $\hat{A} = \hat{B}^{-1}$ and apply the following steps at each iteration of the sampler, after drawing (A, B) :

1. solve $\min_P \{P \cdot -|\hat{A}B|\}$ using a version of the Hungarian algorithm, where P is the usual permutation matrix and $|\hat{A}B|$ stands for the matrix of absolute values of all entries in $\hat{A}B$ (taken element-wise);
2. if the i -th diagonal element in $P\hat{A}B$ is negative, change the 1 in the i -th row of P to -1 , otherwise do nothing. As a result, we get P_s that solves (E-169);
3. Let the Gibbs sampler proceed from B , but store the normalized BP_s , $P'_s A$, $P' D_t P$ for every t , $P' \mathbf{v}$, and any other shock-specific parameters relabeled

accordingly.

We stress that this normalization rule can be applied to any non-Gaussian SVAR, and not just to SVAR models with t -distributed shocks.

Recently, [Jarociński \(2024\)](#) proposed an alternative but similar rule to normalize the draws of a SVAR model with independent t -distributed structural shocks. He used the Gaussian approximation to the likelihood function of A as a criterion. In particular, using our notation, his method amounts to solving

$$\min_{P_s} (\text{vec}(A'P_s) - \text{vec}(\hat{A}'))' \hat{V}^{-1} (\text{vec}(A'P_s) - \text{vec}(\hat{A}')), \quad (\text{E-174})$$

where $\hat{V}$ is the asymptotic variance of $\text{vec}(A')$ i.e. the corresponding block of the inverse of the (minus) Hessian of the likelihood evaluated at the mode. For better comparison, let us write our criterion ([E-169](#)) as

$$\min_{P_s} [(\text{vec}(BP_s) - \text{vec}(\hat{B}))'(I_k \otimes \hat{\Omega})^{-1} (\text{vec}(BP_s) - \text{vec}(\hat{B}))], \quad (\text{E-175})$$

where $\hat{\Omega} = \hat{A}^{-1} \hat{A}'^{-1} = \hat{B} \hat{B}'$. Hence one difference between our method and his is that we normalize B draws, whereas [Jarociński \(2024\)](#) normalizes A draws. He then assumes large sample approximation of the covariance of $\text{vec}(A')$ as the weighting function, whereas we assume block diagonal covariance for $\text{vec}(B)$ with the same block $\hat{\Omega}$ (which however follows directly from the LP normalization approach). However, the main difference lies in how we solve the underlying minimization problem. We use the highly efficient Hungarian algorithm, whereas [Jarociński \(2024\)](#) evaluates all $k!$ permutation matrices, see Algorithm 2 in his Online Appendix. As documented above, the case of $k = 20$ requires about 0.0001 seconds to find the optimal permutation matrix. Using the approach by [Jarociński \(2024\)](#) requires computing $20! = 2.432902 \cdot 10^{18}$ permutation matrices to find the one that solves ([E-174](#)).

F A Comparison with the [Keweloh & Wang \(2026\)](#)

Normalization

Let $\mathcal{P}_s$ denote the set of signed permutation matrices. Fix an “anchor” (target) matrix $\bar{B}$ (e.g., a preliminary maximum likelihood estimate). In this comparison, we will use the same target matrix for both approaches.

A normalization selects, for each draw B , a representative BP_s from the equivalence class $\{BP_s : P_s \in \mathcal{P}_s\}$. With the *same anchor* $\bar{B}$ as in our normalization, [Keweloh & Wang \(2026\)](#) define $C = \bar{B}^{-1}B$ and pick the signed permutation that corresponds to the specific choice of C with the following properties:

$$|C_{jj}| > |C_{j\ell}| \quad \forall \ell > j, \quad C_{jj} > 0 \quad \text{for } j = 1, \dots, k. \quad (\text{F-176})$$

Operationally, (F-176) behaves like a *sequential/lexicographic* rule: row 1 determines which column is labeled as shock 1; row 2 chooses among remaining columns for shock 2; etc.

The dominance rule (F-176) does *not* solve the global minimization (E-169). Therefore, it can select a different signed permutation than the LP objective, even when both procedures use the *same anchor* $\bar{B}$.

F.0.0.1 Example (3 shocks). Consider the following example for $C = \bar{B}^{-1}B$:

$$|C| = \begin{pmatrix} 0.86 & 0.97 & 0.10 \\ 0.10 & 0.88 & 0.65 \\ 0.24 & 0.87 & 0.79 \end{pmatrix}. \quad (\text{F-177})$$

The [Keweloh & Wang \(2026\)](#) normalization assigns shock 1 to column 2 (largest entry in row 1), then shock 2 to column 3 (largest among remaining in row 2), leaving column 1 for

shock 3. This corresponds to the permutation $(2, 3, 1)$ and yields the diagonal “match” sum

$$0.97 + 0.65 + 0.24 = 1.86.$$

Our approach instead uses the permutation $(1, 2, 3)$, with diagonal sum

$$0.86 + 0.88 + 0.79 = 2.53,$$

yielding a strictly smaller objective function (E-169).

Because our P_s *minimizes* Equation (E-169), any alternative normalization that selects a different P_s produces a representative that is *weakly farther* from the anchor in the likelihood-preserving metric (strictly farther if the minimizer is unique), increasing the scope for label instability/mode switching in a posterior sampler.

G Summary of the algorithm

All in all, our Gibbs sampling method for sampling from the joint posterior distribution of a SVAR model with independent, t -distributed structural shocks with a flat prior on B can be summarized as follows:

Algorithm 2: Gibbs sampler for SVAR models with independent t -distributed structural shocks:

0. in a preliminary step to the sampler, estimate a target matrix $\hat{B} = \hat{A}^{-1}$ using e.g. the maximum likelihood estimator.⁶ Then, at each iteration of the sampler:
1. draw ϕ from the Normal conditional posterior from equation (C-120);

⁶To estimate the preliminary target matrix $\hat{B}$ we found it convenient to use the 3-step maximum likelihood estimator suggested by Lanne et al. (2017). This makes the procedure quite fast also for our ten variable application from section 4 of the paper.

2. draw L from the Normal conditional posterior from equation (A-32);
3. draw U from the Normal conditional posterior from equation (A-41) (possibly under the additional identifying restrictions on A);
4. draw x_i 's from the Gamma conditional posteriors from equation (A-52);
5. compute Λ by setting each (i, i) entry either as $\lambda_i = -\sqrt{x_i}$ or $\lambda_i = \sqrt{x_i}$ (with equal probability);
6. compute $\bar{A}$ associated with (Λ, L, U) using equation (A-10) and set $\bar{B} = \bar{A}^{-1}$;
7. draw D from the inverse Gamma conditional posterior from equation (C-130);
8. draw $\mathbf{v}$ from the discretized conditional posterior from equation (C-138);
9. if the burn-in draws are completed, compute P_s using the generalized LP normalization from Algorithm 1, Appendix E. Given $P = |P_s|$, store $\bar{B}P_s$ as the normalized matrix B , $P'_s\bar{A}$ as A , $D_t = P'D_tP$ as D for every t , and $\mathbf{v} = P'\mathbf{v}$ for $\mathbf{v}$. Compute the impulse responses associated with the sign-permuted shocks;
10. repeat from step 1.

In the paper, we found it helpful to initialize the sampler as follows

1. Π : equal to the OLS estimate on the full sample;
2. Λ, L, U : equal to the ΛLU decomposition of $A_0 = C^{-1}$, where C is the Cholesky factor of the OLS estimate of Σ , the reduced-form covariance matrix of the VAR innovations (so that $B_0 = C$ and $U_0 = I$);

3. $\mathbf{v}$: equal to $v_i = 4 + 6/\hat{\kappa}_i$ for each equation i , where $\hat{\kappa}_i$ is the sample excess kurtosis of the OLS residuals implied by Π , solving the Student- t excess-kurtosis formula $6/(v-4)$ for v ; the resulting value is bounded to lie in $[5, 50]$, with $v_i = 50$ used whenever $\hat{\kappa}_i \leq 0$.

We end this section with two final remarks. First, jointly relabeling $(B, A, D, \mathbf{v})$ keeps all shock-specific quantities in the same order without affecting the next conditional draws. Second, the stored values of B are associated with a matrix A that may lie outside the unpivoted chart in equation (A-10). This fact is without loss of generality because the decomposition is used only as an operational device for the Gibbs sampler.

H Additional material on the simulation exercise

H.1 Bivariate model

The parameter values of the data generating process are

$$B = \begin{pmatrix} 0.60 & 0.40 \\ 0.70 & -0.70 \end{pmatrix}, \quad (\text{H-178})$$

$$\mathbf{v} = \begin{pmatrix} 8 \\ 8 \end{pmatrix}, \quad (\text{H-179})$$

$$\Pi_1 = \begin{pmatrix} 0.9793 & -0.0057 \\ -0.0443 & 0.9639 \end{pmatrix}, \quad (\text{H-180})$$

$$\Pi_2 = \begin{pmatrix} -0.0392 & -0.0104 \\ -0.0697 & -0.0608 \end{pmatrix}, \quad (\text{H-181})$$

$$\Pi_3 = \begin{pmatrix} -0.0336 & -0.0081 \\ -0.0283 & -0.0356 \end{pmatrix}, \quad (\text{H-182})$$

$$\Pi_4 = \begin{pmatrix} -0.0264 & -0.0053 \\ 0.0079 & -0.0127 \end{pmatrix}, \quad (\text{H-183})$$

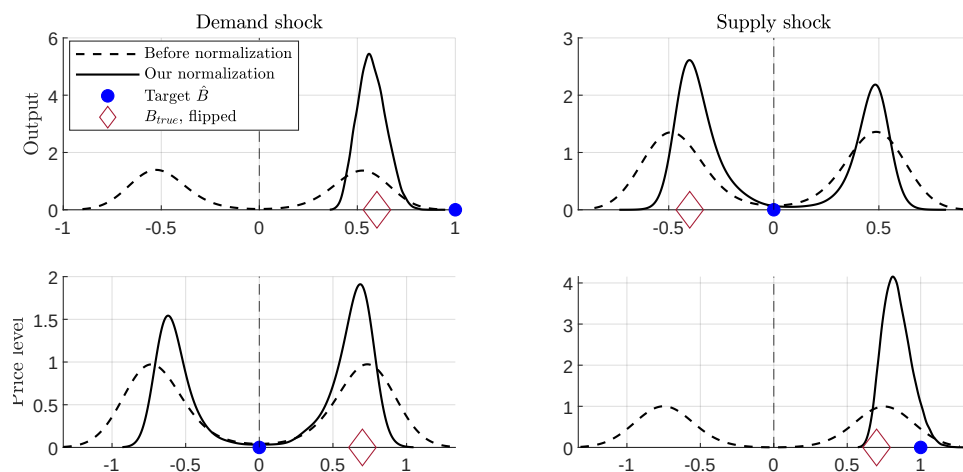
$$\Pi_5 = \begin{pmatrix} -0.0187 & -0.0024 \\ 0.0235 & -0.0010 \end{pmatrix}, \quad (\text{H-184})$$

$$\Pi_6 = \begin{pmatrix} -0.0116 & -0.0000 \\ 0.0209 & 0.0005 \end{pmatrix}. \quad (\text{H-185})$$

The target matrix used for the generalized LP normalization was estimated to

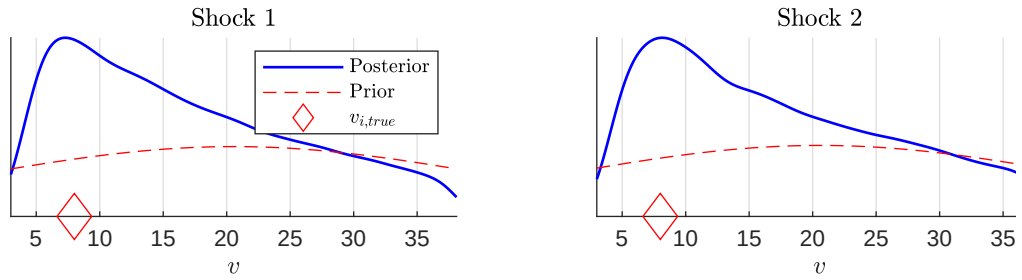
$$\hat{B} = \begin{pmatrix} 0.5272 & -0.4319 \\ 0.7235 & 0.6983 \end{pmatrix}. \quad (\text{H-186})$$

Figure H-2: Impact effect of the shocks (B) when the normalization targets an arbitrary matrix: $\hat{B} = I_2$



Note: The red diamond indicates B_{true} , after flipping the sign of the response to a supply shock. The blue dot indicates the target matrix $\hat{B}$ used for the normalization (here, the identity matrix). The continuous line shows the marginal posterior of the entries of B from 50,000 posterior draws when applying the generalized LP normalization.

Figure H-3: Degrees of freedom, $p(v|Y)$



Note: The red diamond indicates the true value of the degrees of freedom. The dashed red line shows the prior distribution, which is a $N(20.55, 10^2)$ truncated to be positive in the support $[3, 38]$. The solid blue line shows the posterior distribution obtained from 50,000 posterior draws.

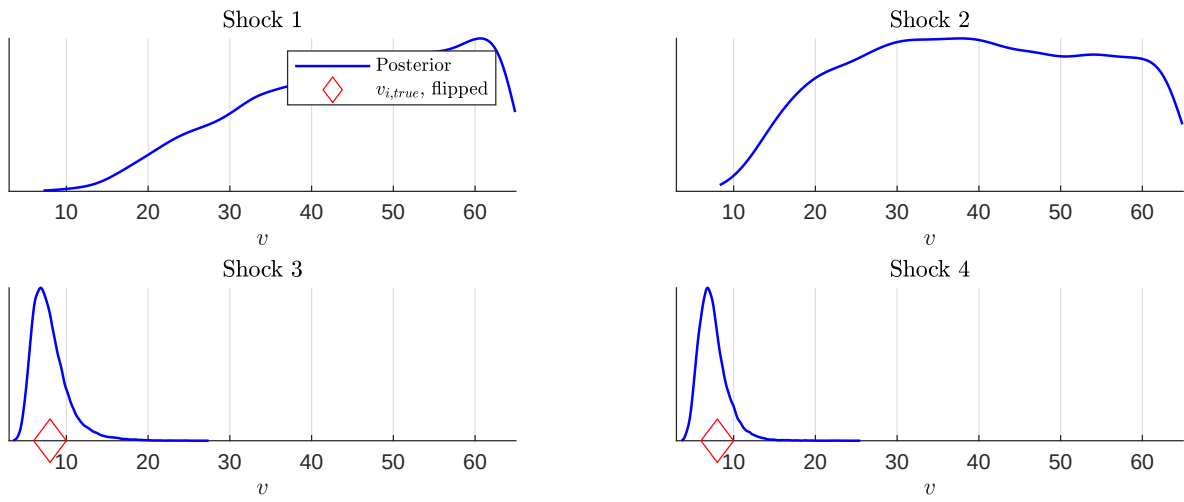
H.2 Partly Gaussian model

We use an extension of the framework in [subsection 3.2](#) of the paper to explore how our Gibbs sampler and normalization perform when a subset of shocks is close to Gaussian, so that their identification is weak in a sample of this size. We set $k = 4$, then set the true degrees of freedom equal to 8 for the first two shocks and equal to 200 for the remaining 2 shocks. We run our sampler exactly as in [subsection 3.2](#), with a flat prior on $\boldsymbol{v}$ in the support indicated by the length of the dataset. With $v = 200$ the last two shocks are non-Gaussian in population and therefore formally identified, but their excess kurtosis of about 0.03 is far below what a sample of 800 observations can detect, and the true value lies above the upper bound of the grid. The exercise therefore asks whether the sampler recovers the strongly non-Gaussian block when the remaining shocks are, for practical purposes, Gaussian.

[Figure H-4](#) reports the posterior distribution for the degrees of freedom from a single pseudo dataset. As in [subsection 3.2](#), the reported true parameters refer to after applying a sign-permutation that makes the true values approach the posterior mean of B . The posteriors correctly single out 2 shocks out of 4 as clearly non-Gaussian, namely the shocks ordered by the sampler as third and fourth. [Figure H-5](#) reports the marginal posterior on B , after applying our normalization. We only report the response to shocks 3 and 4, as the remaining shocks are only weakly identified. Again, the truth refers to the appropriately sign-permuted true values of B . The impact effects of the identified shocks are estimated with precision.

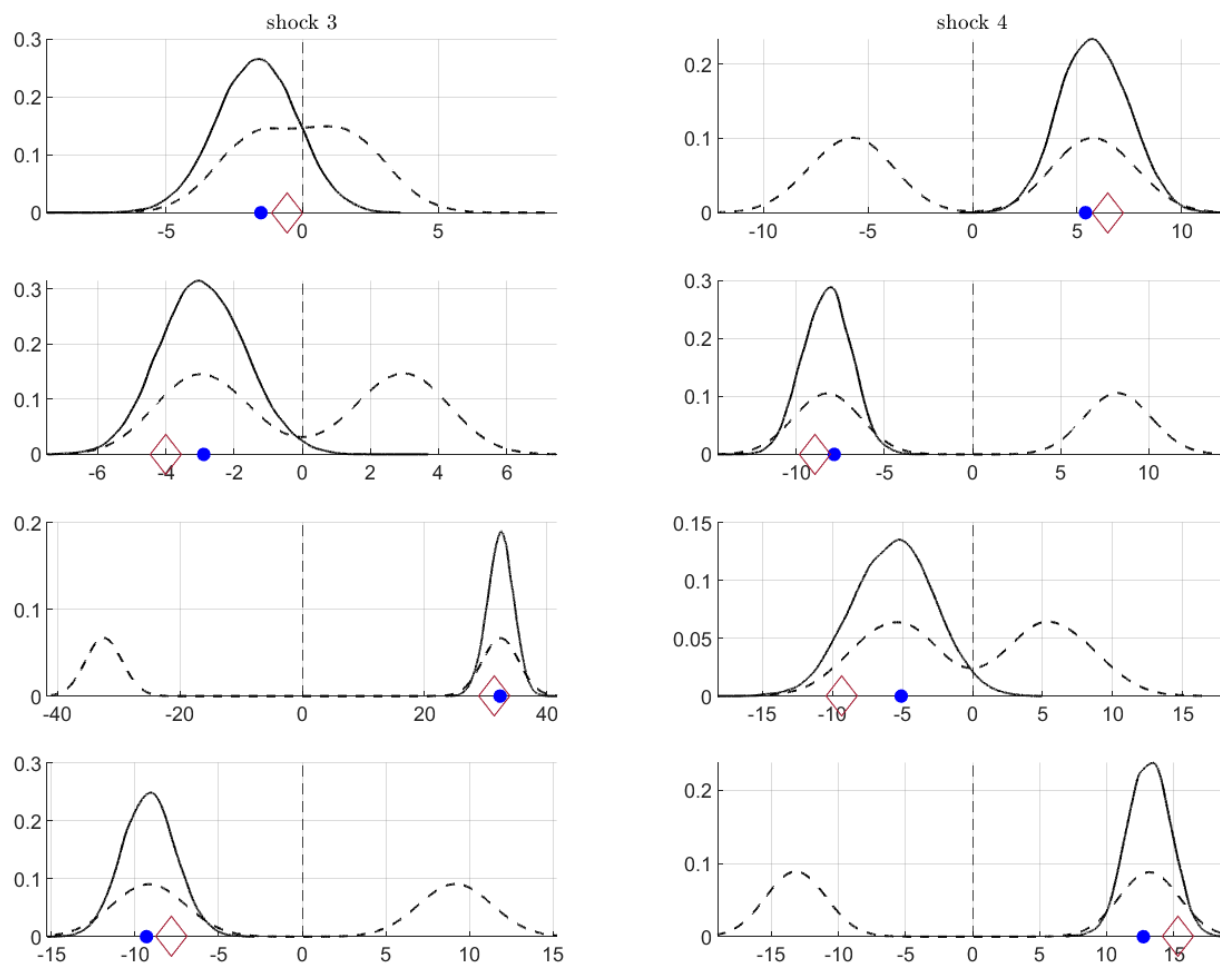
We then extend the analysis across multiple datasets, as done in the simulation exercises in [subsection 3.2](#) of the paper. We simulate 5 datasets. For each, we check that indeed the sampler correctly recovers the low degrees of freedom for two shocks out of 4. We pool the estimates of such shocks across all pseudo datasets. [Figure H-6](#) reports that the sampler is correctly identifying the response of the identified shocks.

Figure H-4: Partly Gaussian model: degrees of freedom, $p(\boldsymbol{v}|Y)$



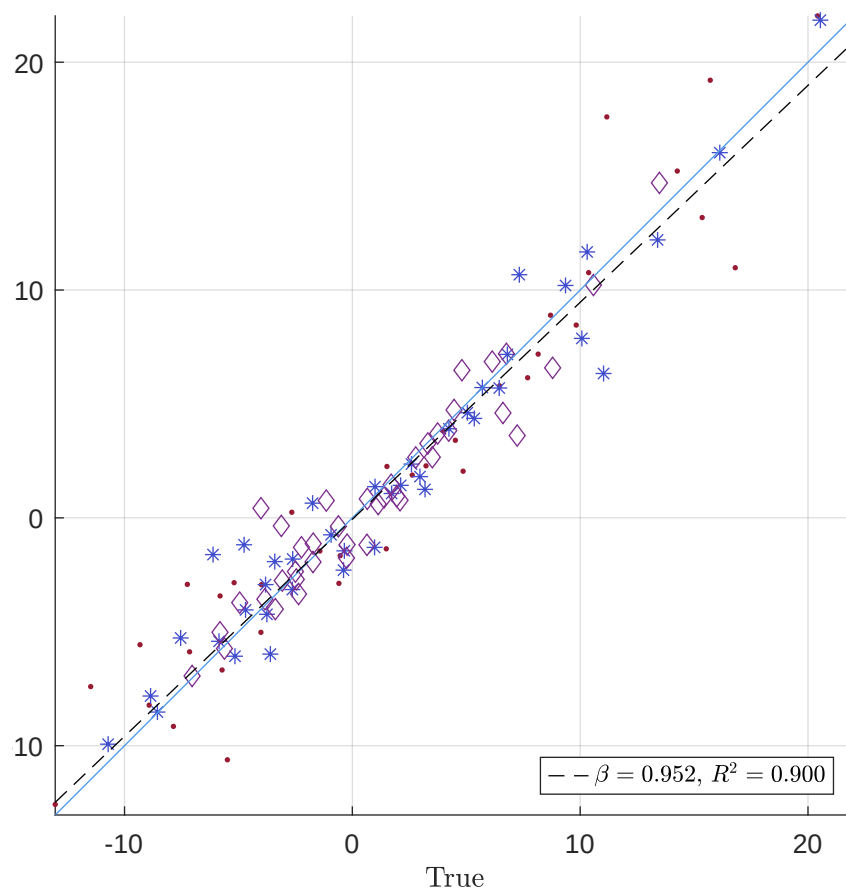
Note: The red diamonds indicate the true degrees of freedom, after applying the sign-permutation that brings the true B closest to the posterior mean of B . The solid blue lines show the posterior distributions obtained from 50,000 posterior draws, after applying the permutation required by the normalization.

Figure H-5: Marginal posterior distributions, $p(B|Y)$



Note: the rows in the figure indicate the variables in the model, the columns indicate the shocks. The blue dots indicate the target matrix $\hat{B}$ used for the normalization. The normalization is successful in removing multimodality from the marginal posterior of B .

Figure H-6: Replication across 5 datasets, identified shocks only



Note: red dots refer to impact effects, stars and diamonds refer to the response after 4 and 8 horizons, respectively.

I Additional material on the application to the US GDP

Table I-2: Test for Gaussianity of reduced-form residuals

Equation	p -value	Reject Gaussianity (5%)
1	0.010	Yes
2	0.001	Yes
3	0.000	Yes
4	0.577	No
5	0.015	Yes
6	0.002	Yes
7	0.000	Yes
8	0.006	Yes
9	0.918	No
10	0.702	No

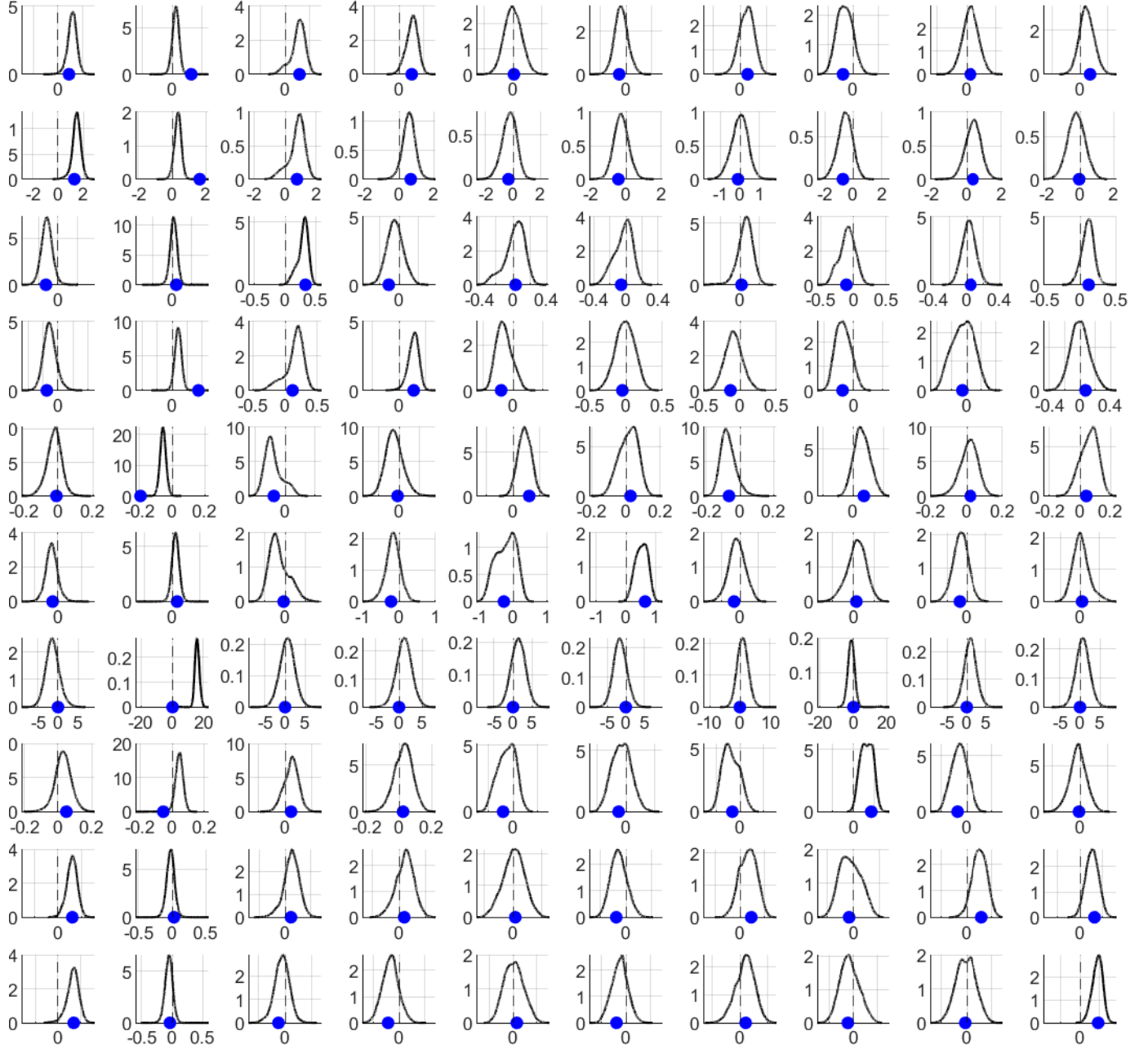
Notes: The table reports, for each equation of the reduced-form VAR, the bootstrap p -value of a Jarque–Bera test for normality of the residuals, following [Kilian & Demiroglu \(2000\)](#). The null distribution of each equation’s Jarque–Bera statistic is obtained by parametric bootstrap: residuals are simulated from a Gaussian distribution matching the estimated reduced-form covariance $\hat{\Sigma}$, Π is re-estimated by OLS on each simulated dataset as on the real data, and the statistic is recomputed on the resulting bootstrap residuals. The p -value for each equation is the share of 1,000 bootstrap replications with a statistic at least as extreme as the one estimated in the data. A small p -value is evidence against Gaussian reduced-form innovations for that equation.

Table I-3: Test for pairwise independence of estimated structural shocks

	1	2	3	4	5	6	7	8	9	10
1	–	0.968	0.478	0.180	0.932	0.084	0.878	0.818	0.984	0.260
2	0.968	–	0.320	0.586	0.824	0.552	0.270	0.390	0.990	0.842
3	0.478	0.320	–	0.656	0.456	0.128	0.238	0.984	0.826	0.110
4	0.180	0.586	0.656	–	0.662	0.522	0.480	0.578	0.952	0.910
5	0.932	0.824	0.456	0.662	–	0.592	0.124	0.398	0.142	0.672
6	0.084	0.552	0.128	0.522	0.592	–	0.456	0.650	0.564	0.766
7	0.878	0.270	0.238	0.480	0.124	0.456	–	0.560	0.924	0.646
8	0.818	0.390	0.984	0.578	0.398	0.650	0.560	–	0.912	0.892
9	0.984	0.990	0.826	0.952	0.142	0.564	0.924	0.912	–	0.598
10	0.260	0.842	0.110	0.910	0.672	0.766	0.646	0.892	0.598	–

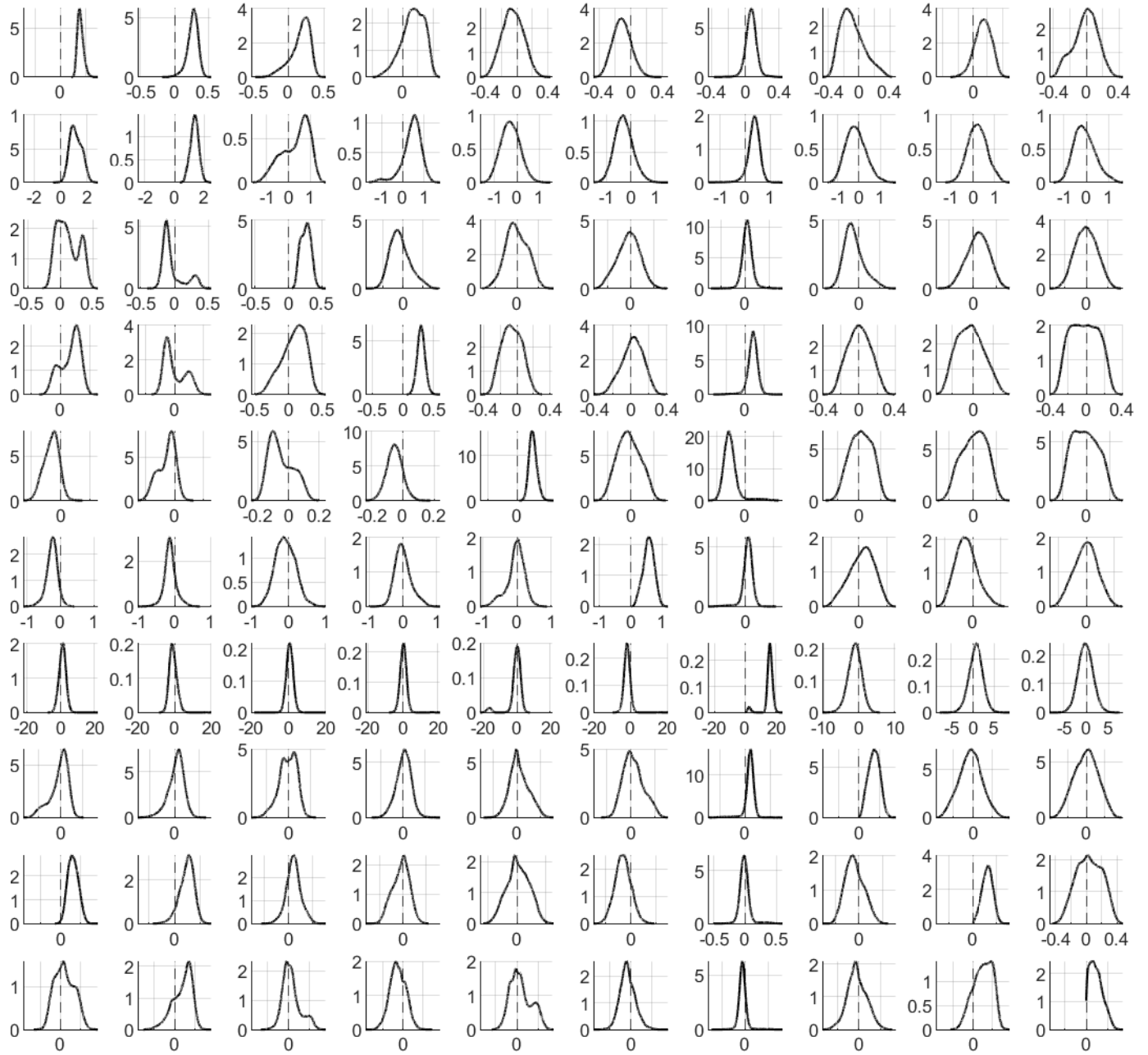
Notes: The table reports pairwise p -values from the discrete-grid independence test of [Amengual et al. \(2022, 2024\)](#), applied to all $\binom{10}{2} = 45$ pairs of estimated structural shocks, evaluated at the posterior mean of A and Π . For each pair, shocks are sorted into $H = 4$ equal-probability quantile bins, a Pearson-type contingency-table statistic is computed, and the p -value is obtained from 500 permutations of one shock relative to the other. A small p -value is evidence against pairwise independence for that pair of shocks. With 500 permutations the smallest attainable p -value is $1/501 \approx 0.002$. No pair rejects the null at the 5 percent level, and the smallest observed p -value, 0.084, exceeds any multiple-testing adjustment one might apply to the 45 tests. The tests are pairwise: nonrejection for all pairs is necessary for the mutual independence assumed by the model, but does not establish it.

Figure I-7: Marginal posterior distributions, $p(B|Y)$
(our normalization)



Note: the rows in the figure indicate the variables in the model, the columns indicate the shocks. The blue dots indicate the target matrix $\hat{B}$ used for the normalization. The normalization is successful in removing multimodality from the marginal posterior of B .

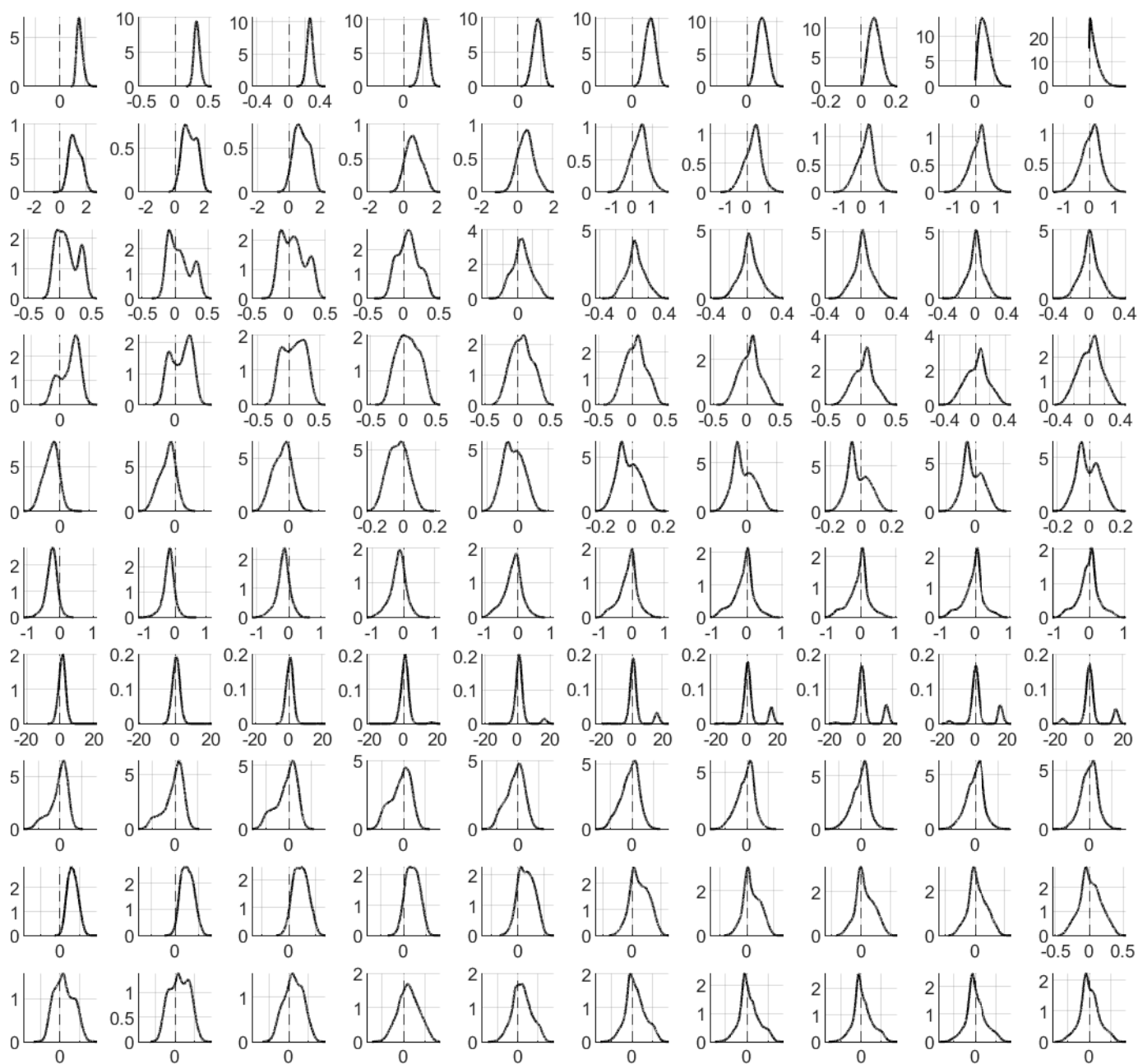
Figure I-8: Marginal posterior distributions, $p(B|Y)$
(normalization by [Lanne et al. 2017](#))



Note: the rows in the figure indicate the variables in the model, the columns indicate the shocks.

Figure I-9: Marginal posterior distributions, $p(B|Y)$

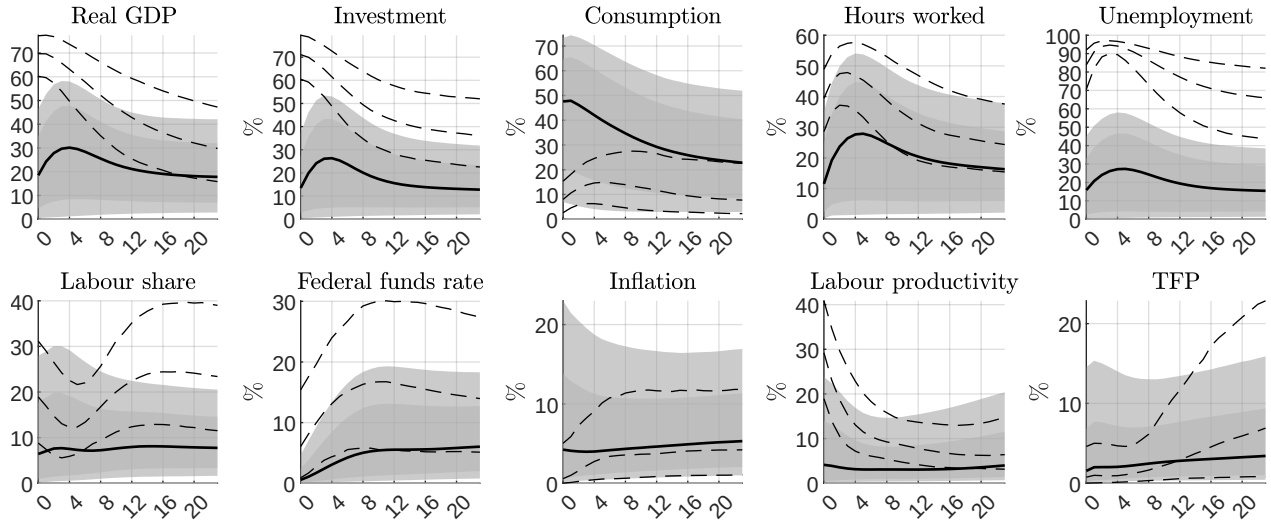
(normalization by [Gouriéroux et al. 2020](#))



Note: the rows in the figure indicate the variables in the model, the columns indicate the shocks.

Figure I-10: Forecast error variance decomposition

Shock 3: demand shock



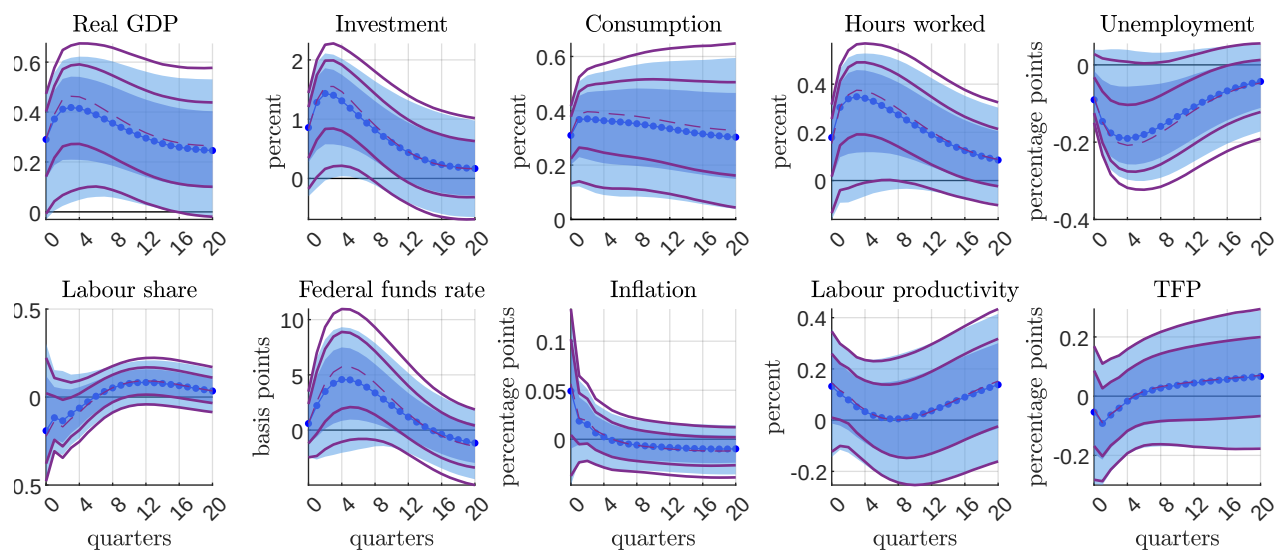
Shock 2: supply shock



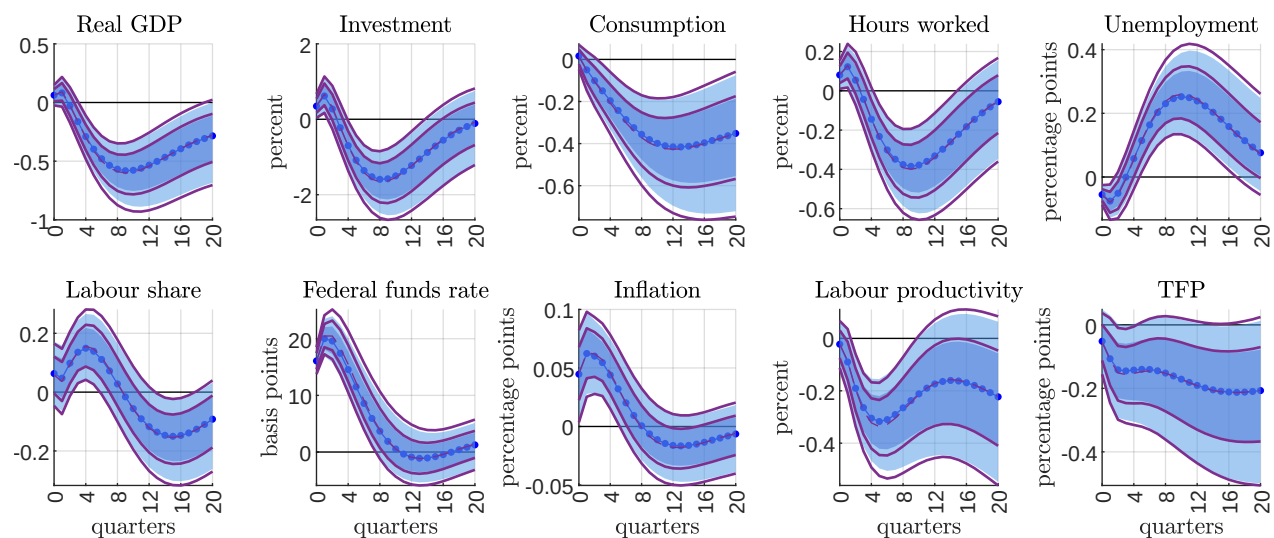
Note: The black solid lines show the pointwise posterior median from our estimation, with corresponding 68% and 90% pointwise posterior credible sets shown with shaded areas. The black dashed lines in the top figure show the pointwise median and 90% credible sets estimated by [Angeletos et al. \(2020\)](#) for the Main Business Cycle shock.

Figure I-11: Robustness for looser prior on ϕ : impulse responses

A) Demand shock



B) supply shock



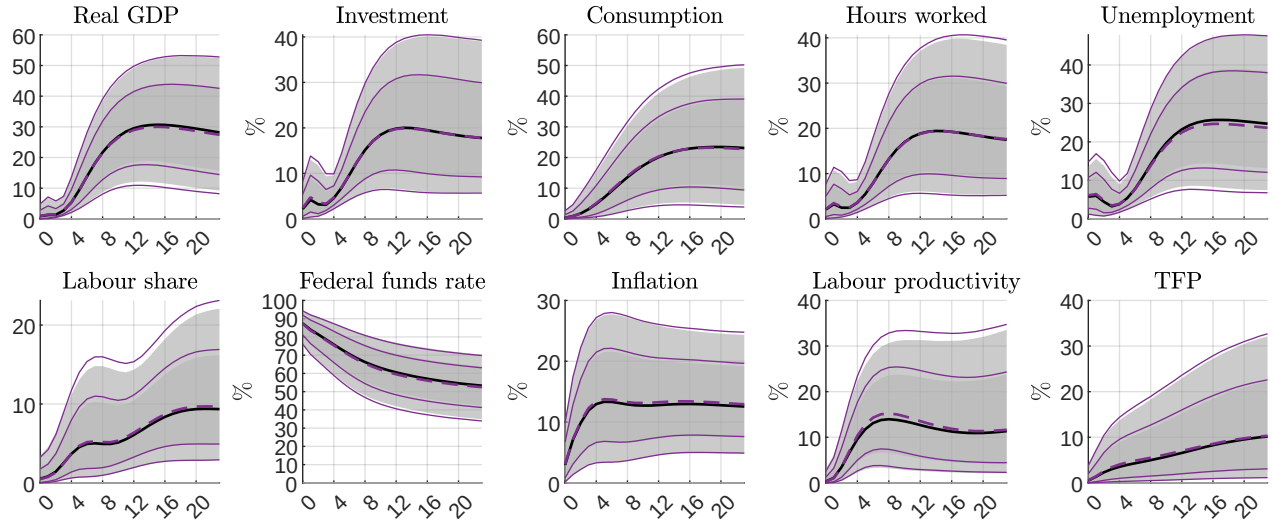
Note: The dotted line and shaded areas show the pointwise median and credible sets from the baseline specification. The solid lines show the pointwise credible sets from the alternative specification.

Figure I-12: Robustness for looser prior on ϕ : forecast error variance decomposition

A) Demand shock



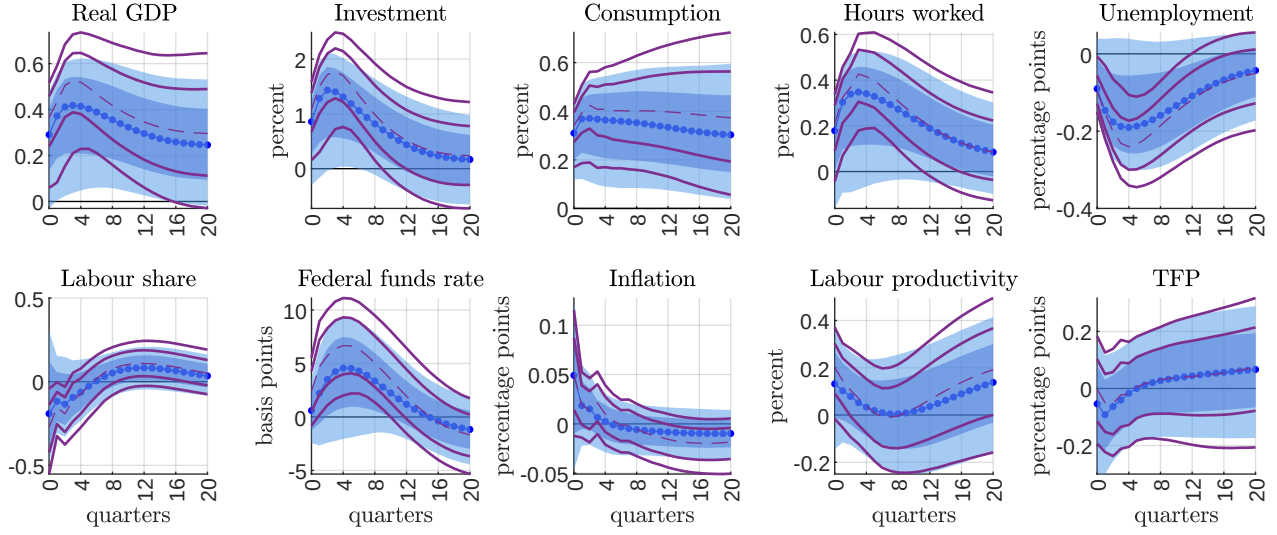
B) supply shock



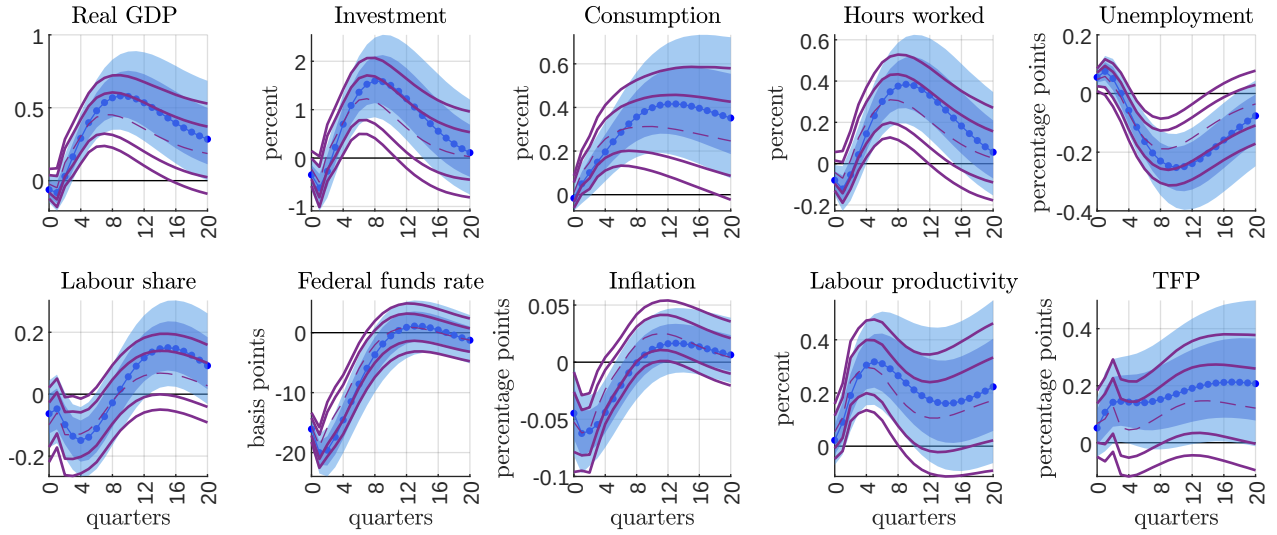
Note: The dotted line and shaded areas show the pointwise median and credible sets from the baseline specification. The dashed and solid lines show the pointwise median and credible sets from the alternative specification.

Figure I-13: Robustness for $p = 4$: impulse responses

A) Demand shock



B) Supply shock



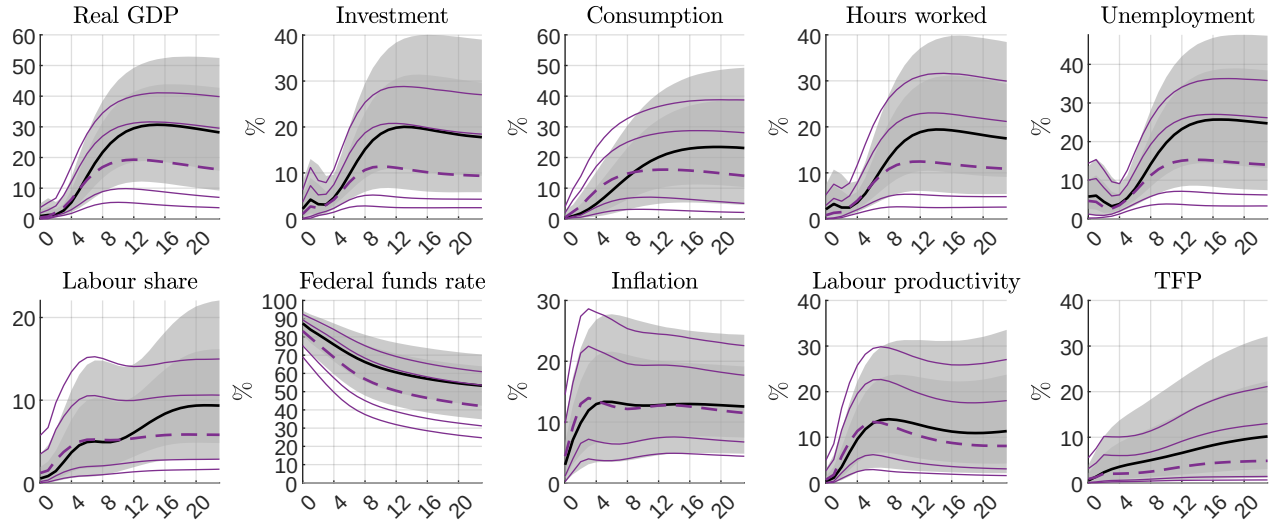
Note: The dotted line and shaded areas show the pointwise median and credible sets from the baseline specification. The solid lines show the pointwise credible sets from the alternative specification.

Figure I-14: Robustness for $p = 4$: forecast error variance decomposition

A) Demand shock



B) Supply shock



Note: The dotted line and shaded areas show the pointwise median and credible sets from the baseline specification. The dashed and solid lines show the pointwise median and credible sets from the alternative specification.

J Comparison to normalization by Lanne et al. (2017) and Gouriéroux et al. (2020)

Figure J-15: Comparing normalizations

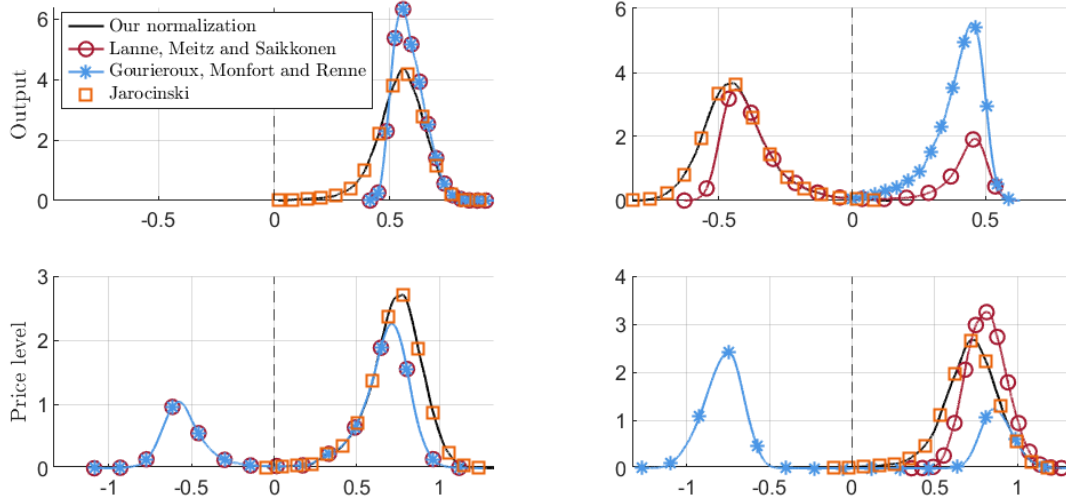
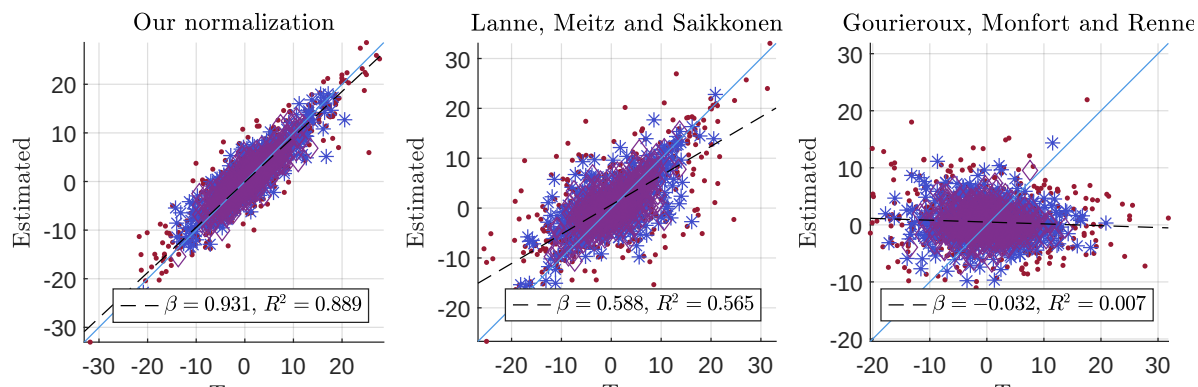


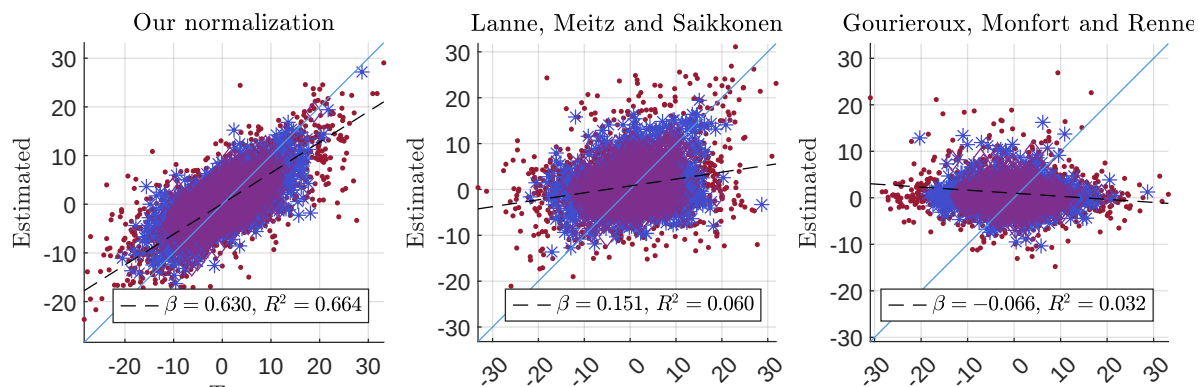
Figure J-15 complements the analysis from the bivariate simulation exercise from Section 3.1 by comparing the posterior on B from our methodology with the posterior obtained using the normalizations suggested by Jarociński (2024), Lanne et al. (2017) and Gouriéroux et al. (2020). The normalization from Jarociński (2024) successfully recovers marginal posterior distributions on B that feature no multimodality, and its only limitation remains scalability. The normalizations by Lanne et al. (2017) and Gouriéroux et al. (2020) lead to marginal posteriors on B that still display multimodality. We further explore this result in the rest of the next section.

Figure J-16: Larger model: posterior mean estimates of B and impulse responses, true versus estimated

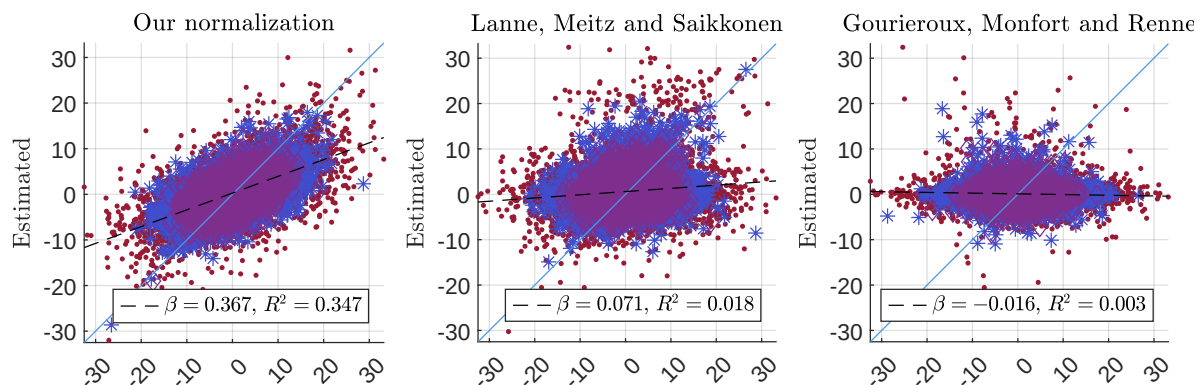
A) $k = 10$



B) $k = 20$



C) $k = 30$



Note: red dots refer to impact effects, stars and diamonds refer to the response after 4 and 8 horizons, respectively.

References

- Amengual, D., Fiorentini, G. & Sentana, E. (2022), ‘Moment tests of independent components’, *SERIEs: Journal of the Spanish Economic Association* **13**, 429–474.
- Amengual, D., Fiorentini, G. & Sentana, E. (2024), ‘Specification tests for non-Gaussian structural vector autoregressions’, *Journal of Econometrics* **244**(2), 105803.
- Angeletos, G.-M., Collard, F. & Dellas, H. (2020), ‘Business-cycle anatomy’, *American Economic Review* **110**(10), 3030–3070.
- Arias, J. E., Rubio-Ramirez, J. F. & Shin, M. (2023), ‘Macroeconomic forecasting and variable ordering in multivariate stochastic volatility models’, *Journal of Econometrics* **235**(2), 1054–1086.
- Arias, J. E., Rubio-Ramírez, J. F. & Shin, M. (2026), Large SVARs, Working Paper 26-04, Federal Reserve Bank of Philadelphia.
- Brunnermeier, M., Palia, D., Sastry, K. A. & Sims, C. A. (2021), ‘Feedbacks: financial markets and economic activity’, *American Economic Review* **111**(6), 1845–1879.
- Canova, F. & De Nicoló, G. (2002), ‘Monetary disturbances matter for business fluctuations in the G-7’, *Journal of Monetary Economics* **49**(6), 1131–1159.
- Chan, J. C., Koop, G. & Yu, X. (2024), ‘Large order-invariant Bayesian VARs with stochastic volatility’, *Journal of Business & Economic Statistics* **42**(2), 825–837.
- Eriksson, J. & Koivunen, V. (2004), ‘Identifiability, separability, and uniqueness of linear ICA models’, *IEEE Signal Processing Letters* **11**(7), 601–604.
- Fonseca, T. C. O., Ferreira, M. A. R. & Migon, H. S. (2008), ‘Objective Bayesian analysis for the Student-*t* regression model’, *Biometrika* **95**(2), 325–333.

- Geweke, J. (1993), ‘Bayesian treatment of the independent Student-t linear model’, *Journal of Applied Econometrics* **8**(S1), S19–S40.
- Gouriéroux, C., Monfort, A. & Renne, J.-P. (2020), ‘Identification and estimation in non-fundamental structural VARMA models’, *The Review of Economic Studies* **87**(4), 1915–1953.
- Hamilton, J. D., Waggoner, D. F. & Zha, T. (2007), ‘Normalization in Econometrics’, *Econometric Reviews* **26**(2-4), 221–252.
- Hoesch, L., Lee, A. & Mesters, G. (2024), ‘Locally robust inference for non-Gaussian SVAR models’, *Quantitative Economics* **15**(2), 523–570.
- Hosszejni, D. (2021), ‘Bayesian estimation of the degrees of freedom parameter of the Student- t distribution—A beneficial re-parameterization’, *arXiv preprint arXiv:2109.01726* .
- Jarociński, M. (2024), ‘Estimating the FED’s unconventional policy shocks’, *Journal of Monetary Economics* **144**, 103548.
- Kagan, A., Linnik, Y. V. & Rao, C. R. (1973), *Characterization Problems in Mathematical Statistics*, John Wiley & Sons Ltd.
- Keweloh, S. A. & Wang, S. (2026), ‘Uncertain Short-Run Restrictions and Statistically Identified Structural Vector Autoregressions’, *Journal of Applied Econometrics* **41**(1), 12–25.
- Kilian, L. & Demiroglu, U. (2000), ‘Residual-based tests for normality in autoregressions: Asymptotic theory and simulation evidence’, *Journal of Business & Economic Statistics* **18**(1), 40–50.
- Kilian, L. & Lütkepohl, H. (2017), *Structural vector autoregressive analysis*, Cambridge University Press.

- Kocięcki, A. (2010), ‘A prior for impulse responses in Bayesian structural VAR models’, *Journal of Business & Economic Statistics* **28**(1), 115–127.
- Lanne, M., Meitz, M. & Saikkonen, P. (2017), ‘Identification and estimation of non-Gaussian structural vector autoregressions’, *Journal of Econometrics* **196**(2), 288–304.
- Lewis, D. J. (2025), ‘Identification Based on Higher Moments in Macroeconometrics’, *Annual Review of Economics* **17**, 665–693.
- Read, M. & Zhu, D. (2026), ‘Fast posterior sampling in tightly identified SVARs using soft sign restrictions’, *arXiv preprint arXiv:2603.27088*.
- Waggoner, D. F. & Zha, T. (2003a), ‘A Gibbs sampler for structural vector autoregressions’, *Journal of Economic Dynamics and Control* **28**(2), 349–366.
- Waggoner, D. F. & Zha, T. (2003b), ‘Likelihood preserving normalization in multiple equation models’, *Journal of Econometrics* **114**(2), 329–347.